

Applied Precalculus

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University of Rhode Island

July 8, 2026

Website: My Website (pretextbook.org)

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Acknowledgements

The authors would like to thank the many instructors of MTH 103 at the University of Rhode Island, including full-time faculty, part-time faculty, and graduate teaching assistants, whose feedback has contributed to the improvement of both *Applied Precalculus* the course and *Applied Precalculus* the textbook over the years. Very special acknowledgment and boundless gratitude is owed to Kyle Monette, who worked closely with us on reformatting and editing as we prepared the second edition of *Applied Precalculus*. We also thank Faith Bailey for her contribution of additional exercises to our book.

Preface

As the title of the book suggests, our emphasis is on the real-life meaning of precalculus concepts. We want our students to master the basic concepts of precalculus while building a solid understanding of those concepts in practical contexts and the skills to apply mathematical tools to real-life processes.

To illustrate our approach, take for example the family of linear functions. When talking about linear functions, our emphasis is on the practical meaning and the units of the slope as well as on the units and the practical meaning of the horizontal and vertical intercepts. Such practical understanding is built through many applied examples and their careful analysis. We show how to use linear functions to model real-life processes which are described through tabulated data. At the same time, we try to ensure that students build the algebraic proficiency necessary to successfully work with linear functions.

We present all the main precalculus topics while stressing their practical meaning and their real-life applications. Those topics include:

- Functions, their graphs, and their numerical representation;
- Linear functions and their applications;
- Quadratic functions and quadratic equations;
- Algebra of exponential expressions, power functions, and their applications;
- Exponential and logarithmic functions, their algebraic properties and their rich variety of applications in the life sciences;
- Trigonometric functions and their applications to modeling periodic phenomena.

Our intended audience is primarily students in the life sciences and pharmacy. These students can be divided generally into two groups. The first group consists of students for whom this course is a terminal mathematics course, who wish to learn tools of before-calculus mathematics and to develop solid skills of

applying those tools to real-life problems. The second group consists of students who plan to continue their study of mathematics in an applied calculus course.

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Chapter 1

Introduction to Functions

The dose of medication that a pediatrician should prescribe to a child patient depends on the child's weight. The amount of medication remaining in a person's body depends on how much time has passed since they took the most recent dose. The distance between you and your destination depends on the speed at which you are traveling. The amount of practice problems you complete when studying for a mathematics class depends on how much time you spend working on them.

Each of the above describes a special type of dependence relationship between two variables called a *function*. Functions are used to model input—output relationships in a precise manner and are an essential part of the foundation for calculus and other advanced mathematical topics.

This chapter introduces the concept of a function and a variety of different ways that a function may be presented, including algebraic, graphical, and numerical representations. It also explains how functions are used to describe relationships in many real—world phenomena.

1.1 The Concept of a Function

Objectives

After completing this section, you should be able to do the following.

1. First objective.
 2. Second objective.
 3. Third objective.
-

One of the main concepts in mathematics and in the mathematical modeling of applied processes is that of a function. The formal mathematical definition of a function is abstract and general, but in applied precalculus we can think of a function as a special kind of dependence between two numerical variables.

Example 1.1.1

Suppose you are in Rhode Island and intend to buy a gift from an online store. The amount of money, C , that you will have to spend (that is, the total cost to purchase the gift) depends on the list price, p , of the item you select. The cost is equal to the list price plus the state of Rhode Island's 7% sales tax plus the rate the store charges for shipping and handling. Assuming that the online store charges a flat rate of \$4.95 for shipping and handling, give a formula for the cost C to purchase a gift of list price p dollars.

Solution. Given the list price p , you can calculate the cost C , in dollars, as

$$C = p + 0.07p + 4.95.$$

Here, $0.07p$ represents 7% of the list price—that is, the sales tax on the item.

In the above formula, C and p are variables—their values *vary* based on the item you select. They are both *numerical* variables as their possible values are numbers. The cost C depends on p and, moreover, the cost is uniquely determined by the item's list price p . We say in this case that C “is a function of p .” Using *function notation*, we write

$$C = f(p).$$

The notation $C = f(p)$ is read as “ C equals f of p .” Note that f does not stand for the word “function” here; it is merely a letter used to name the function and another letter could be used instead. The notation $f(p)$ represents the *value* of the function f at a given list price p ; that is, $f(p)$ is the *cost* corresponding to

a list price p :

$$f(p) = p + 0.07p + 4.95.$$

For every p , the value of the function $f(p)$ gives the cost C for a list price p . For example, $f(50)$ —the value of the function at $p = 50$ —gives the cost corresponding to the list price 50 dollars. Similarly, $f(60)$ gives the cost corresponding to the list price 60 dollars.

Example 1.1.2

Use the function

$$C = f(p) = p + 0.07p + 4.95$$

whose formula was found in Example 1.1.1 to determine the total amount you will spend to purchase a gift with a list price of $p = 80$ dollars from the online store.

Solution. We are looking for the cost C when $p = 80$. In other words, we are looking for $f(80)$ —read aloud as “ f of 80.” Using the formula for $f(p)$ with $p = 80$, we obtain

$$f(80) = 80 + 0.07 \cdot 80 + 4.95 = 90.55.$$

If you select an item listed at $p = 80$ dollars, you will pay $C = 90.55$ dollars. In function notation we could write $90.55 = f(80)$ or, equivalently, $f(80) = 90.55$.

Above, the cost variable, C , is called the *dependent variable* or the *output variable*. The list price, p , is the *independent variable* or the *input variable*. For each list price or input value—input for short—we have exactly one cost value. So we have exactly one output value—output for short—for each input. The formula for the function f gives the rule of how to obtain the output corresponding to each possible input p .

In general, we define a function as follows.

Definition 1.1.3 Informal Definition of a Function.

A function is a correspondence between two numerical variables, the input variable and the output variable, that takes input numbers from a certain subset of the real numbers called the *domain* of the function and prescribes to each ***exactly one*** output.

From the mathematical point of view, the domain of the formula for the function $f(p) = p + 0.07p + 4.95$ consists of all real numbers p . This is because $f(p)$ is unique and defined for each individual real number inputted for p . When we consider the applied context of this formula as presented in Example 1.1.1 and Example 1.1.2, however, the input p is a list price and therefore cannot be negative. For practical reasons, we would restrict the domain of f to positive inputs p .

Functions in General

In an abstract setting, when variables are not associated any specific real-life meaning, we commonly denote the independent variable by x , the dependent variable by y , and a function by f :

$$y = f(x).$$

Example 1.1.4

Let $y = f(x)$ where $f(x) = 0.2x + 1$.

1. What is the domain of f ?
2. Find $f(2)$ and $f(3)$.
3. Find all inputs x for which the output y is equal to 0.

Solution.

1. For every input number x , we can multiply x by 0.2 and add 1. So the output $f(x) = 0.2x + 1$ is defined and is a single number for each x . Hence, the domain of f consists of all real numbers x .
2. The quantity $f(2)$ is the value of the function or the output corresponding to the input $x = 2$. Substituting $x = 2$ into the formula for f , we find

$$f(2) = 0.2 \cdot 2 + 1 = 1.4.$$

Hence, the value of y corresponding to $x = 2$ is $y = 1.4$. Similarly,

$$f(3) = 0.2 \cdot 3 + 1 = 1.6.$$

3. We are looking for inputs x for which the output $y = 0$. In other words, we are looking for the value (or the values) of x which are solutions to the equation $0 = 0.2x + 1$ or equivalently $0.2x + 1 = 0$. To find x , we use the standard techniques of solving equations: we subtract 1 from both sides of the equation and then we divide both sides of the equation by 0.2. These steps are shown below.

$$\begin{aligned} 0.2x + 1 &= 0 \\ 0.2x + 1 - 1 &= 0 - 1 \\ \frac{0.2x}{0.2} &= \frac{-1}{0.2} \\ x &= -\frac{1}{0.2} \\ x &= -5 \end{aligned}$$

Functions in Applied Settings

The previous example illustrates that a function need not be associated with an applied scenario. When an application is involved, it is important to be able to interpret the practical meaning.

Example 1.1.5

The amount of nicotine in a person's bloodstream, $N = f(t)$, in milligrams, is a function of time t , in hours, that have passed since the person finished smoking a single cigarette.

1. Identify the independent variable and the dependent variable of this function.
2. In the statement $f(3) = 0.71$, what is the meaning of 3 and 0.71 in terms of time and nicotine? Include units in the answer.
3. Explain the meaning of the statement $f(4) = 0.5$ in practical terms.

Solution.

1. Here, N depends on t , making N a function of t . Thus N is the dependent variable and t is the independent variable.
2. The number 3 is a value of t , so its units are hours; 0.71 is a value of N , so its units are milligrams. In practical terms, $f(3) = 0.71$ means that 3 hours after a person finishes smoking a single cigarette there will be 0.71 milligrams of nicotine left in their bloodstream.
3. Similarly to the prior solution, $f(4) = 0.5$ means that 4 hours after a person finishes smoking a single cigarette there will be 0.5 milligrams of nicotine left in their bloodstream.

Example 1.1.6

In cases of strep throat, a daily pediatric dose of Amoxicillin, D , in milligrams, depends on the weight of a child, w , in kilograms; that is, D is a function of w . Denoting the function by g results in

$$D = g(w).$$

As a general rule, the daily dose should be 50 milligrams for each kilogram of weight.

1. Write a formula for the function $g(w)$.
2. Calculate the daily dose for a child who weighs 8 kilograms.

3. What does $g(11) = 550$ represent in practical terms?

Solution.

1. The value $g(w)$ represents the dose for the weight of w kilograms. Since the dose is 50 milligrams for each kilogram, the dose $g(w)$ is given by

$$50 \frac{\text{milligrams}}{\text{kilogram}} \times w \text{ kilograms.}$$

Multiplying and simplifying units results in $g(w) = 50w$ milligrams.

2. We calculate

$$g(8) = 50 \cdot 8 = 400$$

and arrive at the conclusion that the dose for a child who weighs 8 kilograms is 400 milligrams.

3. To answer questions like this, it can be useful to assign the appropriate units to each number. Since 11 is a value for w , it is measured in kilograms. 550 is a value for $g(w)$ or D , so it is measured in milligrams. Combining this information with the fact that $g(w)$ gives the dose for a child who weighs w kilograms, we conclude that the practical meaning of the statement $g(11) = 550$ is that a child weighing 11 kilograms should receive a daily dose of 550 milligrams of Amoxicillin when being treated for strep throat.

Example 1.1.7

A man leaves home and drives at 40 miles per hour toward a hospital located 150 miles away from his home. At time t hours after he began driving, his distance from the hospital, d , in miles, is given by

$$d = 150 - 40t.$$

1. Let $d = f(t)$. What is the formula for the function $f(t)$?
2. How far from the hospital is the man after 1.5 hours?
3. When will the man arrive at the hospital?

Solution.

1. Since $d = f(t)$ and the formula for d was given, $f(t) = 150 - 40t$.

2. The man's distance from the hospital at $t = 1.5$ is given by

$$f(1.5) = 150 - 40 \cdot 1.5 = 90.$$

After 1.5 hours, the man is 90 miles from the hospital.

3. The man will arrive at the hospital when his distance d from the hospital is 0 miles. We are looking for t such that $0 = 150 - 40t$ or equivalently $150 - 40t = 0$. Solving for t :

$$150 - 40t + 40t = 0 + 40t$$

$$150 = 40t$$

$$\frac{150}{40} = t$$

$$t = 3.75.$$

The man will arrive at the hospital 3 hours and 45 minutes after he left home.

Increasing and Decreasing Functions

In Example 1.1.6 we considered the function $D = g(w)$ that gives the daily dose D of Amoxicillin for a child weighing w kilograms. Notice that $g(w)$ increases as the weight w increases. In Example 1.1.7, the distance $d = f(t)$ from the hospital decreases as the number of hours t spent driving toward the hospital increases.

Definition 1.1.8 Increasing and Decreasing Functions.

- A function $f(x)$ is *increasing* if the values of $f(x)$ increase as x increases.
- A function $f(x)$ is *decreasing* if the values of $f(x)$ decrease as x increases.

Above, the dose of Amoxicillin would be an illustration of an increasing function of weight while the distance from the hospital would be an illustration of a decreasing function of time. We will see later that many functions are neither increasing nor decreasing through their domain; they may increase on some intervals and decrease on other intervals.

Functions as a Sequence of Operations

A function that is presented as a formula can be thought of as sequence of operations performed on an input in order to obtain the corresponding output. For example, the function $f(x) = 0.2x + 1$ of Example 1.1.4 takes an input, multiplies the input by 0.2, and then adds 1 to the result. The function performs this

sequence of operations no matter what the input is. We can write this symbolically as

$$f(\square) = 0.2 \cdot \square + 1.$$

Whatever input we feed into the function f , a number, an expression, whatever we like, the function will multiply the input by 0.2 and add 1.

By the way... In Chapter 7 we'll even consider putting *functions* into functions.

Example 1.1.9

Let $g(x) = 3 - 4x$. Evaluate and simplify each of the following.

1. $g(2 + h)$
2. $g(1 - h) + g(1)$

Solution. The function $g(x)$ takes any input $\square$ —whatever it might be—multiplies the input by 4, and subtracts the result from 3:

$$g(\square) = 3 - 4 \cdot \square.$$

1. In $g(2 + h)$, the input is an *algebraic expression*: $2 + h$. In any given context, h may stand for a number or for a numerical variable. No matter what h represents here, the function g takes the input $2 + h$, multiplies it by 4, and then subtracts the result from 3. Thus,

$$g(2 + h) = 3 - 4(2 + h).$$

To simplify, we use the distributive law to expand $4(2 + h)$:

$$g(2 + h) = 3 - 8 - 4h = -5 - 4h.$$

We cannot simplify further. The final answer is $g(2 + h) = -5 - 4h$.

2. We consider the whole expression all at once below, though you may find it helpful to separately find each of $g(1 - h)$ and $g(1)$ and then subtract the two results.

$$\begin{aligned} g(1 - h) + g(1) &= \left(3 - 4(1 - h)\right) + \left(3 - 4 \cdot 1\right) \\ &= \left(3 - 4 - 4(-h)\right) + \left(-1\right) \\ &= -1 + 4h - 1 \\ &= -2 + 4h \end{aligned}$$

Whatever h might be, $g(1 - h) + g(1) = -2 + 4h$.

Example 1.1.10

Find a formula for $f(x)$ if...

1. ... f takes the square root of the input, multiplies the result by 3, and subtracts 1.
2. ... f squares the input, adds 2, and takes the reciprocal of the result.

Solution.

1. We begin with an input x . Taking the square root of this input results in $\sqrt{x}$. Multiplying the result by 3 results in $3\sqrt{x}$. Finally, subtracting 1 results in $3\sqrt{x} - 1$. Therefore, the formula for f is

$$f(x) = 3\sqrt{x} - 1.$$

2. We begin with an input x . Squaring the input first results in x^2 . Adding 2 next results in $x^2 + 2$. Now, we take the reciprocal of $x^2 + 2$, which is $\frac{1}{x^2 + 2}$. Therefore, the formula for f is

$$f(x) = \frac{1}{x^2 + 2}.$$

Example 1.1.11

Find the domain of each of the following functions.

1. $f(x) = \frac{1}{x^2 - 1}$
2. $g(x) = \sqrt{x - 3}$

Solution. Note that neither $f(x)$ and $g(x)$ are assigned any specific practical applied meaning. This means that the domain of each function will be all inputs x for which the output is defined.

1. The function $f(x) = \frac{1}{x^2 - 1}$ is defined at any input that does not lead to division by zero and undefined at any input that does lead to division by zero. To find the value(s) of x for which $f(x)$ is undefined, we set its denominator equal to zero and solve for x :

$$\begin{aligned} x^2 - 1 &= 0 \\ x^2 - 1 + 1 &= 0 + 1 \\ x^2 &= 1 \end{aligned}$$

We can now see that $f(x)$ is undefined when $x = 1$ and when $x = -1$. The domain of $f(x)$ is all real numbers except for these two values. This can be written in a variety of different ways.

- Domain of $f(x) = \frac{1}{x^2 - 1}$ in words:

all real numbers $x \neq \pm 1$

- Domain of $f(x) = \frac{1}{x^2 - 1}$ in *set-builder notation*:

$$\{x \in \mathbb{R} : x \neq \pm 1\}$$

- Domain of $f(x) = \frac{1}{x^2 - 1}$ in *interval notation*:

By the way... The symbol $\cup$ is called a *union*. Essentially, it means “include all these intervals”.

$$(-\infty, -1) \cup (-1, 1) \cup (1, \infty)$$

- The function $g(x) = \sqrt{x - 3}$ is defined provided the number under the radical is not negative. That is, it is defined for those values of x satisfying that $x - 3 \geq 0$. We can isolate x on one side of the inequality

$$\begin{aligned} x - 3 &\geq 0 \\ x - 3 + 3 &\geq 0 + 3 \\ x &\geq 3 \end{aligned}$$

which allows us to conclude that the domain of $g(x)$ is all real numbers x that are greater than or equal to 3. As before, this can be written in a variety of different ways.

- Domain of $g(x) = \sqrt{x - 3}$ in words:

all real numbers $x \geq 3$

- Domain of $g(x) = \sqrt{x - 3}$ in *set-builder notation*:

$$\{x \in \mathbb{R} : x \geq 3\}$$

- Domain of $g(x) = \sqrt{x - 3}$ in *interval notation*:

By the way... Notice that we wrote $[3, \infty)$ and not $(3, \infty)$ to denote that $x = 3$ is included in the domain.

$[3, \infty)$

Practice Problems

1.

a. $f(1)$

b. $f(-1.5)$

c. $f(0)$

2. The value of a car, V , in dollars, is a function of the number of years, t , since it was purchased; that is, $V = g(t)$.

The independent variable is $(\square t \quad \square V \quad \square g)$ with units $(\square \text{ dollars} \quad \square \text{ years} \quad \square \text{ dollars/year})$.

The dependent variable is $(\square t \quad \square V \quad \square g)$ with units $(\square \text{ dollars} \quad \square \text{ years} \quad \square \text{ dollars/year})$.

Exercises

1. The amount of caffeine, C , measured in milligrams, in a person's body t hours after drinking a cup of coffee is given by the function $C = f(t)$. What do each of the following statements tell you in practical terms; that is, in terms of time and caffeine? Answer in complete sentences and include units with each number.

(a) $f(0) = 96$

(b) $f(5) = 48$

(c) $f(24) \approx 0$

2. The total number of units, P , produced after t hours of production is given by the function $P = h(t)$. What do each of the following statements tell you in practical terms; that is, in terms of units of product produced and time? Answer in complete sentences and include units with each number.

(a) $h(1) = 10$

(b) $0 = h(0)$

(c) $h(3) \approx 29$

3. The total cost of a meal in a restaurant, C , in dollars, as a function of the menu price of the meal, p , also in dollars, is given by

$$C = p + 0.20p$$

where the term $0.20p$ corresponds to a 20% tip.

(a) What is the input variable and what units is it measured in?

(b) What is the output variable and what units is it measured in?

(c) Calculate the total cost of a meal whose menu price is \$25.

4. A bakery makes 6 batches of cookies an hour, with a batch typically consisting of 12 cookies. The total number of batches baked, B , is measured as function of time t , in hours, so that $B = f(t) = 6t$.

(a) What is the independent variable?

(b) What is the dependent variable?

(c) Evaluate $f(3)$. What is the practical meaning of $f(3)$ in the applied context of this problem?

5. A driver is heading to a faraway town. The amount of fuel, G , in gallons, left in the fuel tank is a function of the number of miles m driven during the trip; that is, $G = f(m)$.

(a) What does the statement $f(70) = 6$ tell you in practical terms? What are units of the numbers 70 and 6?

(b) What does the statement $f(200) = 1$ tell you in practical terms?

6. Let $f(x) = 4 - 3x$. Evaluate and simplify if possible:

(a) $f(3b)$

(b) $f(b + 1)$

(c) $f(b + 3) - f(3)$

7. Let $g(x) = 3x - x^2$. Evaluate:

(a) $g(2)$

(b) $g(-2)$

(c) $g(\sqrt{2})$

8. Let $g(x) = x^2 + 3$. Evaluate and simplify if possible:

(a) $g(h - 1)$

(b) $g(h + 1)$

(c) $g(b + 1) - g(1)$

9. Let $f(x) = \frac{1}{x + 1}$. Evaluate. If the value is undefined, say so.

(a) $f(0)$

(b) $f(-1)$

(c) $f(-2)$

10. Let $f(x) = \frac{3}{x + 2}$. Evaluate and simplify if possible.

(a) $f(h - 1)$

(b) $f(h + 1)$

(c) $f(b) - 4f(1)$

11. Let $f(t) = t^2 + \frac{1}{t}$. Evaluate and simplify if possible.

(a) $f(2)$

(b) $f(h + 1)$

(c) $f(h + 1) - f(2)$

12. Let $g(x) = 2x^2 + \frac{3}{2}x$ and $f(x) = x + 3$. Evaluate and simplify if possible.

(a) $f(2) + g(1)$

(b) $g(1) + 2f(x + 1)$

(c) $f(2x) - g(4 + x)$

13. A URI student tutors for MTH 103. The cost of each session in terms of hours, t , is given by the function $c(t) = 10t + 15$.

(a) What is the cost of a 3 hour session?

(b) If $c(t) = \$60$, how many hours was the session?

14. The value of a car, V , in dollars, t years after purchase is given by the function $V = g(t)$, where $g(t) = 16500 - 1500t$.
- (a) What is the value of the car at the time of purchase?
 - (b) What is the value of the car 5 years after purchase?
 - (c) After how many years is the car worth nothing?
15. Since you see lightning immediately and it takes the sound of thunder about 5 seconds to travel a mile, you can calculate the distance between you and the lightning. Count the number of seconds, S , between the flash of lightning and the sound of thunder. Then the distance, D , in miles, between you and the lightning is given by the function:

$$D = \frac{S}{5}$$

- (a) Identify the independent and the dependent variable.
 - (b) How far is the lightning if you counted 4 seconds between the flash and the thunder?
16. The cost of a medium pizza is \$10 plus \$1.75 per topping. Write a function $f(x)$ to describe the cost of a medium pizza with x number of toppings.
17. Find a formula for a function $h(x)$ for each of the following scenarios:
- (a) h multiplies the input by 7 then adds 5 to the result.
 - (b) h adds 5 to the input and then multiplies the result by 7.
18. Find a formula for a function $f(x)$ for each of the following scenarios:
- (a) f takes the square of the input, multiplies the result by 5, then subtracts 8.
 - (b) f multiplies the input by 5, takes the square of the result, then subtracts 8.
 - (c) f subtracts 8 from the input, takes the square of the result, then multiplies by 5.
19. Find a formula for a function $g(x)$ for each of the following scenarios:
- (a) g takes the input, divides it by 3, then adds this to the square root of the input times 4.
 - (b) g takes the reciprocal of the square root of the input, adds 1 to this, and then adds the input squared.
20. Find the domain of each of the following functions.
- (a) $f(x) = x - 1$

$$(b) \ f(x) = \frac{1}{x-1}$$

$$(c) \ f(x) = \frac{2}{x^2-4}$$

$$(d) \ f(x) = \sqrt{x}$$

$$(e) \ f(x) = x^2 - 4$$

21. Find the domain of each of the following functions.

$$(a) \ f(x) = x^n, \text{ where } n \text{ is a positive integer}$$

$$(b) \ f(x) = x^2 + x + \frac{2}{x}$$

$$(c) \ f(x) = \frac{1}{\sqrt{x}}$$

1.2 The Graph of a Function

Objectives

After completing this section, you should be able to do the following.

1. First objective.
2. Second objective.
3. Third objective.

One way to visualize a function is by graphing it. The graph of a function $y = f(x)$ is the set of all ordered pairs or points $(x, f(x))$ in the xy -plane such that x is in the domain of the function f and $y = f(x)$ is the output assigned to the input x by f .

This graph is often a curve in the xy -plane. Specifically, it is the set of all ordered pairs $(x, f(x))$ for which the second coordinate y is the value of the function f at the first coordinate x ; that is, the graph is the set of all points (x, y) that satisfy the equation $y = f(x)$:

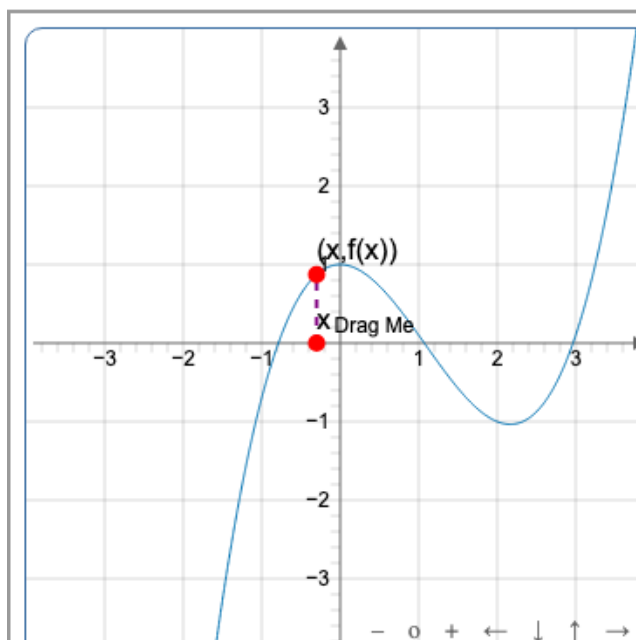
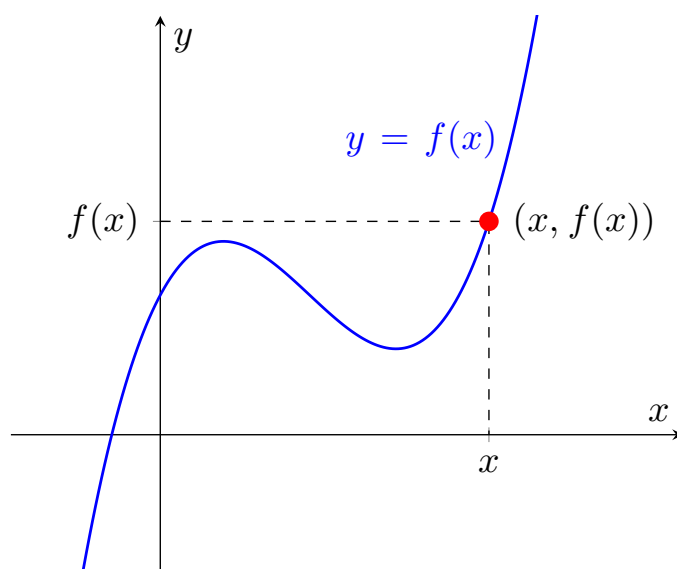


Figure 1.2.1



Standalone



It is not always reasonable or even possible to plot the ordered pair associated with each x in the domain of a given function $f(x)$. This is because functions can have many (sometimes even *infinitely* many) possible inputs. When drawing the graph of a function by hand, we customarily take relatively few points x , plot the corresponding ordered pairs $(x, f(x))$ in the xy -plane, and then join neighboring ordered pairs with straight line segments. The result is an approximation of the graph of the function. Graphing calculators and other graphing tools do exactly this—they just use a greater number of inputs x than we typically would when graphing by hand.

It is useful to state a precise definition of the graph of a function as similar definitions appear in other contexts in mathematics.

Definition 1.2.2 Graph of a Function.

The graph of a function $y = f(x)$ is the collection of all points (x, y) in the xy -plane for which $y = f(x)$. In other words, the graph is the collection of points $(x, f(x))$ for all inputs x in the domain of f .

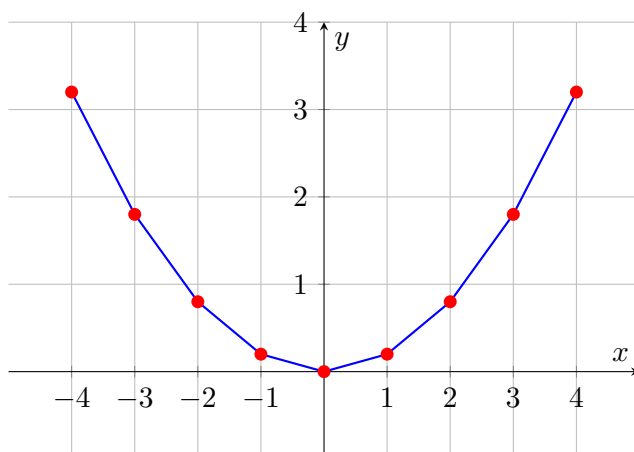
Example 1.2.3

Sketch the graph of the function $y = f(x)$ where $f(x) = 0.2x^2$ by hand. Plot points on the graph for $x = 0, \pm 1, \pm 2, \pm 3, \pm 4$.

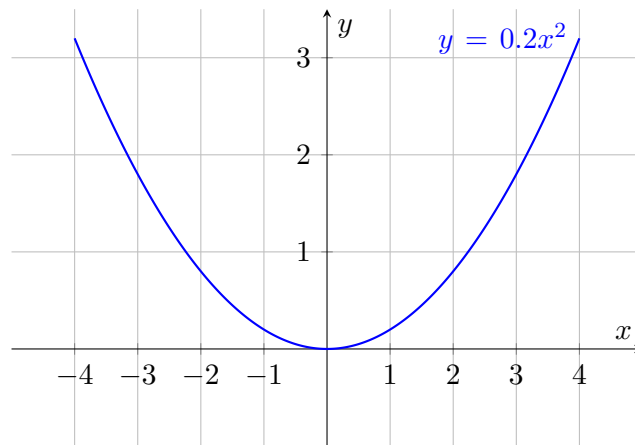
Solution. The below table gives each input x , the corresponding output y obtained by calculating $f(x)$, and then the associated point $(x, f(x))$.

x	$y = f(x)$	$(x, f(x))$ pair
-4	$0.2 \cdot (-4)^2 = 3.2$	$(-4, 3.2)$
-3	$0.2 \cdot (-3)^2 = 1.8$	$(-3, 1.8)$
-2	$0.2 \cdot (-2)^2 = 0.8$	$(-2, 0.8)$
-1	$0.2 \cdot (-1)^2 = 0.2$	$(-1, 0.2)$
0	$0.2 \cdot (0)^2 = 0$	$(0, 0)$
1	$0.2 \cdot (1)^2 = 0.2$	$(1, 0.2)$
2	$0.2 \cdot (2)^2 = 0.8$	$(2, 0.8)$
3	$0.2 \cdot (3)^2 = 1.8$	$(3, 1.8)$
4	$0.2 \cdot (4)^2 = 3.2$	$(4, 3.2)$

Plotting these points on the xy -plane and joining neighboring points by straight line segments results in the following rough sketch of $f(x) = 0.2x^2$:



This is a decent approximation of the graph of $y = 0.2x^2$. A better graph of $y = 0.2x^2$ can be obtained from a graphing calculator, graphing software package, or other similar tool.



In Example 1.2.3, the function $f(x)$ was given *algebraically*. That is, we were provided the algebraic formula $f(x) = 0.2x^2$ for the function.

Sometimes, a function is given *graphically*. That is, we are given only the graph of the function. The following examples illustrate that a great deal of information can be extracted from the graph of a function.

Example 1.2.4

The amount of nicotine in a person's bloodstream, $N = f(t)$, in milligrams, is a function of the time t , in hours, that has passed since the person smoked a single cigarette. The graph of the function $f(t)$ is given in Figure 1.2.5.

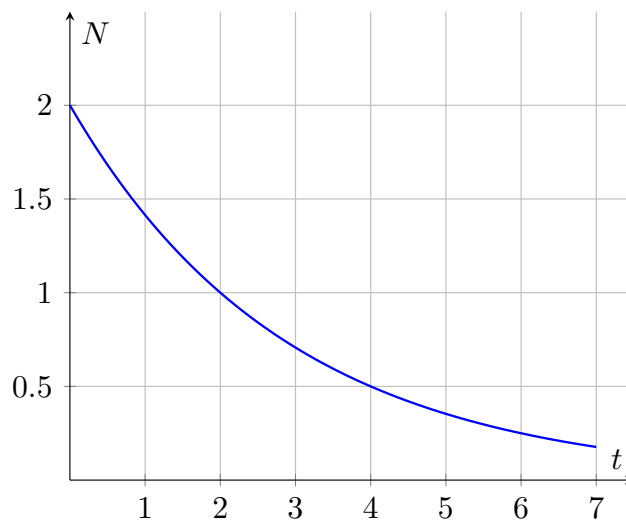


Figure 1.2.5

1. How much nicotine is absorbed from a single cigarette?

- How much nicotine is left in the person's bloodstream 2 hours after smoking a single cigarette? What about after 4 hours?

Solution.

- Note that $f(0) = 2$ (as the point $(0, 2)$ belongs to the graph of $f(t)$). Thus, at 0 hours—so immediately after a person finishes smoking a cigarette—the amount of nicotine in their bloodstream is 2 milligrams. This is the amount absorbed from the cigarette.
- As time goes on, the amount of nicotine decreases. We see from the grid on the tN -plane that $f(2) = 1$; in other words, the point $(2, 1)$ lies on the graph. This tells us that 2 hours after smoking a single cigarette there is 1 milligram of nicotine left in the bloodstream.

Similarly, we observe that $f(4) = 0.5$ and conclude that 4 hours after smoking a single cigarette there is only 0.5 milligram of nicotine left in the bloodstream.

Example 1.2.6

A woman is driving to visit with her family in a town 120 miles from her home. Let t be the time, in hours, since she left her home. Let d be the distance, in miles, to her destination. Here, d is a function of t : $d = g(t)$. The graph of the function $g(t)$ is given in Figure 1.2.7.

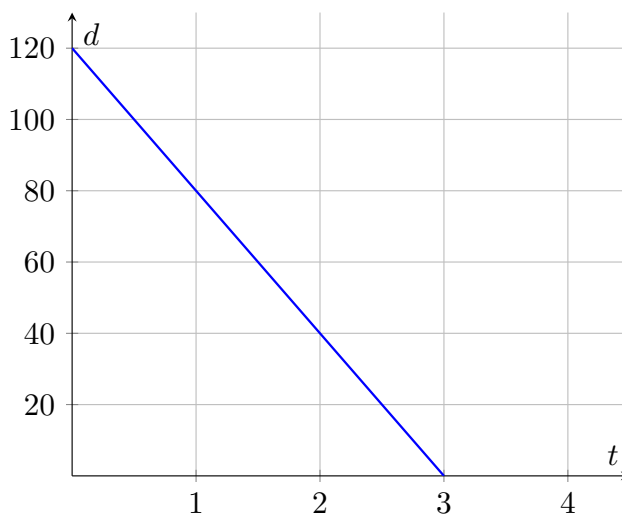


Figure 1.2.7

- What is her distance from her destination 1 hour after she leaves home?
- Estimate the time at which her distance from her destination is 60 miles.

3. When will she reach her destination?
4. How fast is she driving?

Solution.

1. The point on the graph above $t = 1$ is $(1, 80)$. In function notation, this can be written as $g(1) = 80$. The practical meaning of this notation is that the distance to her destination after 1 hour of driving is 80 miles.
2. We look for the point on the graph for which the second coordinate is $d = 60$. The t -coordinate of that point seems to be at $t = 1.5$. We conclude that the point $(1.5, 60)$ lies on the graph, and that the associated function notation is $g(1.5) = 60$. This means that after 1.5 hours, the woman is 60 miles from her destination.
3. The woman reaches her destination when $d = 0$; that is, when $g(t) = 0$. From the graph, it is clear that $g(3) = 0$. Hence the woman arrives at her destination after 3 hours.
4. In the first hour of driving, her distance from her destination drops from 120 miles to 80 miles. After the next hour, it drops from 80 miles to 40 miles, and then from 40 miles to 0 miles during the hour after that. This pattern reveals that the woman is traveling at 40 miles per hour.

Graphs of Increasing and Decreasing Functions

In graphical terms:

- A function is increasing if its graph **climbs** as the independent variable increases; that is, as we move from left to right.
- A function is decreasing if its graph **falls** as the independent variable increases; that is, as we move from left to right.

The function $N = f(t)$ in Example 1.2.3 is decreasing—its graph is falling as t increases. The amount of nicotine decreases as the amount of time after smoking a cigarette increases. In Example 1.2.6 the distance to the destination $d = g(t)$ decreases as the time t spent driving increases. The function in Example 1.2.3 is neither increasing nor decreasing in its domain. At first, over the negative x -axis the graph is falling; the values $f(x)$ decrease as x increases. Then, over the positive x -axis, the graph is climbing; the values

$f(x)$ increase as x increases. We can say that f is decreasing on the interval $x < 0$ and increasing on the interval $x > 0$.

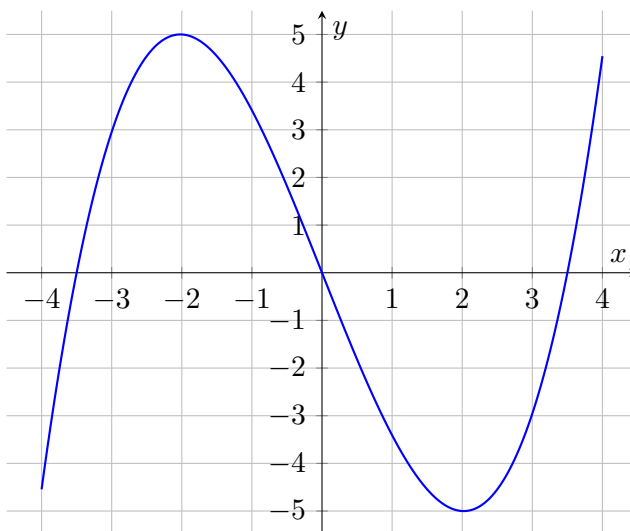
By the way... We typically require that these intervals I be *open*: $(\cdot, \cdot)$. Why? You'll see in calculus.

Definition 1.2.8 Functions Increasing or Decreasing on Intervals.

- A function $f(x)$ is increasing on an interval I if the values $f(x)$ increase as x increases along I .
- A function $f(x)$ is decreasing on an interval I if the values $f(x)$ decrease as x increases along I .

Example 1.2.9

Identify the interval(s) on which the function $f(x)$ depicted below is increasing and the interval(s) on which it is decreasing.



Solution. The graph of $f(x)$ is climbing on the interval $-4 < x < -2$ and on the interval $2 < x < 4$. Hence, the function is increasing on these intervals. In interval notation, we would write that the function is increasing on $(-4, -2) \cup (2, 4)$.

The graph of $f(x)$ is falling and thus $f(x)$ is decreasing on the interval $-2 < x < 2$. In interval notation, we would write that the function is decreasing on $(-2, 2)$.

By the way... Note that we do not use a bracket [or] around any of the endpoints $x = -4$, $x = 2$, or $x = 4$ that appear in the interval notation here. This is because a function is neither increasing nor decreasing at these endpoints—it either “stops” at them, or “turns around.”

Vertical Line Test

We have established that the graph of a function $y = f(x)$ is often a curve in the xy -plane. However, not every curve in the xy -plane is the graph of a function. The definition of a function requires that for each input x in the domain there is *exactly one* output y . That is, there cannot be two different outputs corresponding to the same input.

The Vertical Line Test is a simple visual way of determining if a given curve is or is not the graph of a function.

Result 1.2.10 Vertical Line Test.

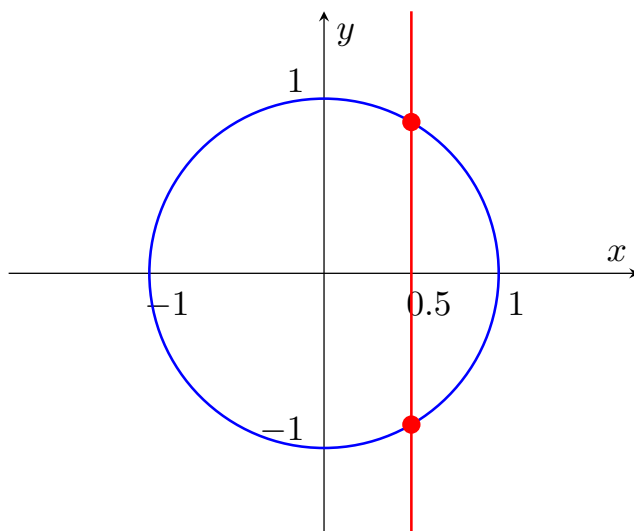
Let a curve in the xy -plane be given. If there is a vertical line that intersects the curve more than once, then the curve does not represent a function. If every vertical line intersects the curve at most once, then the curve represents the graph of a function.

By the way... This test clearly requires having the *graph* of a curve. In the next section, when a table of values is given, we cannot use this test.

Example 1.2.11

Is the circle of radius 1 centered at the origin $(0, 0)$ in the xy -plane the graph of a function?

Solution. Consider the vertical line corresponding to $x = 0.5$; that is, the vertical line passing through the point $(0.5, 0)$ on the x -axis.



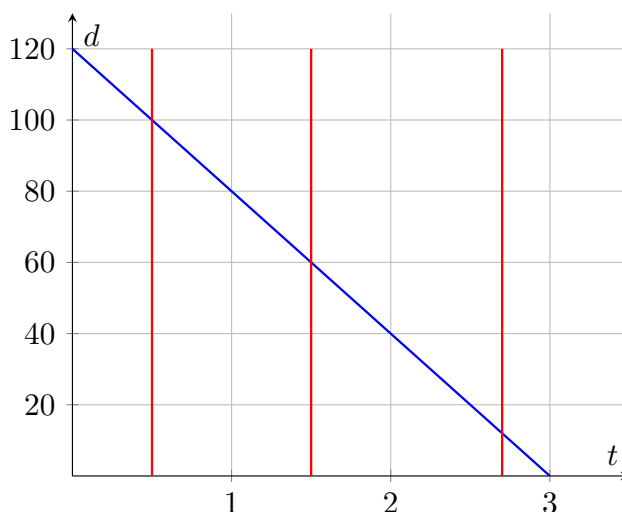
By the way... It takes only *one* vertical line intersecting the curve twice to conclude that the curve is not a function!

This vertical line intersects the circle at two points with two different values of y , meaning there are two outputs corresponding to the input $x = 0.5$. This violates the definition of a function. Hence, the circle is NOT the graph of a function.

In utilizing the Vertical Line Test, it doesn't matter if the coordinates on the plane are labeled with x , y , or other letters. The idea is the same.

Example 1.2.12

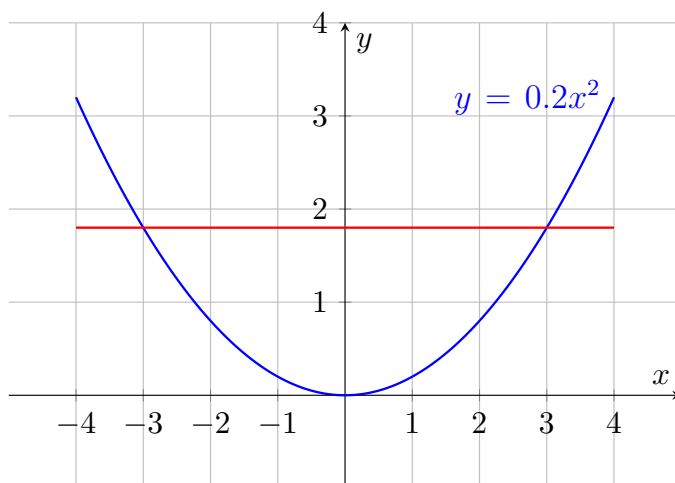
Is the graph of a straight line segment between the ordered pairs $(0, 120)$ and $(3, 0)$ in the td -plane the graph of a function?



Solution. The line segment *is* the graph of a function whose domain is the interval $0 \leq t \leq 3$. Using the Vertical Line Test, we can see that each individual vertical line that passes through a specific value of t with $0 \leq t \leq 3$ on the horizontal axis crosses the line segment at exactly one point. Any vertical line that passes through a value of t outside the interval $0 \leq t \leq 3$ does not intersect the segment.

Note that, in general, a function can have the same output for two different inputs. In Example 1.2.3, the function $f(x) = 0.2x^2$ gives the same output $y = 1.8$ for both $x = 3$ and $x = -3$. Still, for each x , we have only one value of y .

Having the same output for two different inputs means that some *horizontal* lines intersect the graph more than once which is perfectly fine for a function:



We will revisit graphs of functions in Section 1.4.

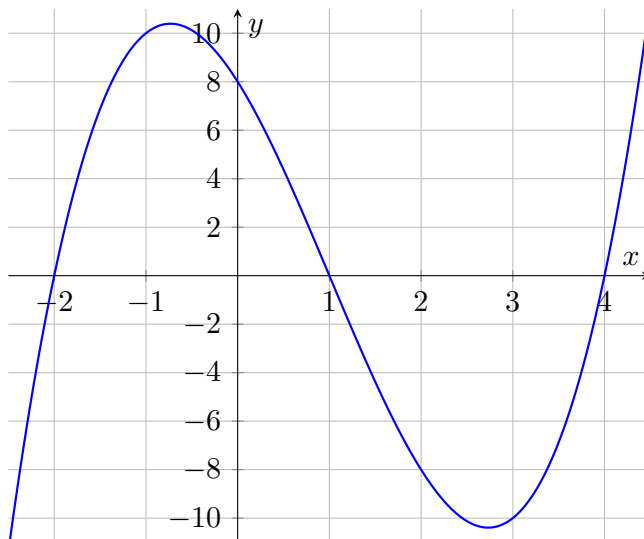
Exercises

1. Create a table of values and sketch the graph of the function $y = x^3$ for $-2 \leq x \leq 2$. Then use a graphing calculator or other graphing utility to check your graph.
2. Create a table of values and sketch the graph of the function $f(t) = t^2 + t$ for $-2 \leq t \leq 1$. Then use your calculator or any other graphing utility to check your graph.
3. Create a table of values and sketch the graph of the function $g(x) = \sqrt{x}$ for $x \geq 0$. Then use a graphing calculator or other graphing utility to check your graph.
4. The total cost of a meal in a restaurant, C , in dollars, as a function of the price of the meal, p , in dollars is given by:

$$C = p + 0.20p$$

where the term $0.20p$ corresponds to the 20% tip. Create a table of values and sketch the graph of the function $C = p + 0.20p$ for $10 \leq p \leq 20$. Then use your calculator or any other graphing utility to check your graph.

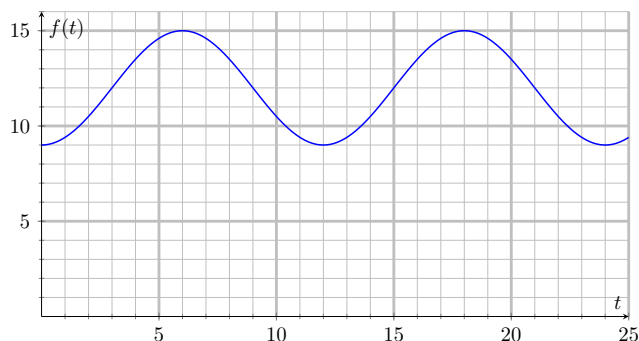
5. Use the graph of the function $y = f(x)$ below to estimate each of the following.



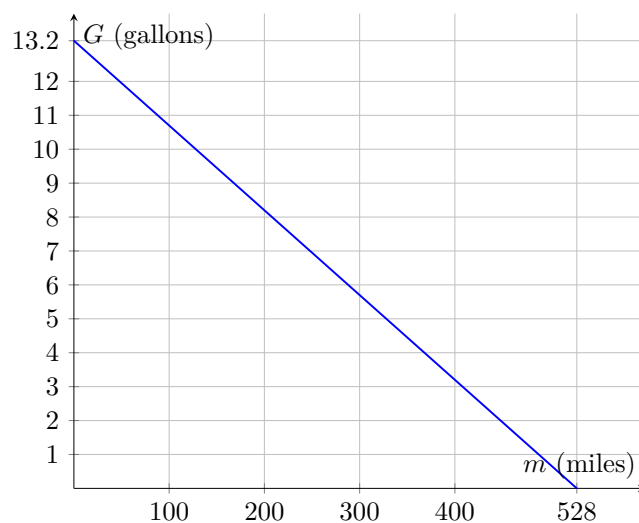
- (a) $f(2)$
 - (b) $f(3)$
 - (c) $f(-1)$
6. For the function $y = f(x)$ whose graph is given in Exercise 5, estimate all values x for which $f(x) = 0$.
 7. For the function $y = f(x)$ whose graph is given in Exercise 5, estimate all values of x for which $f(x) = 8$.

8. Use the graph of the function $y = f(t)$ shown below to estimate each of the following.

- (a) $f(0)$
- (b) $f(12)$
- (c) $f(10)$

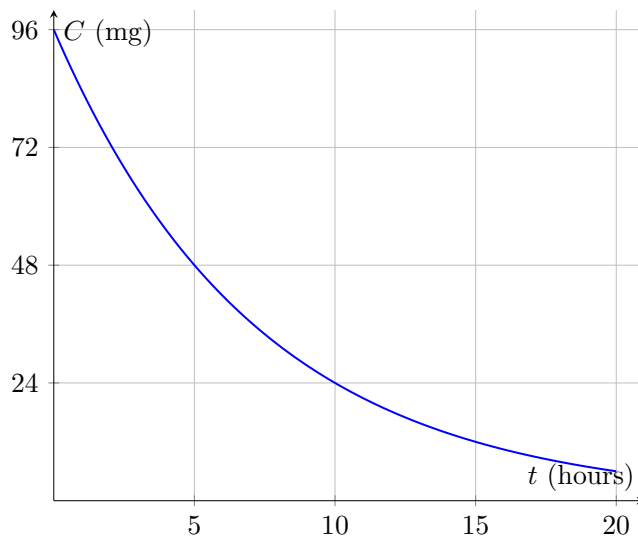


9. For the function $y = f(t)$ whose graph is given in Exercise 8, estimate all values of t for which $f(t) = 15$.
10. For the function $y = f(t)$ whose graph is given in Exercise 8, estimate all values of t for which $f(t) = 12$.
11. A driver of a 2019 Toyota Corolla fills his gas tank and embarks on a highway trip. The amount of gas left in the tank, G , in gallons, is a function of the number of miles m driven, $G = g(m)$. Use the graph of $g(m)$ given below to answer the following questions.

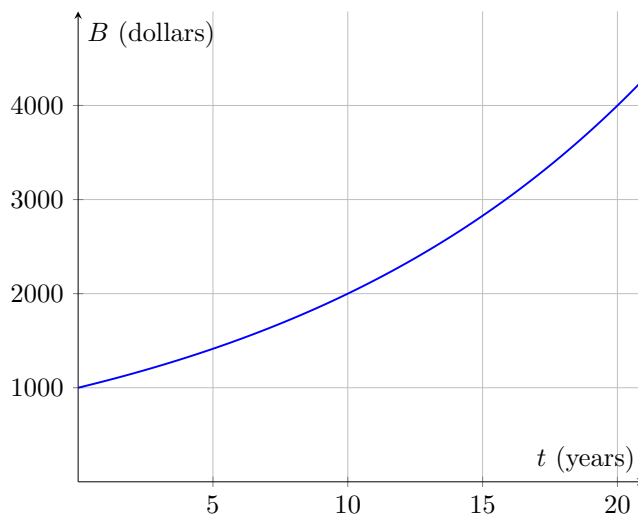


- (a) What is the fuel tank capacity of the 2019 Toyota Corolla?
- (b) How much fuel is left after 200 miles?

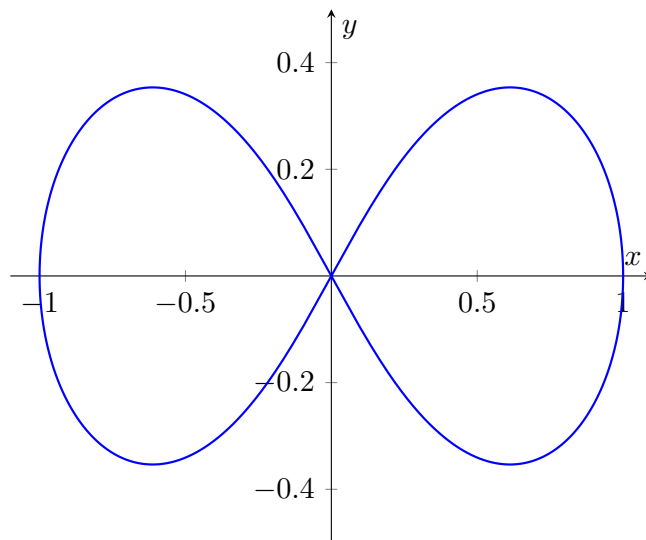
- (c) What happens after 528 miles?
- (d) What is the fuel efficiency of the 2019 Toyota Corolla on the highway?
- 12.** The amount of caffeine remaining in the body, C , in milligrams, t hours after drinking a cup of coffee, $C = C(t)$ is given by the graph below.



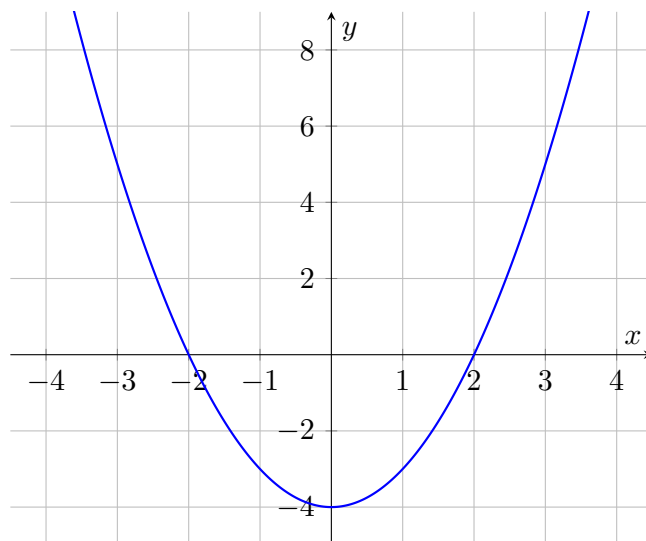
- (a) How much caffeine was absorbed into the bloodstream from the cup of coffee?
- (b) How much caffeine is left after 5 hours? After 10 hours?
- (c) Is the function $C(t)$ increasing, decreasing or neither on the interval $t \geq 0$?
- 13.** A man deposited money into a savings account. His balance $B(t)$, in dollars, after t years is given by the graph below.



- (a) What was his initial deposit?
- (b) How much money was in his account after 10 years? After 20 years?
- (c) Is the function $B(t)$ increasing, decreasing or neither in the interval $t \geq 0$?
14. Is the curve below the graph of a function $y = f(x)$? Explain your answer.

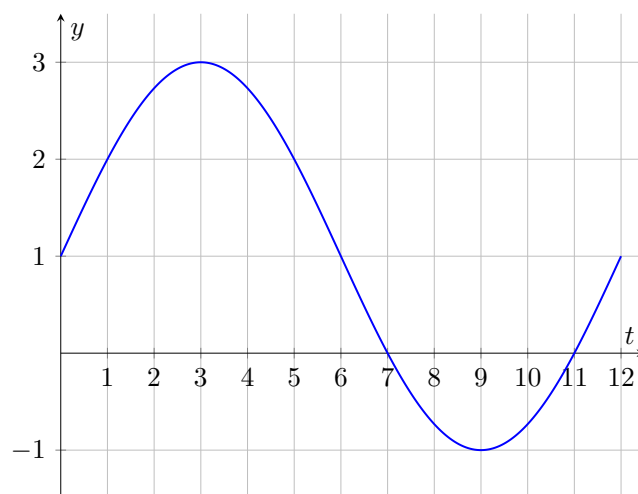


15. The graph of a function $y = f(x)$ is given below. Use it to find the following.



- (a) Estimate $f(0)$.
- (b) Estimate all values of x for which $f(x) = 0$.
- (c) Estimate all values of x for which $f(x) = 5$.

- 16.** For the function $y = f(t)$ whose graph is depicted below, identify the intervals on the t -axis for which the function is increasing and for which the function is decreasing.



1.3 Functions Given Numerically

Objectives

After completing this section, you should be able to do the following.

1. First objective.
 2. Second objective.
 3. Third objective.
-

In the previous section, we saw that if a function is given algebraically—by a formula—we can easily create a table of values for that function. That table of values can then be used to sketch the graph of the function, which in turn provides a visual representation of the function that can allow us to see behavior that may not be readily apparent from the formula alone.

When we study a real-life process, which inevitably involves a function of some kind, we rarely begin with a formula for a function that represents—or *models*—the process. Typically, we take measurements and collect and tabulate numerical data for the process we are studying. We then try to find a formula for a function that “fits” the numerical information we collected. That is, we *start* with a table of values and *then* try to find a formula for a function, not the other way around.

Finding a mathematical description of a real-life process is called *mathematical modeling*. A function that is given by its table of values is said to be given *numerically*.

If you ever kept overripe bananas in your house a bit too long, you probably experienced an invasion by fruit flies. And you probably noticed how quickly the population of fruit flies in your house was increasing. It is difficult to count the number of fruit flies in a house, but it can be done in a laboratory.

Suppose that a population of fruit flies in a laboratory experiment is initially 100 flies and it begins to grow. The size of the population, P , measured in *thousands* of flies, is a function of the time t since the experiment began; that is, $P = f(t)$. Given how quickly fruit flies multiply, it is reasonable to measure t in days. The scientists conducting the experiment do not have a formula for $f(t)$ that describes the growth of this fruit fly population and, in general, that growth (and therefore the function $f(t)$) depends on many factors controlled by the lab—the temperature, food provided, etc. Taking into consideration the length of the fruit fly reproductive cycle, the scientists observe the population and take measurements every 14 days. The data they collect is provided in the table below.

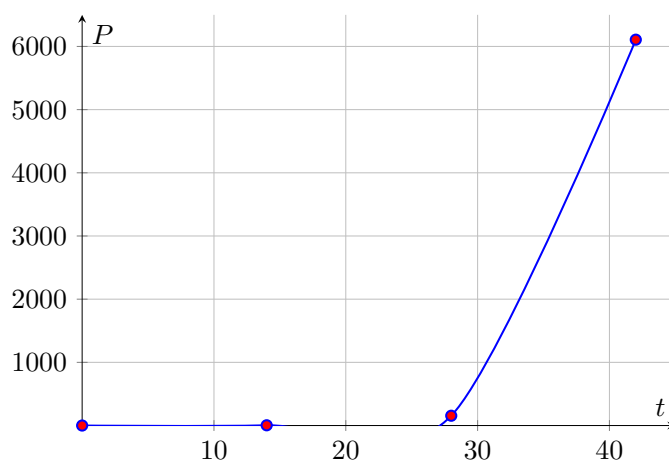
Table 1.3.1

t (in days)	0	14	28	42	56
$P = f(t)$ (in thousands)	0.1	3.94	155.12	6106.99	240432.20

We see that $f(t)$ is an increasing function and it increases very fast. After 56 days, the initial population of 100 flies grew to 240432.2 thousand flies; that is, 240,432,200 flies!

Can we find a formula for a function $f(t)$ so that Table 1.3.1 is a table of values for $f(t)$? In other words, can we find a mathematical model for the population growth exhibited in the fruit fly experiment? It turns out that finding a formula for a function that corresponds to experimental data, even if only approximately, may be difficult. In fact, it can be impossible. A formula corresponding to Table 1.3.1 *can* be found, as the growth of the fruit fly population is exponential. You will see how this is done when we cover exponential functions in Chapter 5.

One can obtain a visual representation of a function from a numerical representation by sketching its approximate graph. The function $f(t)$ associated with Table 1.3.1 for t between 0 and 42 is plotted below. The value at $t = 56$ is so large that it squeezes the graph vertically too much and makes it hard to read, which is why we excluded it.



Example 1.3.2

On December 31, 2019 the price of ExxonMobil stock (NYSE: XOM) was changing during the day. The price of the stock each hour from 10 am to 4 pm is recorded in the table below.

Table 1.3.3

time	10 am	11 am	12 pm	1 pm	2 pm	3 pm	4 pm
price (dollars)	69.35	69.17	69.41	69.38	69.32	69.32	69.78

1. Let $E = p(t)$ be the function that gives the price of ExxonMobil stock t hours after 10 am on

December 31, 2019. Rewrite Table 1.3.3 with the first row giving values of t and the second values of $E = p(t)$.

2. Sketch an approximate graph of $p(t)$.
3. Does it appear simple to find a formula or mathematical model for the function $p(t)$?

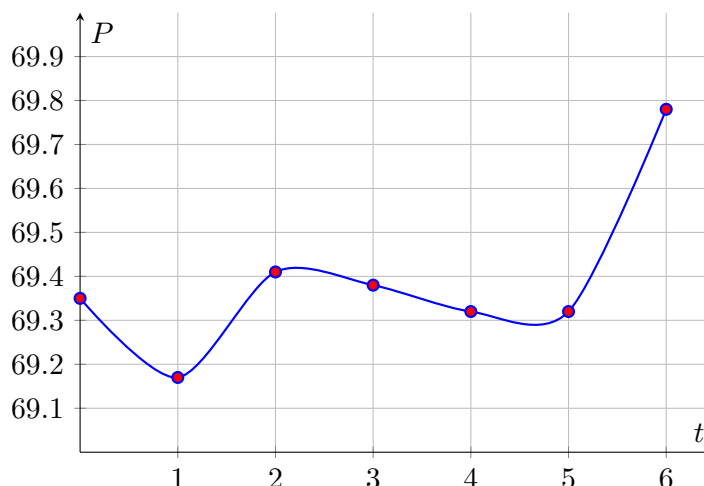
Solution.

1. We denote by t the time, in hours, since 10 am on December 31, 2019. This means that 10 am is $t = 0$, 11 am is $t = 1$, 4 pm is $t = 6$, and so on. As the function $E = p(t)$ gives the price of ExxonMobil stock at time t , we can express $E = p(t)$ numerically by:

Table 1.3.4

t	0	1	2	3	4	5	6
$p(t)$	69.35	69.17	69.41	69.38	69.32	69.32	69.78

2. We use Table 1.3.4 to sketch an approximation of the graph of $E = p(t)$.



Note that the red ordered pairs are guaranteed to be on the graph of $p(t)$, while the line segments used to connect them are not.

3. The function $p(t)$ is neither increasing nor decreasing as the graph sometimes climbs and sometimes falls as t increases. Finding a formula for the function $p(t)$ that models fluctuations of the price of a stock does not seem simple and we will not attempt to do so.

Example 1.3.5

A woman goes to a gym to exercise. After t minutes on a treadmill, her pulse (heart rate), H , in beats per minute, is:

Table 1.3.6

t (minutes)	0	2	4	6	8	10
H (bpm)	80	84	88	92	96	100

1. Express the above table as a function.
2. Does it appear simple to find a formula or mathematical model for this function?
3. Sketch an approximate graph of the function.

Solution.

1. The woman's pulse, H , is a function of time t , which, based on the table, we will measure in minutes since her treadmill workout began. We could write $H = f(t)$ if we chose to name the associated function f . For the sake of simplicity, we often denote the dependent variable and the function by the same letter. In this case, we will instead write $H = H(t)$.
2. Can we find a formula for $H(t)$? We haven't developed any specific techniques to do so yet, and hence to answer this question we can only attempt to observe patterns and make educated guesses based on common sense.

Looking at the data in Table 1.3.6, we can see that the woman's pulse H increases as time t increases, which we might expect based on our own experience with exercise. How does H increase? The initial value of H — $H(0)$ in function notation—is 80. During the first two minutes from $t = 0$ to $t = 2$, her pulse increases by 4 bpm from 80 to 84. During the next two minutes from $t = 2$ to $t = 4$, her pulse increases again by 4 bpm from 84 to 88. Looking at each of the remaining two-minute intervals, we see that her pulse increases by 4 bpm every two minutes: 88 to 92, 92 to 96, and 96 to 100.

As we attempt to find a possible formula for $H(t)$, we observe that if her pulse is increasing by 4 bpm every two minutes, we could perhaps think of this as 2 bpm every 1 minute instead. So her pulse starts at an initial value of 80 bpm and then we add 2 bpm for every one-minute change in t . The formula that reflects that is:

$$H(t) = 80 + 2t.$$

Let's check if the values of $H(t)$ reflect the data in Table 1.3.6:

$$H(0) = 80 + 2 \cdot 0 = 80$$

$$H(2) = 80 + 2 \cdot 2 = 84$$

$$H(4) = 80 + 2 \cdot 4 = 88$$

$$H(6) = 80 + 2 \cdot 6 = 92$$

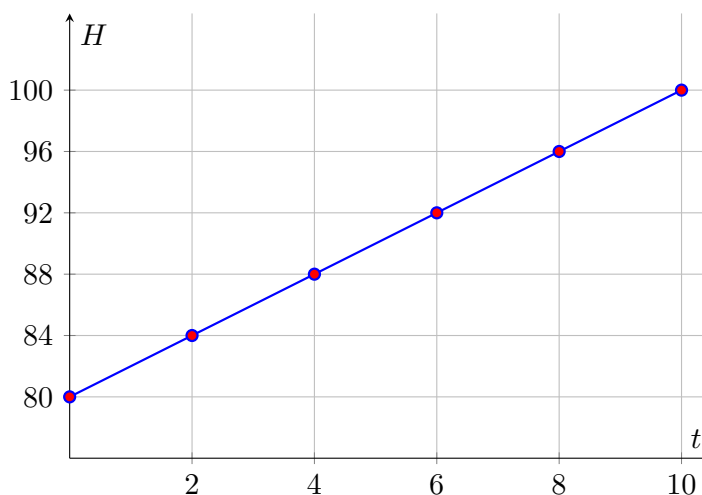
$$H(8) = 80 + 2 \cdot 8 = 96$$

$$H(10) = 80 + 2 \cdot 10 = 100.$$

Yes! Our formula produces all the right values and matches the information given in Table 1.3.6.

Note that the formula $H(t) = 80 + 2t$ gives a *possible* formula for $H(t)$. It is entirely possible that the woman's actual heart rate at time $t = 1.5$ minutes does not match what the formula $H(t) = 80 + 2t$ suggests it should be. We have no way of knowing, as the table does not tell us anything about the values of $H(t)$ at any times other than $t = 0, 2, 4, 6, 8, 10$ minutes.

3. Here is an approximate graph of the function $H = H(t)$ based on the numerical data:



Example 1.3.7

Functions $y = f(x)$ and $y = g(x)$ are given numerically below. Find possible formulas for $f(x)$ and $g(x)$.

Table 1.3.8

x	0	1	2	3	4	5
$f(x)$	0	-1	-4	-9	-16	-25

1.

Table 1.3.9

x	0	1	2	3	4	5
$g(x)$	80	78	76	74	72	70

2.

Solution.

1. We notice that the values of $f(x)$ for each x given in the table are negative and have the magnitude equal to x^2 . Hence, a possible function represented by Table 1.3.8 is $f(x) = -x^2$. We can easily check that $f(x) = -x^2$ works by substituting $x = 0, 1, 2, 3, 4, 5$ into $f(x)$ and calculating the corresponding values. Those are indeed 0, -1 , -4 , -9 , -16 , and -25 , respectively.
2. At $x = 0$, $g(0) = 80$. Then, as x increases, $g(x)$ decreases in a very regular fashion: for each increase of 1 in x , $g(x)$ decreases by 2. A formula that gives exactly such behavior is:

$$g(x) = 80 - 2x.$$

We can verify that this function produces the values in Table 1.3.9:

$$\begin{array}{ll}
 g(0) = 80 - 2 \cdot 0 = 80 & g(1) = 80 - 2 \cdot 1 = 78 \\
 g(2) = 80 - 2 \cdot 2 = 76 & g(3) = 80 - 2 \cdot 3 = 74 \\
 g(4) = 80 - 2 \cdot 4 = 72 & g(5) = 80 - 2 \cdot 5 = 70
 \end{array}$$

For now, we can only try to observe patterns and “guess” formulas for numerically given functions. As we study various families of functions, we will develop more systematic methods for finding formulas of functions given numerically.

Exercises

1. Use the numerically given function $f(x)$ below to find each of the following.

x	0	5	10	15	20	25
$f(x)$	0	2.23	3.16	3.87	4.47	5

(a) $f(5)$ (b) $f(15)$

(c) $f(20)$

2. Let $f(x)$ be the function given in Exercise 1.3.1.

(a) Find all values of x for which $f(x) = 0$ (b) Find all values of x for which $f(x) = 2.23$

3. Use the numerically given function $g(t)$ below to find each of the following.

t	0	2	4	6	8	10
$g(t)$	1.5	3.7	1.5	4.2	1.5	0

(a) $g(4)$ (b) $g(2)$ (c) $g(8)$

4. Let $g(t)$ be the function given in Exercise 1.3.3.

(a) Find all values of t for which $g(t) = 1.5$.(b) Find all values of t for which $g(t) = 3.7$.

5. Determine a formula for the function $h(x)$ given numerically in the table below.

x	-2	-1	0	1	2	3	4
$h(x)$	-8	-1	0	1	8	27	64

6. Determine a formula for the function $F(t)$ given numerically in the table below and fill in the missing values.

t	-2	-1	0	1	2	3	4	5	6
$F(t)$	4	2	0	-2	-4	-6	-8	?	?

7. A driver of a Volkswagen Passat fills up his gas tank and starts a highway trip to a faraway city. Let G , in gallons, be the amount of gas left in the tank after driving d miles. Fill in the missing numbers and find the gas mileage of the VW Passat; that is, the number of miles the SUV gets per gallon.

d	0	36	72	108	144	180	216	252
$G(d)$	18.5	17	15.5	14	12.5	?	?	?

8. Data regarding the world population between 2010 and 2018, in billions, is recorded in the table below.

Year	2010	2011	2012	2013	2014	2015	2016	2017	2018
Population	6.957	7.041	7.126	7.211	7.296	7.380	7.464	7.548	7.631

The information in the table can be rewritten so that the population, P , in billions, is a function of the number of years, t , since 2010. This rewriting has been partially completed for you. Fill in the missing values.

t	0	1	?	3	4	?	6	7	8
$P(t)$	6.957	7.041	?	7.211	7.296	7.380	7.464	?	7.631

9. Sketch an approximate graph of the population function $P(t)$ whose table you completed in Exercise 1.3.8.
10. The amount of power a dragon must output to carry a knight depends on how fast it is flying, as shown in the table below.

speed x (mph)	60	70	80	90	100
power y (kW)	23	25	30	36	45

- (a) Is the amount of power required for the dragon to carry the knight increasing or decreasing as its speed increases? Explain how you arrive at your conclusion from the data in the table.
 - (b) Is the amount of power required for the dragon to carry the knight changing faster and faster or slower and slower as its speed increases? Explain how you arrive at your conclusion from the data in the table.
11. Sketch the graph of the function given numerically in Exercise 1.3.10 and use it to answer the following questions.
 - (a) Is the amount of power required for the dragon to carry the knight increasing or decreasing as its speed increases? Explain how you arrive at your conclusion from the graph.
 - (b) Is the amount of power required for the dragon to carry the knight changing faster and faster or slower and slower as its speed increases? Explain how you arrive at your conclusion from the graph.

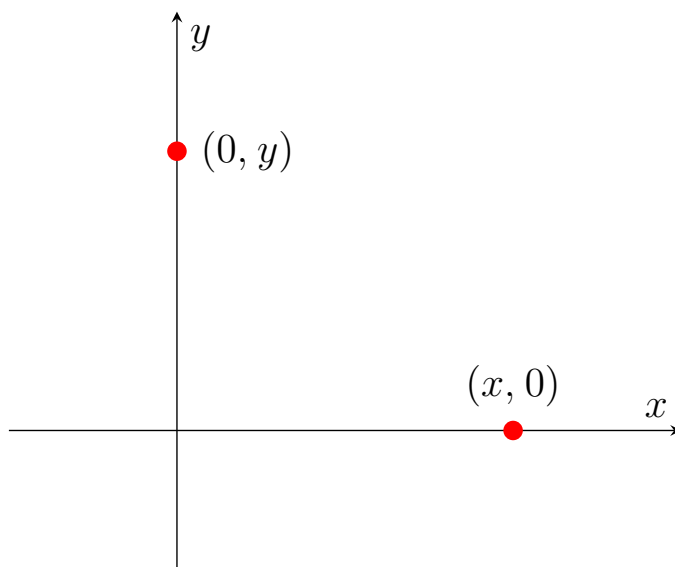
1.4 Vertical and Horizontal Intercepts

Objectives

After completing this section, you should be able to do the following.

1. First objective.
 2. Second objective.
 3. Third objective.
-

The points where the graph of a function crosses the horizontal axis or the vertical axis are of special interest. They are called the horizontal and the vertical intercepts. If the independent variable is named x and the dependent variable is named y , the horizontal and vertical intercepts are also called x -intercepts and y -intercepts, respectively.



The plot above illustrates that every point in the xy -plane that lies on the y -axis has the x -coordinate 0, and every point that lies on x -axis has the y -coordinate 0. We will use this observation to find the vertical and horizontal intercepts (if they exist).

Definition 1.4.1 Vertical Intercept.

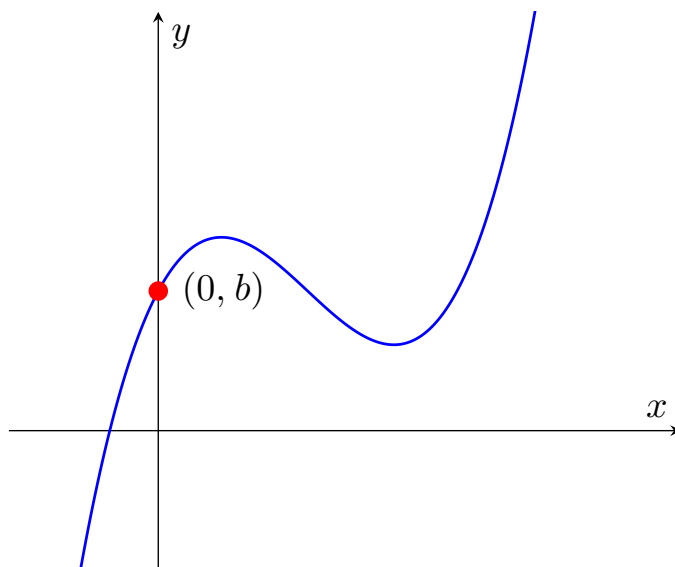
Suppose the graph of the function $y = f(x)$ crosses the y -axis at $y = b$; that is, at the point $(0, b)$. Then b is called the y -intercept of the function f .

Since $(0, b)$ is on the graph of the function, $b = f(0)$. So the y -intercept is simply the value of the function at $x = 0$.

By the way... Why can a function not have more than one vertical intercept? Hint: Consider the vertical line test.

If $x = 0$ is in the domain of f , we have exactly one y -intercept. If $x = 0$ is not in the domain, the graph of f does not cross the y -axis and there is no y -intercept.

That is: *To find the vertical intercept of $f(x)$, evaluate $f(0)$.*

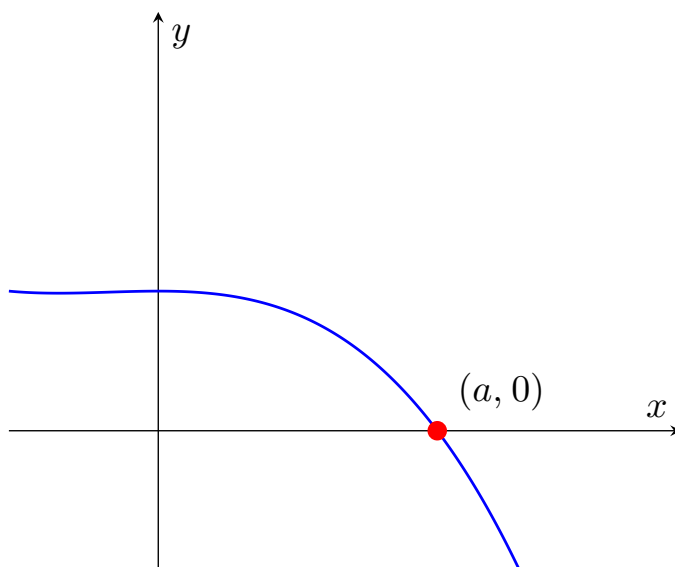


Definition 1.4.2 Horizontal Intercepts.

Suppose that the graph of the function $y = f(x)$ crosses the x -axis at $x = a$; that is, at the point $(a, 0)$. Then a is called an x -intercept of the function f .

Since the point $(a, 0)$ is on the graph of the function, $f(a) = 0$. So the x -intercepts are the real values of x at which $f(x) = 0$. A function may have many x -intercepts. They are often called the *zeros* or *roots* of a graphed function f .

That is: *To find horizontal intercepts of $f(x)$, solve $f(x) = 0$ for x .*

**Example 1.4.3**

Consider the function $y = f(x)$ where $f(x) = x^2 - 4$. Find the vertical intercept and the horizontal intercepts.

Solution. To find the y -intercept, we evaluate $f(0) = 0^2 - 4 = -4$. Therefore, $f(x)$ crosses the y -axis at $y = -4$; that is, the vertical intercept is the point $(0, -4)$.

By the way... In describing the vertical intercept of a function, we might say it is at $y = -4$ or at $(0, -4)$. Keep in mind that these represent the same concept.

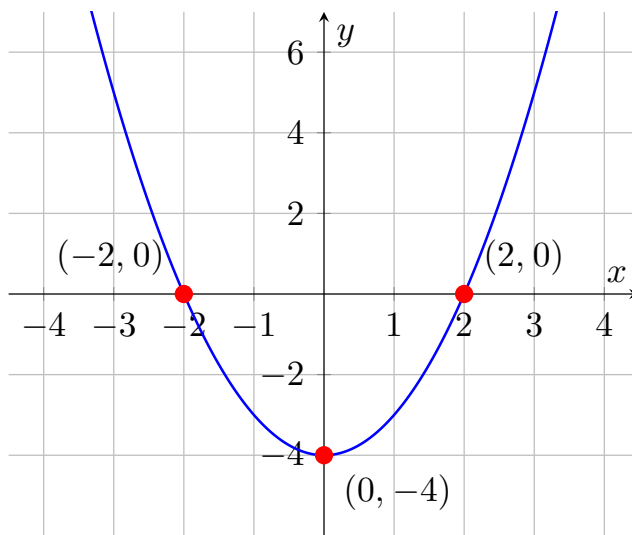
To find x -intercepts, we have to solve the equation $f(x) = 0$. For our function, the equation we must solve is $x^2 - 4 = 0$. We use the standard techniques for solving equations.

$$\begin{aligned} x^2 - 4 &= 0 \\ x^2 &= 4 \\ x &= \pm\sqrt{4} \\ x &= \pm 2 \end{aligned}$$

There are two solutions, $x = -2$ and $x = 2$. These are the two horizontal intercepts or *zeros* of the function. Since $f(-2) = 0$ and $f(2) = 0$, the associated ordered pairs are $(-2, 0)$ and $(2, 0)$.

By the way... In describing a horizontal intercept of a function, we might say it is at $x = 2$ or at $(2, 0)$. Keep in mind that these represent the same concept.

The graph of the function clearly shows the intercepts and the corresponding points on the xy -plane:



Example 1.4.4

A full tank of water springs a leak. Water is leaking out at the rate of 3 gallons per hour. Let $A = w(t)$ be the amount of water in the tank, in gallons, t hours after the leak started. The function $w(t)$ is given by

$$A = w(t) = 60 - 3t.$$

Find vertical and horizontal intercepts of the function $A = w(t)$ and interpret them in practical terms. Also graph the function.

Solution. For the vertical intercept—or the A -intercept—of the function $A = w(t)$, we evaluate $w(0)$, which results in $A = 60$. Since 60 is the value of A at $t = 0$, it is measured in gallons and it gives the initial amount of water in the tank. We conclude that the vertical intercept of 60 gallons is the initial amount of water in the tank. Note that 60 gallons is also the capacity of the tank as the tank was full when the leak began at time $t = 0$.

To find horizontal intercepts, we have to solve the equation $w(t) = 0$ for t ; that is,

$$60 - 3t = 0$$

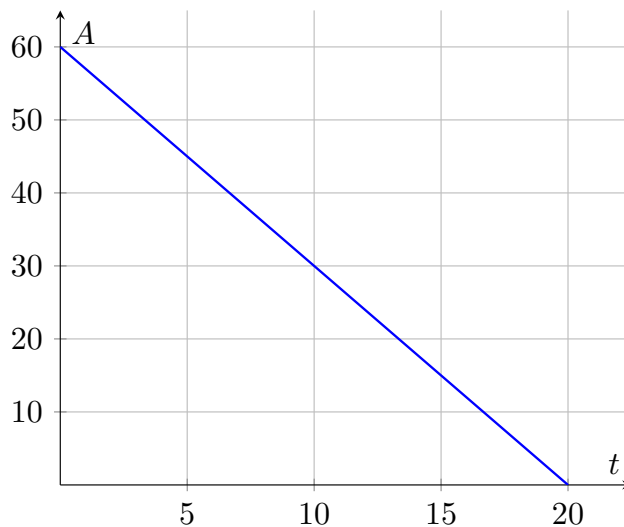
$$60 = 3t$$

$$20 = t$$

$$t = 20$$

The function has one horizontal intercept at $t = 20$. Thus, 20 hours after the leak started, the amount of water $A(20)$ left in the tank is $A = w(20) = 0$. In practical terms, it will take 20 hours until the tank is empty.

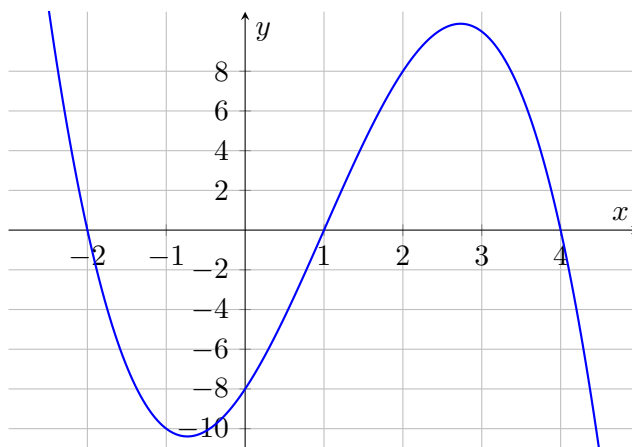
Here is the graph of the function $A = w(t)$. The intercepts are clearly visible on the graph and their meaning is clear as well.



Given the graph of a function, with its formula unknown, we are still able to find the intercepts.

Example 1.4.5

Use the graph of a function $y = f(x)$ below to estimate its vertical and horizontal intercepts.



Solution. The vertical intercept is where the graph of a function crosses the vertical axis, which is the y -axis for the function shown. We can see that this is the ordered pair $(0, -8)$. Thus, the

vertical intercept occurs at $y = -8$.

The graph crosses the x -axis at each of $x = -2, 1, 4$. These are the x -intercepts (the zeros) of $f(x)$, meaning $f(x) = 0$ for each of $x = -2, 1, 4$. The corresponding points are $(-2, 0)$, $(1, 0)$, and $(4, 0)$.

Similarly, given a function $y = f(x)$ that is described numerically (i.e., by a table of values), we can find the intercepts by looking for when $y = 0$ or $x = 0$.

Example 1.4.6

Every summer the depth of water in a reservoir is measured weekly over the 10-week period beginning with July 1. Let D be the depth of water, in meters, and t denote the time, in weeks, since July 1. Here, D is a function of t , which can be written as $D = D(t)$. The readings from 2019 are provided in the table below.

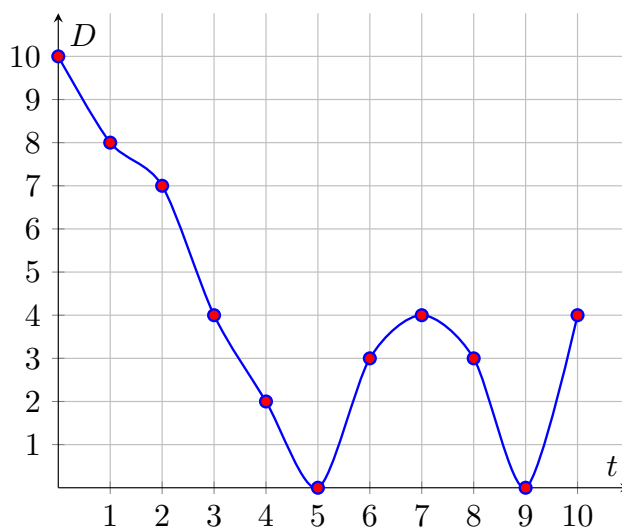
Table 1.4.7

t (weeks)	0	1	2	3	4	5	6	7	8	9	10
$D(t)$ (meters)	10	8	7	4	2	0	3	4	3	0	4

What are the vertical and horizontal intercepts given by the table? What is their practical meaning? Verify your answer on a graph of $D(t)$.

Solution. The vertical intercept occurs when $t = 0$ and hence has value given by $D(0)$. Per the table, $D(0) = 10$. Hence, the vertical intercept, or D -intercept, is 10. In practical terms, the vertical intercept tells us that the water depth on July 1, 2019 was 10 meters.

The horizontal intercepts are values of t for which $D(t) = 0$. The depth is 0 at $t = 5$ and $t = 9$. In practical terms, the horizontal intercepts tell us that at both 5 and 9 weeks after July 1, 2019, the reservoir was empty. Below is a graph that fits the data from Table 1.4.7. We can clearly see the intercepts.



Exercises

1. Find the horizontal and vertical intercepts or state that they don't exist for each function given below.

(a) $y = 3x - 1.5$

(b) $f(x) = 1 - x^2$

(c) $g(t) = 5.4 - 0.2t$

(d) $y = 2x^3 + 16$

(e) $h(m) = 2\sqrt{m} - 3$

(f) $y = \frac{1}{x}$

(g) $y = \frac{1}{x - 1}$

2. Identify the horizontal and vertical intercepts for each function given numerically below.

(a)

x	-2	-1	0	1	2	3
y	3	0	2	2.5	0	4

(b)

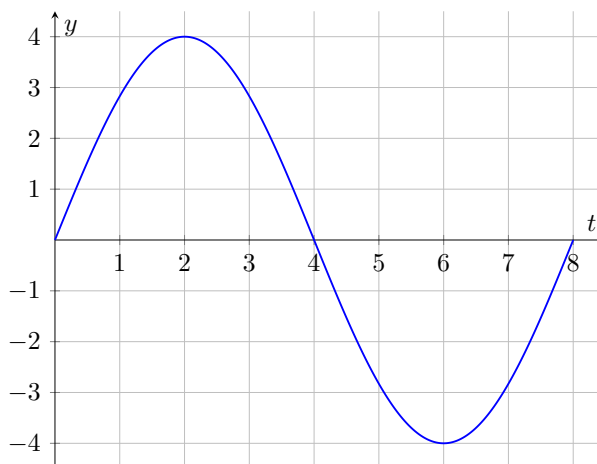
t	0	2	4	6	8	10
$f(t)$	5	6	8	4	8	4

(c)

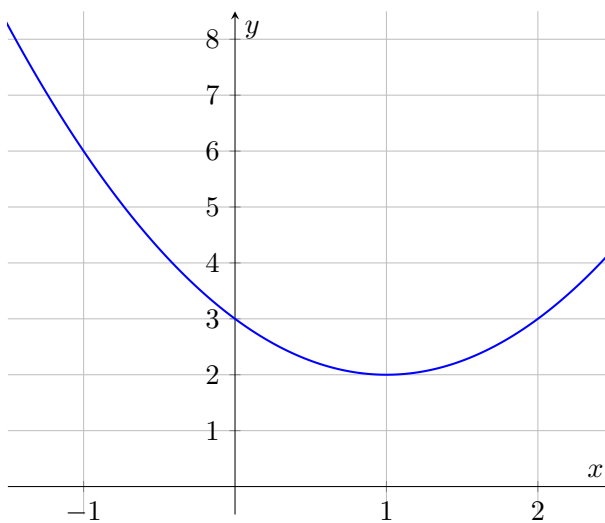
x	3	4	5	6	7	8
$g(x)$	-5	-4	-5	-2	-2	0

3. Estimate the horizontal and vertical intercepts for each function given graphically below.

(a)



(b)



4. A ball is dropped from a cliff above a lake. The ball's height above the surface of the lake, $h(t)$, in feet, t seconds after the ball is dropped, is given by

$$h(t) = 300 - 16t^2.$$

Find the vertical intercept of the function $h(t)$ and the horizontal intercept(s) for which $t \geq 0$. Give your answers to two decimal places and interpret the intercepts in practical terms.

5. The value of a car $V(t)$, in dollars, t years after the car was purchased is given by

$$V(t) = 21500 - 1500t.$$

Find the vertical and horizontal intercepts of the function $V(t)$ and interpret them in practical terms.

1.5 The Average Rate of Change

Objectives

After completing this section, you should be able to do the following.

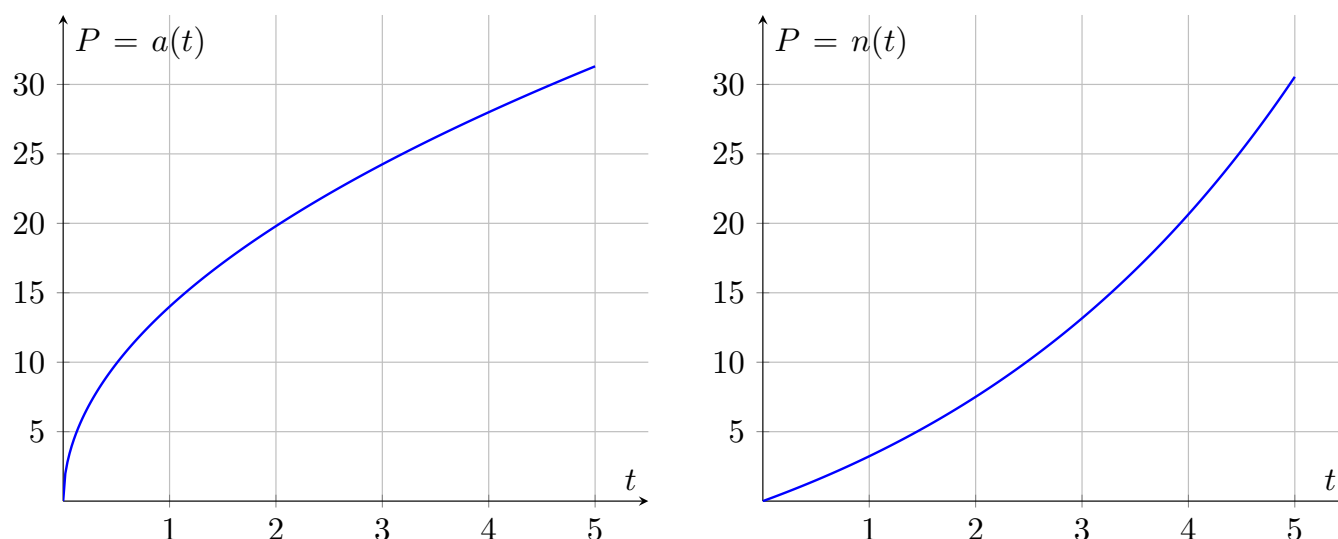
1. First objective.
 2. Second objective.
 3. Third objective.
-

The notion of a rate of change is central to mathematics and its applications. Roughly speaking, a rate of change tells us how fast and in what manner a given quantity is changing with respect to another changing quantity. We will consider three examples below.

- Suppose that a population of a town grows by 200 people each year. Here, the rate of change of the population with respect to time is constant and equal to 200 people/year.
- Recall Example 1.1.6. A daily pediatric dose of Amoxicillin, D , in milligrams, depends on the weight of a patient, w , in kilograms. More precisely, $D = 50w$. How fast does the dose increase as the weight increases? The formula for D clearly shows that for each 1 kilogram increase in weight, the dose increases by 50 milligrams. In this case, the rate of change of D with respect to weight w is 50 milligrams/kilogram.
- Suppose that the value of your car changes at the rate -1500 dollars/year. The rate of change is negative, meaning that the value of the car *decreases* by 1500 dollars each year. In general, a quantity that changes at a negative rate is decreasing while one that changes at a positive rate is increasing.

In the three examples above, the rate of change of the dependent variable with respect to the independent variable is constant. In real-life processes, this is rarely the case.

Consider the following example of two students, Amir and Nick, who went to the library to study for a precalculus exam. After t hours of uninterrupted study, Amir mastered $P = a(t)$ pages of new material from the textbook while Nick mastered $P = n(t)$ pages. The graphs of the two functions $a(t)$ and $n(t)$ are in Figure 1.5.1.

**Figure 1.5.1**

According to the graphs, both students stayed in the library for 5 hours. We can see that during the 5-hour session, both students learned about 30 new pages, as $a(5) \approx 30$ and $n(5) \approx 30$.

During the first hour, Amir mastered approximately

$$a(1) - a(0) = 15 - 0 = 15$$

pages of new material, while Nick mastered approximately

$$n(1) - n(0) = 3 - 0 = 3$$

pages. Comparatively speaking, Amir learned quickly during the first hour, while Nick learned more slowly.

During the last hour, Amir mastered approximately

$$a(5) - a(4) = 30 - 27 = 3$$

pages of new material, while Nick mastered approximately

$$n(5) - n(4) = 30 - 20 = 10$$

pages.

This means that Amir went from learning quickly at the beginning of the study session to learning more slowly as time went on. Nick, on the other hand, got off to a slow start but picked up the pace as time went on. Their learning patterns are clearly not identical.

Does it make sense to ask how quickly—or at what rate in pages per hour—each student learned during the 5-hour session? Not really. The question is ambiguous. When exactly in the study session should we

focus on? The beginning? The end? For both students, the rate at which they were learning during the session changed, with Amir learning more and more slowly over time and Nick learning faster and faster.

What *does* make sense is to consider how quickly each student learned during the 5-hour session ***on average***. Amir learned quickly at first and then more slowly later, but overall he mastered 30 pages of new material in 5 hours. He learned at an ***average*** rate of $\frac{30}{5} = 6$ pages/hour. Nick, on the other hand, learned slowly at first, then faster, but he also mastered 30 pages in 5 hours. So, he too learned at an ***average*** rate of $\frac{30}{5} = 6$ pages/hour.

Readers who proceed to Calculus will study instantaneous rate of change—how quickly a quantity changes at a specific input. For now, the ***average*** rate of change is the best we can do.

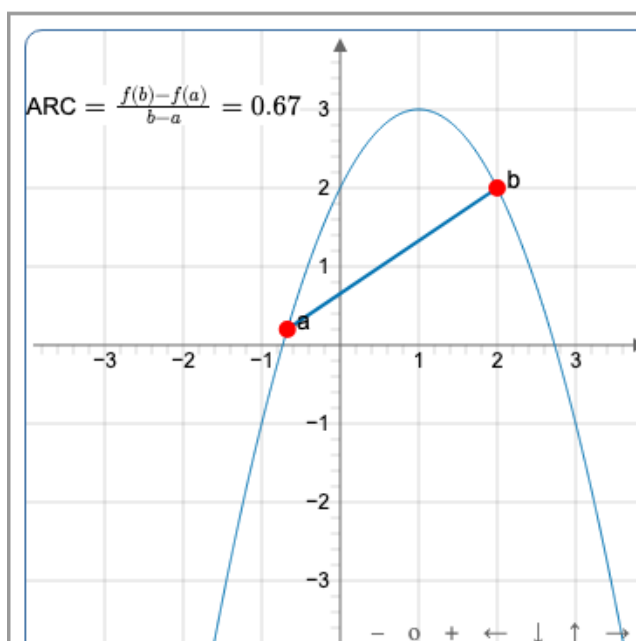
Definition 1.5.2 Average Rate of Change.

Let y be a function of x , $y = f(x)$. Let $x = a$ and $x = b$ be given. The average rate of change of y between $x = a$ and $x = b$ is given by

$$\text{ARC} = \frac{\text{Change in } y}{\text{Change in } x} = \frac{\Delta y}{\Delta x} = \frac{f(b) - f(a)}{b - a}.$$

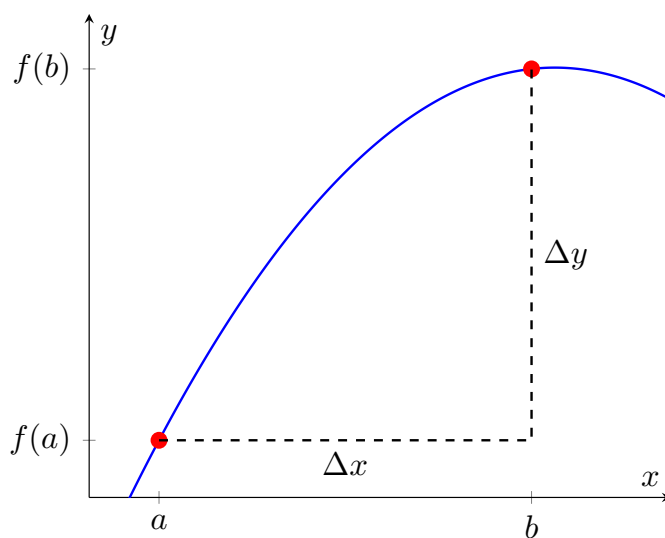
We'll use ARC to abbreviate Average Rate of Change.

Typically, the symbol Δ (the Greek letter Delta) is used to mean “change in.” So, Δy stands for a change in y while Δx stands for a change in x . In the definition above, Δx stands for the change in x between $x = a$ and $x = b$. This can be thought of as the distance between the numbers a and b , which is given by $\Delta x = b - a$. The notation Δy stands for the change in $y = f(x)$ that takes place as x changes from $x = a$ to $x = b$. That change is given by $\Delta y = f(b) - f(a)$. The graph below illustrates the changes in x and y .



Standalone

Figure 1.5.3

**Example 1.5.4**

Let $y = f(x) = x^2 - 2$. Find the average rate of change of y between $x = 1$ and $x = 3$.

Solution. We use the definition of the average rate of change with $a = 1$, $b = 3$ and $y = f(x) = x^2 - 2$:

$$\frac{\Delta y}{\Delta x} = \frac{f(3) - f(1)}{3 - 1} = \frac{(3^2 - 2) - (1^2 - 2)}{3 - 1} = \frac{7 - (-1)}{2} = \frac{8}{2} = 4$$

Therefore, the average rate of change of y between $x = 1$ and $x = 3$ is 4.

This is not an applied example so x and y don't have real-life units. In general, the average rate of change $\frac{\Delta y}{\Delta x}$ is measured in

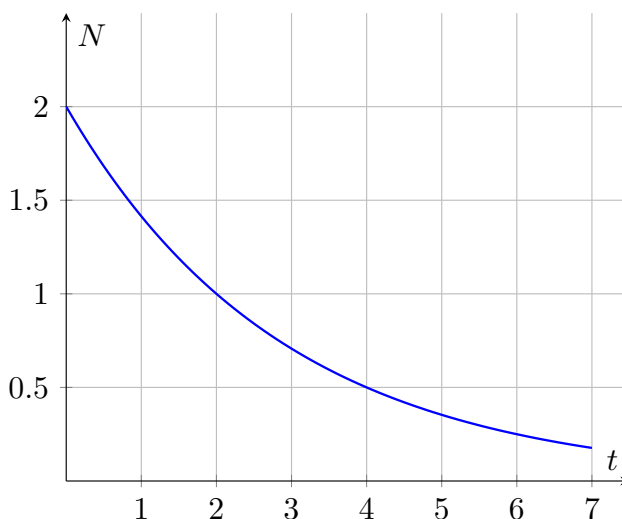
$$\text{units of ARC} = \frac{\text{units for } y}{\text{unit for } x}$$

Were units associated with x and y in the last example, we would write that the average rate of change of y between $x = 1$ and $x = 3$ is $4 \frac{\text{units for } y}{\text{units for } x}$.

Let's revisit Example 1.1.5 about nicotine leaving the body after a cigarette is smoked.

Example 1.5.5

The amount of nicotine in a person's bloodstream, $N = f(t)$, in milligrams, is a function of the time t , in hours, that has passed since the person smoked a single cigarette. The graph of the function $f(t)$ is shown below.



Use the graph to find and interpret the average rate of change of N between:

1. $t = 0$ and $t = 4$;
2. $t = 2$ and $t = 4$.

Include units with your answers.

Solution.

1. To find the average rate of change between $t = 0$ and $t = 4$, we use the graph to find the values of the function $f(t)$ at $t = 0$ and at $t = 4$. At $t = 0$, we have $N = f(0) = 2$ milligrams.

At $t = 4$, $N = f(4) = 0.5$ milligrams. Thus:

$$\text{ARC} = \frac{\Delta N}{\Delta t} = \frac{f(4) - f(0)}{4 - 0} = \frac{0.5 - 2}{4 - 0} = \frac{-1.5}{4} = -0.375.$$

Here the variables are associated with units: N is measured in milligrams while t is measured in hours. Hence, ΔN is measured in milligrams and Δt is measured in hours. The average rate of change has units $\frac{\text{milligrams}}{\text{hour}}$ which can be written as $\frac{\text{mg}}{\text{hr}}$.

Therefore, the average rate of change of N between $t = 0$ and $t = 4$ is

$$-0.375 \frac{\text{mg}}{\text{hr}}.$$

The rate is negative as the amount of nicotine decreased between $t = 0$ and $t = 4$. For the interpretation, we would say that on the interval $0 \leq t \leq 4$, nicotine leaves the body at the average rate of 0.375 milligrams per hour.

2. We need values $f(2)$ and $f(4)$, which are obtained from the graph: $f(2) = 1$ and $f(4) = 0.5$. Hence,

$$\text{ARC} = \frac{\Delta N}{\Delta t} = \frac{f(4) - f(2)}{4 - 2} = \frac{0.5 - 1}{2} = \frac{-0.5}{2} = -0.25 \frac{\text{mg}}{\text{hr}}.$$

On the interval $2 \leq t \leq 4$, nicotine leaves the body at the average rate of 0.25 milligrams per hour.

Example 1.5.6

A population P of a small village at time t , measured in years since the year 2010, is given by $P = P(t)$. Here are the annual readings:

Table 1.5.7

t (years)	0	1	2	3	4	5	6	7	8	9	10
$P(t)$ (people)	150	160	140	180	210	200	250	290	300	350	400

- Find the change in the population ΔP between $t = 2$ and $t = 6$. What are the units of ΔP ?
- Find the average rate of change in the population between $t = 2$ and $t = 6$. What are the units of this rate of change?

Solution.

- The change in P between $t = 2$ and $t = 6$, ΔP , tells us by how much the population changed

from $t = 2$ and $t = 6$:

$$\Delta P = P(6) - P(2) = 250 - 140 = 110.$$

The population increased by 110 people from $t = 2$ to $t = 6$ (from 2012 to 2016).

2. The average **rate** of change tells us how fast the population was changing between $t = 2$ and $t = 6$ and is measured in people/year:

$$\text{ARC} = \frac{\Delta P}{\Delta t} = \frac{P(6) - P(2)}{6 - 2} = \frac{250 - 140}{6 - 2} = \frac{110}{4} = 27.5 \frac{\text{people}}{\text{year}}.$$

Example 1.5.8

A ball is dropped from the rooftop of a building 150 feet tall. The height of the ball above the ground, $h(t)$, in feet, t seconds after the ball is dropped is:

$$h(t) = 150 - 16t^2.$$

1. Find the average rate of change of $h(t)$ between $t = 0$ and $t = 2$.
2. When will the ball hit the ground?

Solution.

1. We use the definition of the average rate of change:

$$\begin{aligned} \text{ARC} &= \frac{\Delta h}{\Delta t} = \frac{h(2) - h(0)}{2 - 0} \\ &= \frac{(150 - 16 \cdot 2^2) - (150 - 16 \cdot 0^2)}{2} \\ &= \frac{86 - 150}{2} = -32 \frac{\text{ft}}{\text{sec}} \end{aligned}$$

This can also be thought of as the average velocity of the ball between $t = 0$ and $t = 2$. The velocity is negative because the height is decreasing.

2. The ball hits the ground when it falls to a height of 0 feet, so we solve the equation $h(t) = 150 - 16t^2 = 0$ for t :

$$\begin{aligned} 150 - 16t^2 &= 0 \\ 150 &= 16t^2 \end{aligned}$$

$$\begin{aligned}\frac{150}{16} &= t^2 \\ t^2 &= 9.375 \\ t &= \pm\sqrt{9.375}\end{aligned}$$

Mathematically, we get two solutions: $t = \sqrt{9.375}$ and $t = -\sqrt{9.375}$. However, only one makes sense in the applied context for our problem. As t is the number of seconds after the ball is dropped, a negative answer does not make sense—the ball can't hit the ground before it was dropped! Hence, the ball hits the ground $t = \sqrt{9.375} \approx 3.062$ seconds after it is dropped.

Exercises

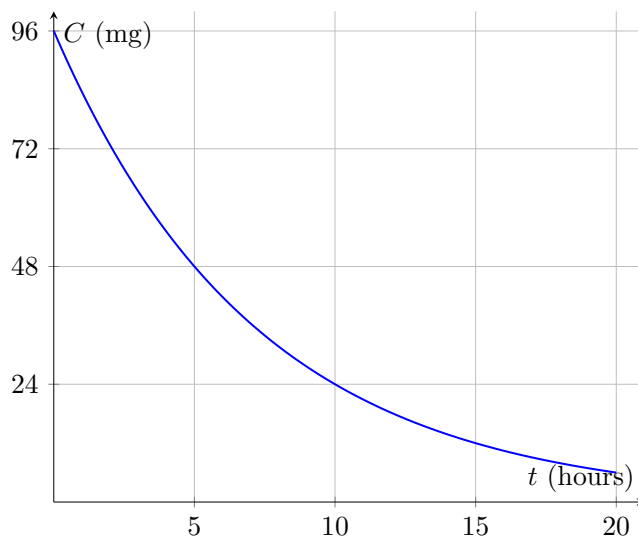
- Find the average rate of change of each function over the indicated interval. Give your answers to two decimal places.
 - $f(x) = 1 + x^2$ between $x = 1$ and $x = 5$.
 - $f(x) = 1 + x^2$ between $x = -3$ and $x = 0$.
 - $g(t) = \frac{1}{t}$ between $t = 1$ and $t = 3$.
 - $p(x) = 2 - 3x$ between $x = 1$ and $x = 5$.
 - $p(x) = 2 - 3x$ between $x = -1$ and $x = 2$.
- In Exercise 1.3.8, data about the world population, in billions, between the years 2010 and 2018, was given in the table shown below. Use the data to answer the following questions.

Year	2010	2011	2012	2013	2014	2015	2016	2017	2018
Population	6.957	7.041	7.126	7.211	7.296	7.380	7.464	7.548	7.631

- What was the average rate of change of the world population between 2010 and 2012? Between 2016 and 2018? Give your answers to four decimal places and include units with your answers.
 - Is there a one-year interval in the time period from 2010 to 2018 during which the average rate of change of the population was negative?
- A ball is dropped from a cliff above a lake. The ball's height above the surface of the lake, $h(t)$, in feet, t seconds after the ball is dropped, is given by

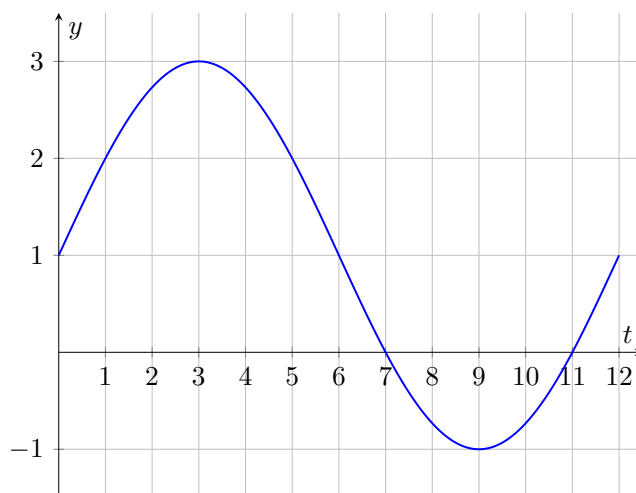
$$h(t) = 300 - 16t^2.$$

- (a) What is the average rate of change in height $h(t)$ between $t = 0$ and $t = 3$? Between $t = 1$ and $t = 4$? Give units with your answers.
- (b) Are your answers in part (a) positive or negative? Explain why.
4. In Exercise 12, the graph of a function $C(t)$ (provided again below) was given.



Recall that this graph that models the amount of caffeine, in milligrams, remaining in the body t hours after drinking a cup of coffee. Use the graph to do the following.

- (a) Estimate the average rate of change of $C(t)$ between $t = 0$ and $t = 5$. Give units with your answer.
- (b) Estimate the average rate of change of $C(t)$ between $t = 10$ and $t = 15$. Give units with your answer.
- (c) Compare the magnitudes of your answers in (a) and (b). Can you explain the difference by what you see on the graph?
5. The graph of a function $y = f(t)$ is shown below.



Without performing any calculations, answer the following questions.

- (a) Is the average rate of change of $f(t)$ between $t = 0$ and $t = 3$ positive or negative? Explain your answer.
 - (b) Is the average rate of change of $f(t)$ between $t = 3$ and $t = 6$ positive or negative? Explain your answer.
6. A person's weight, W , in pounds, depends on the number of minutes m of daily exercise. Hence, $W = W(m)$ is a function of m . Below is a numerical representation of $W(m)$:

m (minutes)	0	20	40	60	80	100	120
W (pounds)	159	152	146	141	137	136	135.6

- (a) Find the average rate of change of $W(m)$ between $m = 0$ and $m = 40$. Give units with your answer.
 - (b) Find the average rate of change of $W(m)$ between $m = 80$ and $m = 120$. Give units with your answer.
 - (c) Based on your answers and the data in the table, what can you say about the rate of weight loss with increasing amount of exercise?
7. Worldwide sales of passenger cars fluctuated between 2012 and 2019, as can be seen in the table below.

Year	2010	2011	2012	2013	2014	2015	2016	2017	2018	2019
Cars Sold (millions)	55.82	57.84	60.94	63.43	65.71	66.31	69.46	70.69	68.68	64.34

- (a) Calculate the average rate of change in sales of passenger cars between the years 2010 and 2017. Give units with your answer.

- (b) Calculate the average rate of change in sales of passenger cars between the years 2017 and 2019. Give units with your answer.
- (c) In what one-year intervals was the average rate of change negative?

Chapter 2

Linear Functions

The daily dose of medication to administer to a child of weight w initially presented in Example 1.1.6 and the amount of water remaining in a tank t minutes after it began to leak presented in Example 1.4.4 are two illustrations of *linear functions*, which will be covered in depth in this chapter.

Linear functions are generally considered the simplest type of function in mathematics. This is largely due to the simplicity of the equations used to describe them when presented algebraically and the simplicity of their graphs. It is also the case that when a linear function appears in the context of a real-life process, the meaning of all elements of the function is clear.

2.1 Properties of Linear Functions

Objectives

After completing this section, you should be able to do the following.

1. First objective.
2. Second objective.
3. Third objective.

Definition 2.1.1 Linear Function.

A function $y = f(x)$ is called a **linear function** if it can be written in the form

$$f(x) = mx + b$$

where m and b are constants. The constant m is called the **slope** of the function $y = f(x)$. The constant b is the **initial value** of $f(x)$; it is also the vertical intercept of the function as $f(0) = m \cdot 0 + b = b$.

A linear function $y = f(x)$ written in the form $f(x) = mx + b$ or $y = mx + b$ is said to be in **slope-intercept form**.

By the way... Another form of linear functions will be presented in the next section.

Practical Interpretation of the Slope and Vertical Intercept

Applied examples of linear functions will help us understand the practical meaning of the slope m and the initial value b . In such examples, the input and output variables will often be denoted by letters that correspond to their practical meaning rather than by x and y .

Example 2.1.2

Maya is saving money to purchase a car. She initially has \$60 in savings and for each hour she works at her summer job she is able to put an additional \$5 into savings. Find a formula for the function $A = f(n)$ that gives the total amount of money, A , that Maya has in savings after working n hours.

Solution. We begin with a table of values reflecting the total amount of money Maya will have in savings after working a certain number of hours.

Table 2.1.3

<i>If Maya works ...</i>	<i>Maya's savings will be ...</i>
0 hours	$5(0) + 60 = \$60$
1 hour	$5(1) + 60 = \$65$
2 hours	$5(2) + 60 = \$70$
3 hours	$5(3) + 60 = \$75$
$\vdots$	$\vdots$

Recognizing the pattern unfolding, a formula for the amount of money A that Maya has in savings as a function of the number of hours n that she works is

$$A = f(n) = 5n + 60.$$

The function $A = f(n)$ found in the solution to Example 2.1.2 is a linear function with slope $m = 5$ and vertical intercept $b = 60$. Mathematically, the domain of the function $f(n) = 5n + 60$ consists of all inputs n , but in the applied context of the function, it is not possible for Maya to work a negative number of hours. For this reason, we must restrict the domain to $n \geq 0$.

Observe that, as in most applied examples, the input and output variables have units: n is measured in hours worked while A is measured in dollars. Likewise, both constants $m = 5$ and $b = 60$ have units and a practical meaning:

- $b = 60$ dollars is the initial value of A at $n = 0$; that is, the amount of money that Maya has in savings initially.
- The slope $m = 5$ indicates that Maya saves \$5 per each hour worked. Hence, the slope $m = 5$ is measured in dollars per hour and gives the rate at which Maya's savings A grow as the number of hours n she works increases.

We can additionally observe that the average rate of change between any two distinct inputs for the function $f(n) = 5n + 60$ is always the same. The below calculations illustrate this.

By the way... Recall that average rate of change of a function $f(x)$ between $x = a$ and $x = b$ is given by $\frac{f(b)-f(a)}{b-a}$.

$$\text{ARC from } n = 0 \text{ to } n = 1 : \frac{65 - 60}{1 - 0} = \frac{5}{1} = 5 \quad \text{dollars/hour}$$

$$\begin{array}{ll}
\text{ARC from } n = 1 \text{ to } n = 2 : & \frac{70 - 65}{2 - 1} = \frac{5}{1} = 5 \quad \text{dollars/hour} \\
\text{ARC from } n = 2 \text{ to } n = 3 : & \frac{75 - 70}{3 - 2} = \frac{5}{1} = 5 \quad \text{dollars/hour} \\
\text{ARC from } n = 0 \text{ to } n = 3 : & \frac{75 - 60}{3 - 0} = \frac{15}{3} = 5 \quad \text{dollars/hour}
\end{array}$$

Maya's savings grow at a constant rate with respect to the hours that she works, and that constant rate of 5 dollars/hour is the same as the slope of the linear function $f(n) = 5n + 60$. In general, for linear functions—and only for linear functions—the rate of change of the dependent variable with respect to the independent variable is constant and equal to the slope.

Result 2.1.4 Slope as Rate of Change.

For every linear function $y = f(x) = mx + b$, the rate of change of y with respect to x is constant and equal to m , with units

$$\frac{\text{units of } y}{\text{units of } x}.$$

Further, b represents the initial value of the function, $b = f(0)$, and is measured in units of y .

The constancy of the rate of change of a linear function can be proven using algebraic properties of linear functions. Take two distinct inputs x_1, x_2 and the corresponding outputs $y_1 = f(x_1), y_2 = f(x_2)$. The average rate of change of $f(x)$ between x_1 and x_2 is given by:

$$\begin{aligned}
\frac{\Delta y}{\Delta x} &= \frac{f(x_2) - f(x_1)}{x_2 - x_1} = \frac{(mx_2 + b) - (mx_1 + b)}{x_2 - x_1} \\
&= \frac{mx_2 + b - mx_1 - b}{x_2 - x_1} \\
&= \frac{m(x_2 - x_1)}{x_2 - x_1} \\
&= m
\end{aligned}$$

Example 2.1.5

The number of people residing in the town of Linesville t years after 2010 is given by the linear function

$$P(t) = 150t + 4500.$$

Identify the slope and the initial value of the function. Include units and explain their practical meaning.

Solution. From the formula for $P(t)$, we see that the slope of the function is $m = 150$ and the initial value is $b = 4500$.

The slope m is measured in people/year. In practical terms, the slope tells us that the population of Linesville increases at a constant rate of 150 people per year.

The initial value $b = 4500$ is measured in number of people. It represents the initial number of residents in the town of Linesville, i.e., the number of residents at $t = 0$. As time $t = 0$ corresponds to the year 2010, this means that the population of Linesville was 4500 people in 2010.

Example 2.1.6

A container holding 50 gallons of water has sprung a leak. Water is leaking out at a rate of 5 gallons/hour. Let $W(t)$ be the amount of water in the container, in gallons, t hours after the leak began. Find a formula for the function $W(t)$.

Solution. Water is leaking out at a constant rate so $W(t)$ is a linear function. Hence $W(t) = mt + b$.

The amount of water in the container is *decreasing* at a rate 5 gallons/hour. This means that the rate of change or slope of $W(t)$ is -5 gallons/hour. That is, $m = -5$.

The initial amount of water is $W(0) = 50$ gallons, so $b = 50$. The formula for $W(t)$ is:

$$W(t) = -5t + 50.$$

Graphs of Linear Functions

We will next explore the graphs of linear functions.

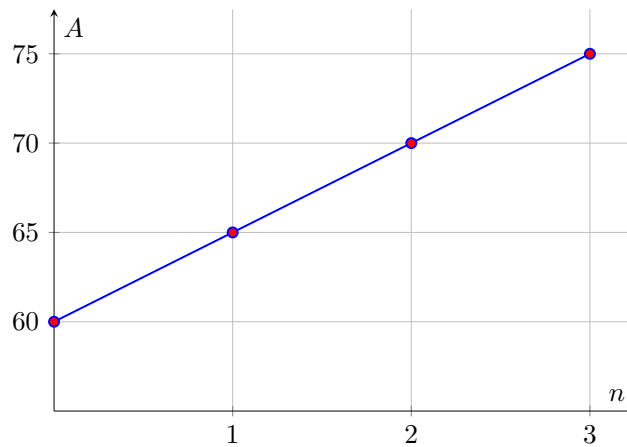
Example 2.1.7

In Example 2.1.2, Maya's savings A , in dollars, as a function of the number of hours, n , was given by

$$A = f(n) = 5n + 60.$$

Use Table 2.1.3 to create a graph of the function $f(n)$.

Solution. Table 2.1.3 contains the points $(0, 60)$, $(1, 65)$, $(2, 70)$, and $(3, 75)$. By plotting these points, we obtain the following graph. It is clear that the points lie along a straight line.

**Result 2.1.8 Graph of a Linear Function.**

The graph of a linear function

$$y = f(x) = mx + b$$

is a straight line with slope m and vertical intercept b . The equation of the line in the **slope-intercept** form is $y = mx + b$.

- If the slope $m > 0$, the function $f(x)$ is increasing and its graph is climbing as x increases.
- If the slope $m < 0$, the function $f(x)$ is decreasing and its graph is falling as x increases.
- If the slope $m = 0$, the function $f(x)$ is the constant function $f(x) = b$, and its graph is the horizontal line $y = b$.

In the following figure, we can see how the slope and vertical intercept change the graph of the function.

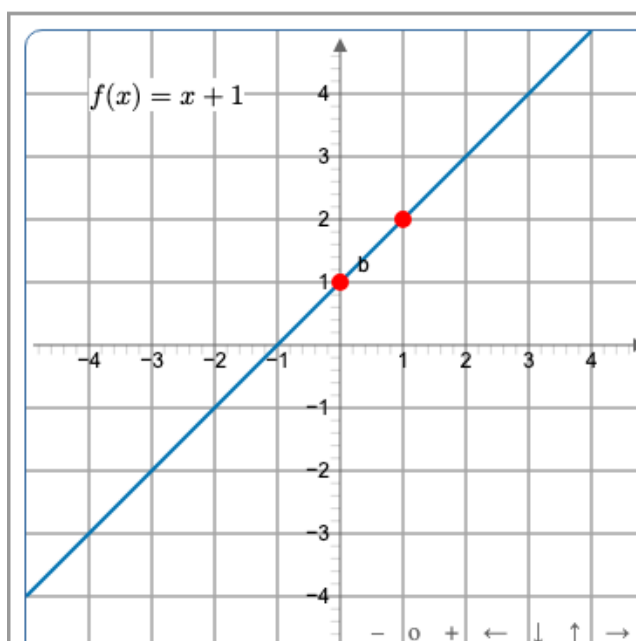


Figure 2.1.9



Standalone

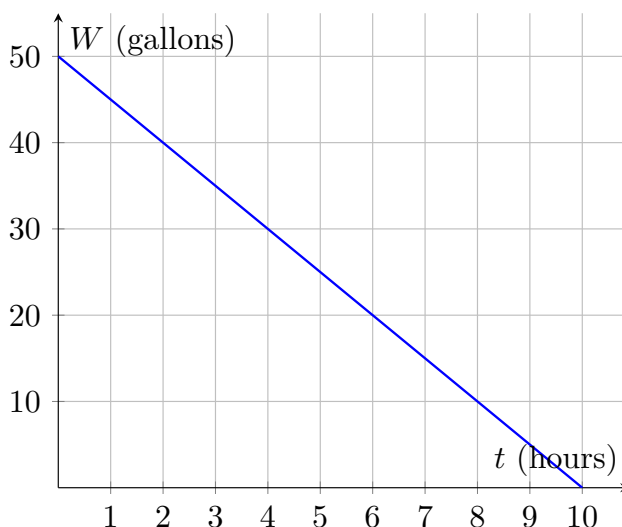
Example 2.1.10

Consider the function $W(t)$ from Example 2.1.6, which gives the amount of water in a container, $W(t)$, in gallons, t hours after the container sprung a leak:

$$W(t) = -5t + 50.$$

Create a graph of the function $W(t)$. What restriction should be placed on the domain of $W(t)$ in the applied context of this exercise?

Solution. The slope $m = -5$ is negative, and thus the graph is decreasing: for each 1 hour increase in t , $W(t)$ decreases by 5. The vertical intercept is $b = 50$ gallons. Using this information, we construct the following graph.



The practical significance of vertical and horizontal intercepts is apparent from the graph. The vertical intercept, $W = 50$, gives the initial amount of water in the container. The horizontal intercept, $t = 10$, is the number of hours that must pass for the tank to empty completely. Based upon this, the domain of $W(t) = -5t + 50$ should be restricted to $0 \leq t \leq 10$ in order for the formula to make sense in the applied context of this exercise.

The Slope Formula

We are often given multiple points on the graph of a linear function but not provided the formula for the function nor explicitly told its slope. In this case, we have enough information to determine both the slope and the slope-intercept form of the function.

Given two distinct points (x_1, y_1) and (x_2, y_2) on the graph of a linear function $y = f(x)$, the average rate of change of $f(x)$ between x_1 and x_2 is given by

$$\frac{f(x_2) - f(x_1)}{x_2 - x_1} = \frac{y_2 - y_1}{x_2 - x_1}.$$

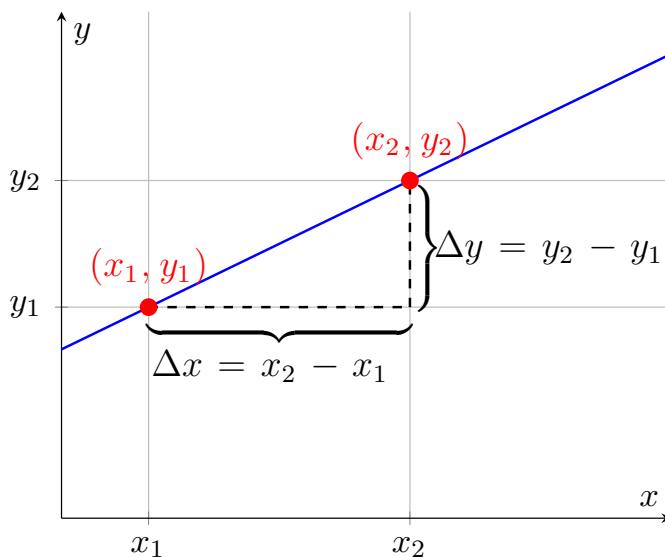
Since the average rate of change between any two points on a linear function is its slope (see Result 2.1.4), the above is the value of m .

Result 2.1.11 Slope Formula.

The slope m of a linear function $y = f(x)$ whose graph passes through two distinct points (x_1, y_1) and (x_2, y_2) is:

$$m = \frac{y_2 - y_1}{x_2 - x_1} = \frac{\Delta y}{\Delta x} = \frac{\text{change in output}}{\text{change in input}}$$

Graphically, the slope formula $m = \frac{y_2 - y_1}{x_2 - x_1}$ is often thought of as “rise over run.” The “rise” is what we denote by Δy or $y_2 - y_1$; it is the change in y -values from the ordered pair (x_1, y_1) to the ordered pair (x_2, y_2) . The “run” is what we denote by Δx or $x_2 - x_1$; it is the change in the corresponding x -values. This is illustrated below.



Example 2.1.12

Find the slope of the line passing through each set of points.

1. $(-3, -1)$ and $(-2, 4)$
2. $(2, 3)$ and $(-1, 3)$
3. $(1/2, 6)$ and $(1, -1/3)$.

Solution.

1. If $(x_1, y_1) = (-3, -1)$ and $(x_2, y_2) = (-2, 4)$ then the slope is

$$m = \frac{y_2 - y_1}{x_2 - x_1} = \frac{4 - (-1)}{-2 - (-3)} = \frac{4 + 1}{-2 + 3} = \frac{5}{1} = 5.$$

2. If $(x_1, y_1) = (2, 3)$ and $(x_2, y_2) = (-1, 3)$ then the slope is

$$m = \frac{y_2 - y_1}{x_2 - x_1} = \frac{3 - 3}{-1 - 2} = \frac{0}{-3} = 0.$$

By the way... Because the slope is 0, the line through the points is *horizontal*. This will be explained in further detail later on in this section.

3. If $(x_1, y_1) = (1/2, 6)$ and $(x_2, y_2) = (1, -1/3)$ then the slope is

By the way... Here, find a common denominator to simplify $\frac{-1}{3} - 6$ and $1 - \frac{1}{2}$.

$$\begin{aligned} m &= \frac{y_2 - y_1}{x_2 - x_1} = \frac{-1/3 - 6}{1 - 1/2} = \frac{-1/3 - 18/3}{2/2 - 1/2} = \frac{-19/3}{1/2} \\ &= -\frac{19}{3} \cdot \frac{2}{1} = -\frac{38}{3}. \end{aligned}$$

Example 2.1.13

Find a linear function $f(x) = mx + b$ such that $f(1) = 2$ and $f(3) = -2$.

Solution. The two associated points on the graph of the function are

$$(x_1, y_1) = (1, 2) \quad \text{and} \quad (x_2, y_2) = (3, -2).$$

Using the slope formula, we find that

$$m = \frac{y_2 - y_1}{x_2 - x_1} = \frac{-2 - 2}{3 - 1} = -2.$$

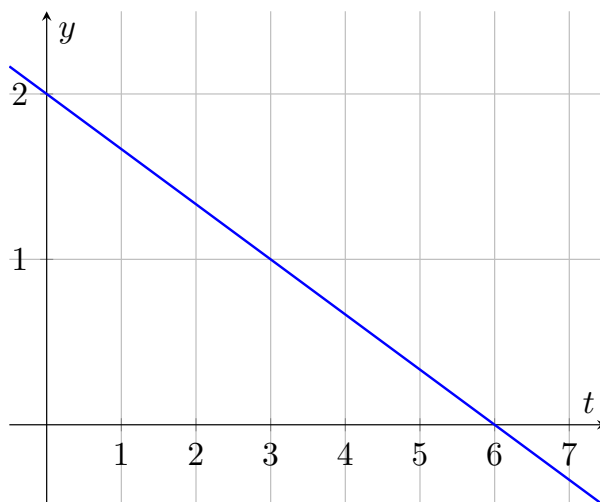
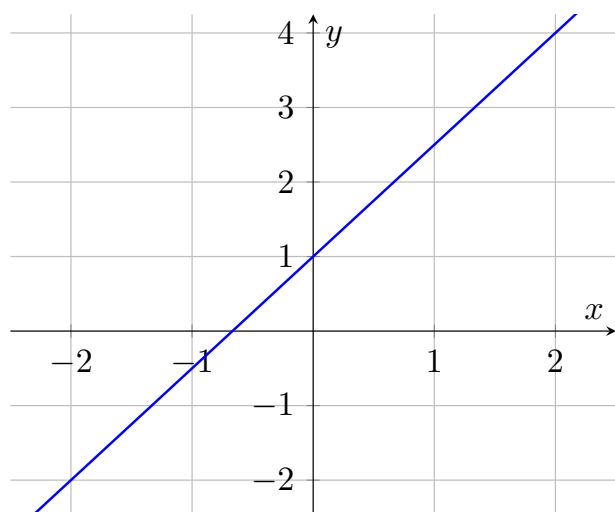
So, $f(x) = -2x + b$. The value of b is not immediately apparent as we were not given the value of the function at $x = 0$. However, we can solve for it by using either one of the function values provided. For instance, we can use the fact that $f(1) = 2$ to set up an equation for b . Substituting $x = 1$ into $f(x) = -2x + b$ results in

$$-2 \cdot 1 + b = 2.$$

Adding 2 to both sides allows us to conclude that $b = 4$. Therefore, $f(x) = -2x + 4$.

Example 2.1.14

Identify the slope and vertical intercept of each line. Then write the equation of the line in slope-intercept form.



Solution.

1. The first line intersects the y -axis at the point $(0, 1)$, so the vertical intercept is $b = 1$. To calculate the slope, find two points on the graph and use the slope formula. We'll use $(x_1, y_1) = (0, 1)$ and $(x_2, y_2) = (2, 4)$. Then

$$m = \frac{y_2 - y_1}{x_2 - x_1} = \frac{4 - 1}{2 - 0} = \frac{3}{2}.$$

The equation of the line is $y = \frac{3}{2}x + 1$. The corresponding linear function is $f(x) = \frac{3}{2}x + 1$.

2. For the second line, note that the variable t is being used instead of x . We must again find two points on the line to substitute into the slope formula. We'll use $(t_1, y_1) = (0, 2)$ and $(t_2, y_2) = (6, 0)$ to find that

$$m = \frac{y_2 - y_1}{t_2 - t_1} = \frac{0 - 2}{6 - 0} = \frac{-2}{6} = -\frac{1}{3}.$$

Since the vertical intercept is $b = 2$, the equation of the line is $y = -\frac{1}{3}t + 2$. The corresponding linear function is $f(t) = -\frac{1}{3}t + 2$.

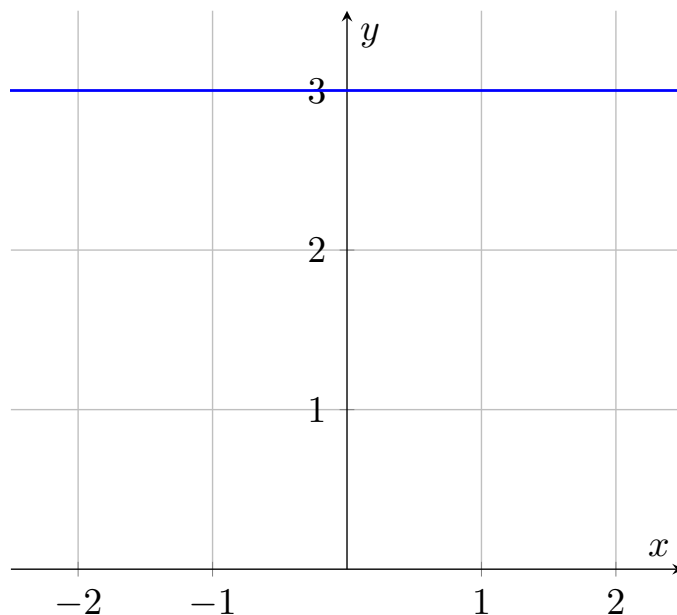
Constant Functions, Horizontal and Vertical Lines

In Example 2.1.12, we encountered a linear function with slope 0 but did not go into great detail about what this means. To further explore the concept, consider the linear function

$$f(x) = 3.$$

The formula for this function can be rewritten as $f(x) = 0 \cdot x + 3$. The slope of this function is 0 and the y -intercept is 3. We can see that for every x , the value of the function is the same; that is, $f(x) = 3$ for all x . We say that the function $f(x) = 3$ is **constant** because it is constantly equal to 3. Using the rate of change interpretation of the slope of a linear function, it is unsurprising that the function is constant — the rate of change or slope of the function being $m = 0$ implies that the function doesn't change.

The graph of the function $f(x) = 3$ consists of all points on the plane for which $y = 3$, resulting in a horizontal line through the y -value 3 as depicted below.



Result 2.1.15 Constant Functions and Horizontal Lines.

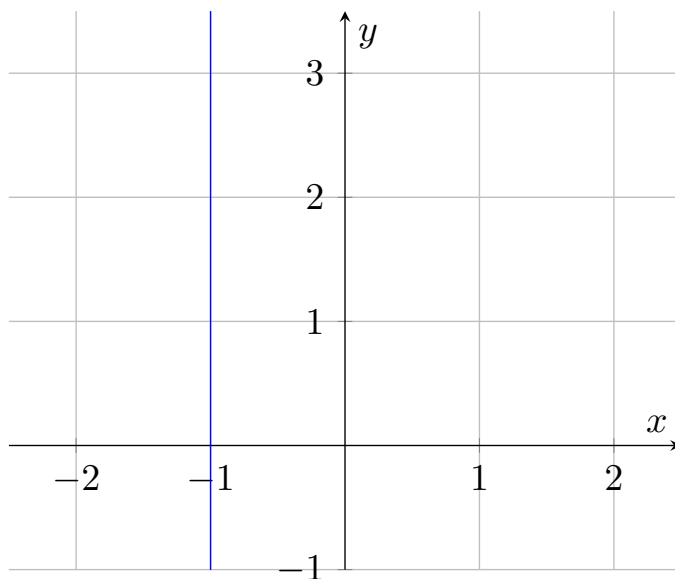
The linear function $y = f(x)$ with slope $m = 0$ and vertical intercept b is a constant function with equation given by

$$f(x) = b \quad \text{or} \quad y = b$$

and its graph is the horizontal line through the y -value b .

Having established that horizontal lines in the plane are graphs of constant functions, it is natural to wonder if vertical lines in the plane are also associated with a special type of linear function.

Vertical lines—lines parallel to the y -axis—are *not* graphs of functions. A vertical line consists of all points for which the x -coordinate is constant. Take for example the line $x = -1$ which is graphed below.



By the way... A key takeaway here is that vertical lines are not functions, but still have *equations*, such as $x = -1$. Equations are *not* the same thing as functions.

This line is not the graph of any function $y = f(x)$ as for the single input $x = -1$ there is more than one output—infinitely many outputs, in fact. You can reach the same conclusion by using the Vertical Line Test. There exists a vertical line (the line $x = -1$ itself) that crosses the graph of $x = -1$ more than once (everywhere, actually). Therefore the graph of $x = -1$ is not a function.

We conclude that vertical lines are not graphs of functions. Furthermore, slopes of vertical lines are undefined. Why? Take two points on the line $x = -1$, say $(-1, 0)$ and $(-1, 1)$. The slope formula gives:

$$m = \frac{1 - 0}{-1 - (-1)} = \frac{1}{-1 + 1} = \frac{1}{0}$$

which is undefined due to division by 0.

Result 2.1.16 Vertical Lines.

The vertical line with horizontal intercept k has equation given by

$$x = k$$

and its slope is undefined.

By the way... This is to say: all linear functions are lines, but not all lines are linear functions.

In summary, the graph of every linear function $y = f(x) = mx + b$ is a line in the xy -plane while every *non-vertical* line has an equation of the form $y = mx + b$ and thus is the graph of a linear function. However, not every line in the plane is the graph of a linear function, since vertical lines are not functions. Sometimes people talk about linear functions and lines in the plane interchangeably. Despite the duality between lines and linear functions, it is important to remember that a linear function is a special dependence between two numerical variables that can be visually represented by a line. The converse is not necessarily true.

Exercises

Slope-Intercept Form. For each of the following, rewrite the linear function in slope-intercept form. Then identify the slope and vertical intercept.

1. $f(x) = 5x - 2(x + 3)$
2. $g(x) = 4(2x - 1) + 2$
3. $h(t) = \frac{18t - 5}{2}$
4. $m(x) = 2(x - 1) + 2$

Computing the Slope. For each of the following, find the slope of the line passing through each pair of points.

5. $(0, 10)$ and $(3, 16)$
6. $(6, -4)$ and $(-15, -1)$
7. $(-0.7, 7.2)$ and $(-0.5, 12.5)$
8. $(\frac{1}{3}, 1)$ and $(-1, \frac{1}{2})$
9. $(3, -5)$ and $(8, -5)$
10. $(0, 0)$ and $(2, 2)$

Finding the Slope. For each of the following, find the slope of the linear function corresponding to the information provided.

11. $f(1) = -2$ and $f(3) = 1$
12. $g(0.5) = 0$ and $g(2) = -2$
13. $h(-1) = -2$ and $h(1) = 2$
14. $w(2) = -3$ and $w(6) = -3$

15. Identify the slope and vertical intercept of each line. Then write an equation of the line.

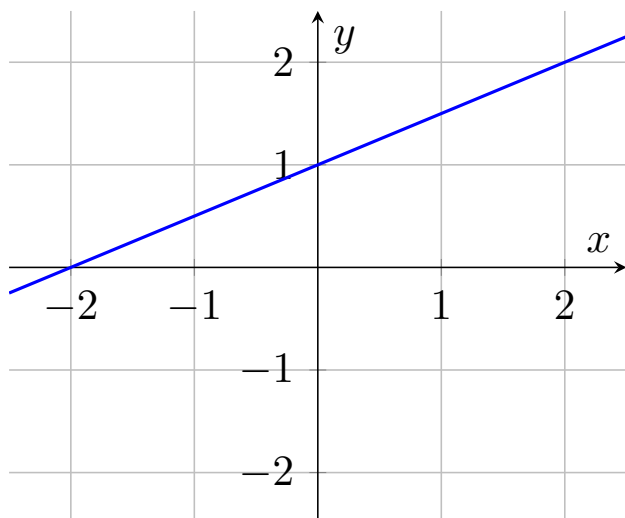


Figure 2.1.17

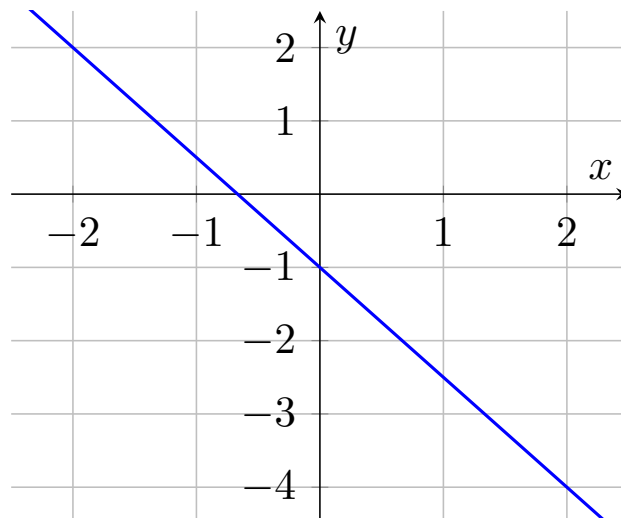


Figure 2.1.18

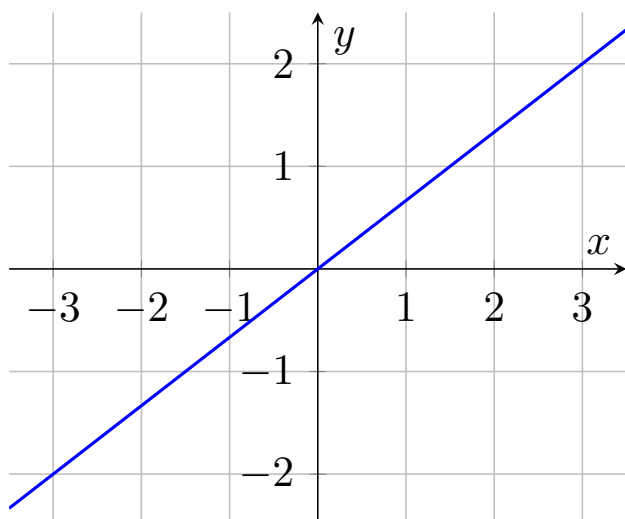


Figure 2.1.19

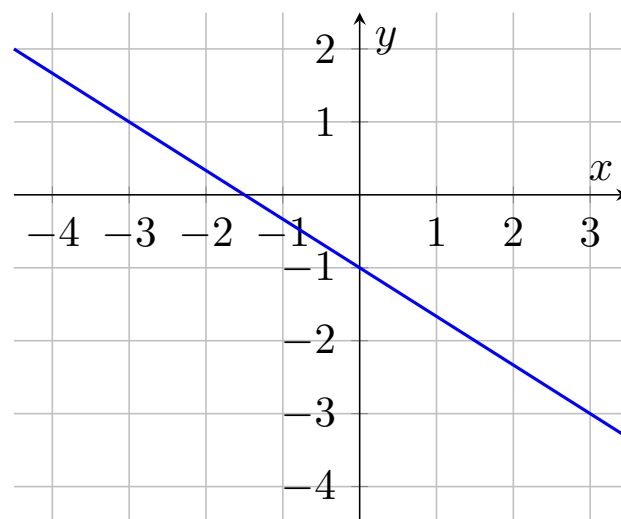
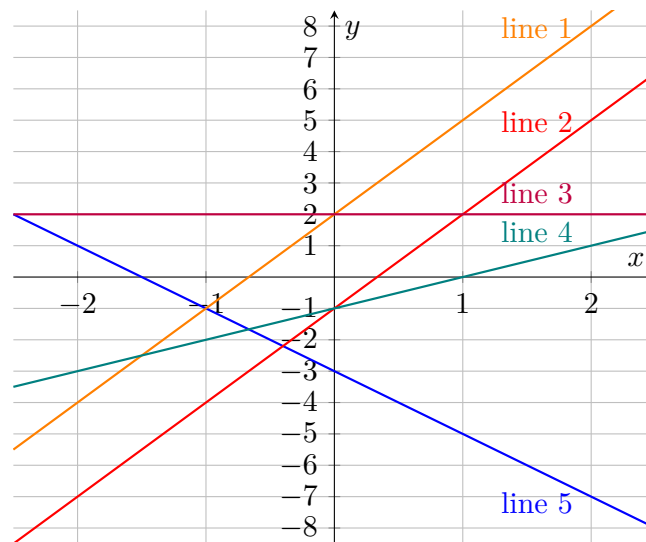


Figure 2.1.20

16. Five linear functions are given below. Drag each function on the left to the corresponding line number on the right.



$y = 3x + 2$	Line 1
$y = 3x - 1$	Line 2
$y = 2$	Line 3
$y = x - 1$	Line 4
$y = -2x - 3$	Line 5

17. During the 2019-2020 academic year, a senior lecturer at URI earned a base salary of \$55,683 for a standard teaching load. For each credit taught in excess of the standard load, \$1,567 were added to his salary. Write a linear equation expressing the total amount A that the senior lecturer earned for teaching n credits in excess of the standard workload during the 2019-2020 academic year.
18. The elevation in feet, $E(t)$, of a hiker t minutes after beginning her hike is given by $E(t) = 35t + 1350$.
- Write a complete sentence explaining the practical meaning of the vertical intercept of this linear function. Include units in your answer.
 - Write a complete sentence explaining the practical meaning of the slope of this linear function. Include units in your answer.
19. The distance in miles from the finish line, $D(t)$, of a bicyclist t hours after beginning a race is given by $D(t) = 50 - 25t$.
- Write a complete sentence explaining the practical meaning of the vertical intercept of this linear function. Include units in your answer.
 - Write a complete sentence explaining the practical meaning of the slope of this linear function. Include units in your answer.

2.2 Working with Linear Functions and Linear Equations

Objectives

After completing this section, you should be able to do the following.

1. First objective.
2. Second objective.
3. Third objective.

In the last section, we introduced the slope-intercept form $y = mx + b$ of a linear function, which is often used when given the slope m and vertical intercept b . When given two points on the linear function, the *point-slope form* is frequently used instead.

Result 2.2.1 Point-Slope Form.

- The point-slope form of the equation of a line with slope m passing through (x_0, y_0) is:

$$y - y_0 = m(x - x_0).$$

- A linear function $y = f(x)$ with slope m for which $f(x_0) = y_0$ can be written in the point-slope form as:

$$f(x) = y_0 + m(x - x_0).$$

Note that $f(x_0) = y_0$ means that the point (x_0, y_0) is on the graph of the function $y = f(x)$.

The point-slope form is very useful for finding formulas for linear functions and equations of lines.

Example 2.2.2

Find an equation in point-slope form for the line passing through each set of points. Then rewrite each equation in the slope-intercept form.

1. $(2, -1)$ and $(4, 5)$
2. $(-1, -2)$ and $(1, 3)$
3. $(0, 0)$ and $(1, 1.5)$

Solution. In each part, we follow the same steps. We first use the two given points to find the slope. Once we have obtained the value of the slope, we then designate one of the two points as (x_0, y_0) and write an equation in the point-slope form. Finally, we rewrite the equation in the slope-intercept form.

1. Using the slope formula, we find that

$$m = \frac{5 - (-1)}{4 - 2} = \frac{6}{2} = 3.$$

We will next let $(x_0, y_0) = (2, -1)$ and use point-slope form. An equation of the line is thus

$$y - (-1) = 3(x - 2) \text{ or } y + 1 = 3(x - 2).$$

To convert to slope-intercept form, we begin by expanding the right-hand side, which results in

$$y + 1 = 3x - 6.$$

By subtracting 1 from both sides, we obtain the slope-intercept form

$$y = 3x - 7.$$

2. The slope is

$$m = \frac{3 - (-2)}{1 - (-1)} = \frac{5}{2}.$$

If we take $(x_0, y_0) = (1, 3)$, an equation of the line in slope-intercept form is

$$y - 3 = \frac{5}{2}(x - 1)$$

which simplifies to

$$y - 3 = \frac{5}{2}x - \frac{5}{2}.$$

Adding 3 to both sides (which can be rewritten as $\frac{6}{2}$ to obtain a common denominator) to results in the slope-intercept form

$$y = \frac{5}{2}x + \frac{1}{2}.$$

3. The slope is

$$m = \frac{1.5 - 0}{1 - 0} = \frac{1.5}{1} = 1.5.$$

If we take $(x_0, y_0) = (0, 0)$, the resulting point-slope form of the line is

$$y - 0 = 1.5(x - 0)$$

which simplifies to the slope-intercept form

$$y = 1.5x.$$

By the way... Note that $y = 1.5x$ is in the form $y = mx + b$ with $b = 0$. This makes sense, as the line passes through the origin $(0, 0)$, meaning that the y -intercept of the line is 0.

Example 2.2.3

Let $y = f(x)$ be a linear function such that $f(1) = -2$ and $f(3) = -5$. Find a formula for the function in the slope-intercept form.

Solution. We are given the values of f at two inputs, which tells us that the two points $(1, -2)$ and $(3, -5)$ are on the graph of the function. Using these to calculate the slope we find that

$$m = \frac{-5 - (-2)}{3 - 1} = -\frac{3}{2}.$$

Recall that the point-slope form for a linear function $f(x)$ can alternatively be written as

$$f(x) = y_0 + m(x - x_0).$$

If we take $(x_0, y_0) = (1, -2)$ we find that

$$f(x) = -2 - \frac{3}{2}(x - 1).$$

The formula simplifies to the slope-intercept form

$$f(x) = -\frac{3}{2}x - \frac{1}{2}.$$

Linear Functions and Linear Equations

When working with linear functions, we frequently find ourselves needing to set the formula for the linear function equal to a specific quantity and then to solve the resulting equation. For instance, when tasked with finding the horizontal intercept of the linear function $f(x) = -5x + 1$, we would set $f(x) = 0$ and solve for x . The resulting equation $-5x + 1 = 0$ is an illustration of a *linear equation in one variable*.

Definition 2.2.4 Linear Equation in One Variable.

An equation that can be written in the form

$$Ax + B = C$$

where A , B , and C are real numbers with $A \neq 0$ and x is the variable is called a ***linear equation in one variable***.

We will explore how such linear equations arise in applications of linear functions and review how to solve them in the examples below.

Example 2.2.5

Mr. Bell is driving through Nevada to Reno at a constant speed of 40 mph. Let $D(t)$ be his distance from Reno, in miles, t hours after he began driving and suppose that Mr. Bell checks his GPS at $t = 2$ to find that his distance from Reno is 150 miles.

1. Find a formula for the function $D(t)$.
2. How far from Reno was Mr. Bell when he began driving?
3. How long from the moment he began driving will it take him to reach Reno?

Solution.

1. At first, it may seem that we don't have enough information to find a formula for $D(t)$ as we only know the value of the function at a single point: $D(2)=150$. However, we also know that Mr. Bell is driving at the constant speed of 40 mph. In the applied context of this example, this tells us that his distance from Reno is decreasing at the constant rate of 40 mph. A decreasing function has a negative rate of change, so we could say that $D(t)$ is changing at the constant rate of -40 mph. Furthermore, since the rate of change of $D(t)$ is constant, the function $D(t)$ is linear and its slope is the constant rate of change; that is, $m = -40$. With both the slope $m = -40$ and the point $(2, 150)$, we can write the point-slope form for $D(t)$:

$$D(t) = 150 - 40(t - 2).$$

This can be simplified to slope-intercept form:

$$D(t) = 150 - 40(t - 2)$$

$$\begin{aligned}
 &= 150 - 40t + 80 \\
 &= -40t + 230
 \end{aligned}$$

or, equivalently, be written as

$$D(t) = 230 - 40t.$$

2. Mr. Bell began driving at $t = 0$ when, according to the formula, his distance from Reno was $D(0) = 230$ miles.
3. Mr. Bell will reach Reno when his distance from Reno is 0 miles; that is, at the value of t such that

$$D(t) = 230 - 40t = 0.$$

To find such t we must solve the equation

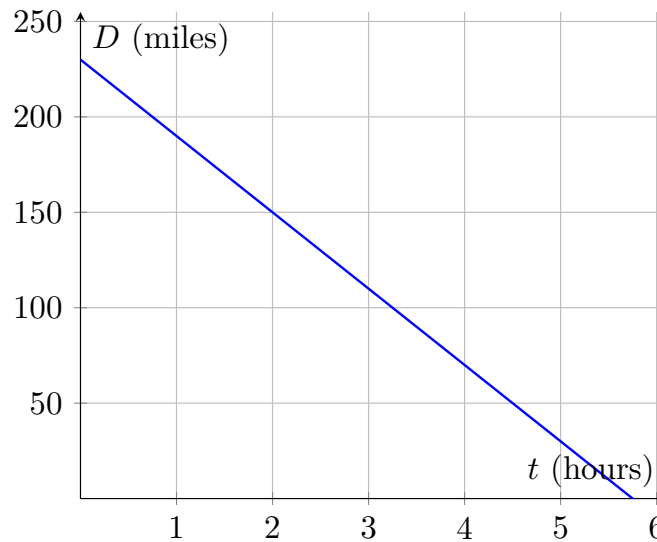
$$230 - 40t = 0.$$

To solve our equation, we apply the standard methods of algebra, which in that case will involve subtracting 230 from both sides of the equation and then dividing both sides by -40 .

$$\begin{aligned}
 230 - 230 - 40t &= 0 - 230 \\
 -40t &= -230 \\
 \frac{-40t}{-40} &= \frac{-230}{-40} \\
 t &= \frac{23}{4} \\
 t &= 5.75
 \end{aligned}$$

It takes Mr. Bell 5.75 hours total to reach Reno. Alternatively, we could translate 5.75 hours into 5 hours and 45 minutes.

Observe that to answer parts (b) and (c) of Example 2.2.5, we had to find the *intercepts* of the function $D(t)$. In applied problems, intercepts have an important practical meaning. Here is a graph of the function $D(t) = 230 - 40t$ that serves as a visual illustration of the meaning of the intercepts that we found:



Example 2.2.6

During the first ten days after hatching, a chick gains weight at the rate of 18.9 grams/day. Let $w(d)$ be the weight of the chick d days after hatching and suppose that three days after hatching the chick weighs 100.6 grams.

1. Find a formula for the function $w(d)$.
2. What was the weight of the chick when it hatched?
3. How many days after it hatched will the chick weigh 120 grams?

Solution.

1. The function $w(d)$ changes at a constant rate so it is linear. As the slope m is equal to the constant rate of change, $m = 18.9$. We are also given the value of the function at $d = 3$: $w(3) = 100.6$. Hence, the point $(d_0, w_0) = (3, 100.6)$ is on the graph of the function $w(d)$. Using the point-slope form of a linear function we find:

$$w(d) = 100.6 + 18.9(d - 3).$$

In slope-intercept form,

$$w(d) = 18.9d + 43.9$$

2. The chick hatched when $d = 0$. Therefore, the weight of the chick when it hatched was

$$w(0) = 18.9 \cdot 0 + 43.9 = 43.9 \approx 44 \text{ grams.}$$

3. The weight of the chick will reach 120 grams for the value of d that satisfies the equation $w(d) = 120$. That is, for d satisfying the equation

$$43.9 + 18.9d = 120.$$

To solve for d , subtract 43.9 from both sides and then divide both sides by 18.9

$$18.9d + 43.9 - 43.9 = 120 - 43.9$$

$$18.9d = 76.1$$

$$\frac{18.9d}{18.9} = \frac{76.1}{18.9}$$

$$d \approx 4.026$$

It will take approximately 4 days for the chick to reach the weight of 120 grams.

Example 2.2.7

Twin towns are municipalities in different countries that are paired to encourage human contact and cultural links. The populations of two twin towns—one in the Netherlands and one in France—are $N(t)$ and $F(t)$, respectively, where t is the number of years since a twin partnership was established.

Both populations grow linearly according to the formulas:

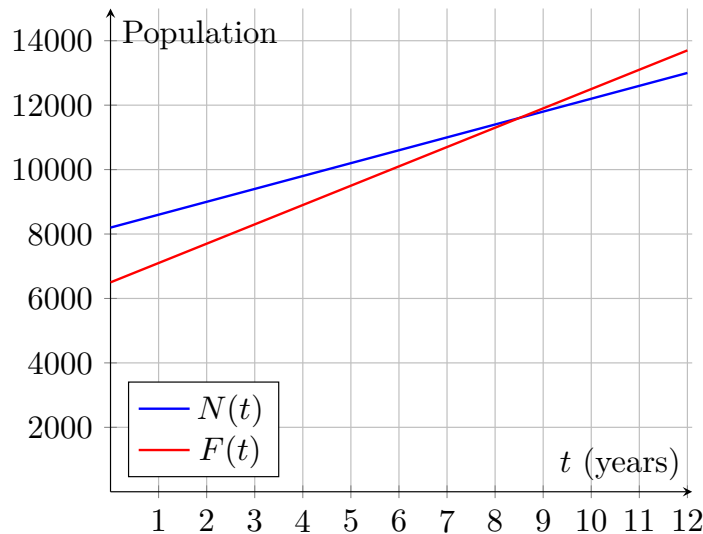
$$N(t) = 400t + 8200 \quad \text{and} \quad F(t) = 600t + 6500.$$

1. Which town had a larger population when a twin partnership was initially established?
2. Which town has a faster growing population? At what rate does it grow?
3. Will the two populations ever be equal? If yes, when?

Solution.

1. The initial population of the Dutch town is $N(0) = 8200$ people, of the French town $F(0) = 6500$. Initially, the population of the Dutch town was larger.
2. It is the slope of each function that gives its rate of increase. The population of the Dutch town grows at the rate of 400 people/year while the population of the French town grows faster at the rate of 600 people/year.

3. The population of the French town is initially lower than the population of the Dutch town. However, as the populations of both towns grow, the population of the French town grows faster and therefore will at some point surpass the population of the Dutch town. This can be seen clearly when we graph both functions.



The population of the French town overtakes the population of the Dutch town where the graphs of $F(t)$ and $N(t)$ cross. This appears to occur between years 8 and 9. As we have the formulas for both $F(t)$ and $N(t)$, we can be more precise than this.

The populations of the two towns are equal when $F(t) = N(t)$. That is, for the value of t satisfying the equation

$$400t + 8200 = 600t + 6500.$$

This is a linear equation in the variable t . To solve it, we begin by moving all constant terms to one side of the equation and all the terms containing t to the other.

$$400t + 8200 - 6500 = 600t + 6500 - 6500$$

$$1700 + 400t = 600t$$

$$400t + 1700 - 400t = 600t - 400t$$

$$1700 = 200t$$

Dividing both sides by 200, we find that $t = 8.5$ years after establishing a twin partnership, the populations of both towns are equal.

Example 2.2.8

The value $V(t)$ of a car, in dollars, t years after the car was purchased is given by

$$V(t) = 15000 - 1250t.$$

1. Find the vertical intercept and the horizontal intercept of the function $V(t)$ and explain their meaning in practical terms.
2. Create a graph of the function $V(t)$. What restriction should be placed on the domain of $V(t)$ in the applied context of this exercise?

Solution.

1. The vertical intercept is the value of the function at $t = 0$:

$$V(0) = 15000 - 1250 \cdot 0 = 15000.$$

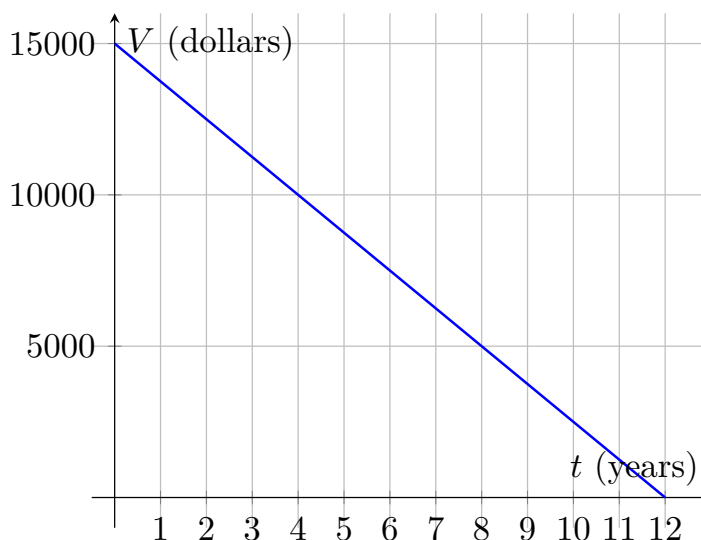
In practical terms, it is the purchase value of the car.

For the horizontal intercept, we must find the value of t for which $V(t) = 0$. That is, we must solve the linear equation $15000 - 1250t = 0$:

$$\begin{aligned} 15000 - 1250t + 1250t &= 0 + 1250t \\ 15000 &= 1250t \\ \frac{15000}{1250} &= \frac{1250t}{1250} \\ 12 &= t \end{aligned}$$

We find that the horizontal intercept is $t = 12$. In practical terms, the horizontal intercept tells us that the car will have no value 12 years after it was purchased.

2. The graph of the function $V(t) = 15000 - 1250t$, with intercepts clearly visible, is provided below.



While the line $y = 15000 - 1250t$ extends forever in both directions and has a domain of all real numbers, the domain of the function $V(t)$ should be limited to $0 \leq t \leq 12$ for practical reasons. This is because the formula begins to model the applied scenario when the car was purchased at time $t = 0$ and ceases to model the applied scenario after 12 years have passed (it does not make sense for the value of the car to be negative).

Number of Solutions to a Linear Equation in One Variable

You may have noticed that each individual linear equation that we set up and solved in the preceding examples resulted in a single solution. It is, however, possible for a linear equation in one variable to result in no solution or an infinite number of solutions. Although you will rarely encounter either of these scenarios when working with linear functions modeling real-life situations, it is important to be familiar with the possibility.

Example 2.2.9

Solve each linear equation, if possible.

1. $5x - 10 = 5(x - 2)$
2. $6z - 14 = z + 4$
3. $4 - 0.5t = -4 - 0.5t$

Solution.

1. To solve, we begin by expanding the left-hand side:

$$5x - 10 = 5(x - 2)$$

$$5x - 10 = 5x - 10$$

You may observe that both sides of the equation are identical. Because of this, the equation is true for all values of x and hence has an ***infinite number of solutions***.

2. To solve, we move all terms involving z to one side of the equation and all constants to the other:

$$6z - 14 = z + 4$$

$$6z - 14 + 14 = z + 4 + 14$$

$$6z = z + 18$$

$$6z - z = z - z + 18$$

$$5z = 18.$$

Dividing both sides by 5 results in the ***single solution*** $z = \frac{18}{5} = 3.6$.

3. To solve, we move all terms involving t to one side of the equation:

$$4 - 0.5t = -4 - 0.5t$$

$$4 - 0.5t + 0.5t = -4 - 0.5t + 0.5t$$

$$4 = -4$$

We end up with a statement that is never true; 4 is not the same value as -4 . The equation has ***no solution***.

Exercises

Slope-Intercept Form. For each of the following, write the equation of the line satisfying the given set of conditions in slope-intercept form.

1. passing through $(2, 6)$ and $(1, 8)$

2. passing through $(-4, 7)$ and $(-10, -8)$
3. with slope 2.1 passing through $(0.1, 0)$
4. with slope 6 passing through $(-5, 2)$
5. with slope $-\frac{1}{2}$ passing through $(\frac{2}{5}, -1)$
6. passing through $(1, 3)$ and $(-7, 3)$
7. passing through $(5, -5)$ and $(5, 2)$
8. with slope 0 and passing through $(-3, -1)$

Slope-Intercept Form. For each of the following, write the formula in slope-intercept form for the linear function $f(x)$ satisfying the given set of conditions.

9. $f(2) = -5$ and $f(4) = 0$
 10. $f(1) = -2$ and $f(3) = -5$
 11. $f(x)$ has slope 2.1 and its graph passes through $(0.1, 0)$
 12. $f(x)$ has slope -1 and its graph passes through $(-1, 2)$
 13. the graph of $f(x)$ passes through $(-1, 3)$ and $(-7, -2)$
 14. the slope of $f(x)$ is 3 and its horizontal intercept is -2
 15. the slope of $f(x)$ is 1 and its vertical intercept is 2
 16. the vertical intercept of $f(x)$ is 3 and its horizontal intercept is 7
17. Which of the following lines is NOT the graph of a linear function?
- (a) $y = -2x - 3x + 1$
 - (b) $y = -5$
 - (c) $x = 7$
18. The distance in miles from the finish line, $D(t)$, of a bicyclist t hours after beginning a race is given by $D(t) = 60 - 20t$
- (a) Write a complete sentence explaining the practical meaning of the vertical intercept of this linear function. Include units in your answer.
 - (b) Write a complete sentence explaining the practical meaning of the horizontal intercept of this linear function. Include units in your answer.
 - (c) Write a complete sentence explaining the practical meaning of the slope of this linear function. Include units in your answer.

19. The value of an antique lamp, $V(t)$, in dollars, t years after its purchase is given by

$$V(t) = 1900 + 250t.$$

- (a) What was the purchase price of the lamp?
 - (b) When will the value of the lamp reach \$4000?
20. Two mobile phone companies sell an international roaming plan. Company A charges a \$75 dollar fixed monthly fee and \$0.20 per minute for talk. Company B charges a \$90 dollar fixed monthly fee and \$0.10 per minute for talk.
- (a) Write the linear functions $A(m)$ and $B(m)$ which give the monthly cost charged by Company A and Company B, respectively, with m minutes spent talking on the phone.
 - (b) For what value of m is the cost the same with either company?
 - (c) Which company gives you a better deal if you plan to talk for hours?

Solving Linear Equations. Solve each linear equation. If there is no solution or infinitely many solutions, say so.

21. $0.1x - (2x - 5) = 2.5$

22. $\frac{-2(x - 3)}{5} = x + 1$

23. $3x - 7 = 5 + 3x$

24. $2x + 2 = 4 + 2(x - 1)$

2.3 Modeling and Numerically Given Functions

Objectives

After completing this section, you should be able to do the following.

1. First objective.
2. Second objective.
3. Third objective.

How do we recognize when it is appropriate for a function given numerically—through a table of values—to be modeled by a linear function? The key is to remember that a function is linear if it changes at a constant rate. We will explore how to determine whether or not a numerically given function changes at a constant rate in the following examples.

Example 2.3.1

In her laboratory, a biologist is studying the growth of the larvae of a certain insect species during the last instar. Her team takes weight measurements of the larvae every 6 hours for 48 hours beginning with $t = 0$. Let $W(t)$ denote the average larval weight, in grams, at time t . Here are the measurements they record rounded off to two decimal places ¹.

t (hours)	0	6	12	18	24	30	36	42	48
$W(t)$ (grams)	7.00	7.45	7.90	8.35	8.80	9.25	9.70	10.15	10.60

Is it appropriate for the growth of the larvae to be modeled by a linear function? If it is, find a formula that matches the data.

Solution. It would be appropriate to model $W(t)$ by a linear function if it changes at a constant rate. A function changes at a constant rate if equal changes in the independent variable correspond to equal changes in the dependent variable.

From the table, we can see that the independent variable t increases by the quantity $\Delta t = 6$ as we move between consecutive inputs. Starting at $t = 0$, we must determine the corresponding change in $W(t)$ each time t increases by $\Delta t = 6$ (or, equivalently, the difference between each pair of consecutive outputs in the table).

$$\begin{array}{llll}
W(6) - W(0) & = & 7.45 - 7.00 & = 0.45 \text{ grams} \\
W(12) - W(6) & = & 7.90 - 7.45 & = 0.45 \text{ grams} \\
W(18) - W(12) & = & 8.35 - 7.90 & = 0.45 \text{ grams} \\
W(24) - W(18) & = & 8.80 - 8.35 & = 0.45 \text{ grams} \\
W(30) - W(24) & = & 9.25 - 8.80 & = 0.45 \text{ grams} \\
W(36) - W(30) & = & 9.70 - 9.25 & = 0.45 \text{ grams} \\
W(42) - W(36) & = & 10.15 - 9.70 & = 0.45 \text{ grams} \\
W(48) - W(42) & = & 10.60 - 10.15 & = 0.45 \text{ grams}
\end{array}$$

These calculations indicate that each time t increases by $\Delta t = 6$, the function $W(t)$ increases by $\Delta W = 0.45$.

Hence the function $W(t)$ changes at a constant rate. Indeed, the average rate of change in $W(t)$ on each of the 6-hour intervals is

$$\frac{\Delta W}{\Delta t} = \frac{0.45}{6} = 0.075 \frac{\text{grams}}{\text{hour}}.$$

Since it changes at a constant rate, it is appropriate to model $W(t)$ by a linear function $W(t) = mt + b$ for some constants m and b . The vertical intercept b is the value of the function at $t = 0$. We have that value in the table:

$$b = W(0) = 7.00 \text{ grams}$$

The slope is the constant rate of change which we have just calculated, so $m = 0.075$ grams/hour. Hence

$$W(t) = 0.075t + 7.00$$

is an appropriate mathematical model for the growth of the larvae.

Example 2.3.2

Decide which of the following tables could represent a linear function. For each table that could represent a linear function, find a formula matching the data in the table.

¹Data source: <http://www.biology.arizona.edu/biomath/tutorials/Linear/LinearModels.html> (www.biology.arizona.edu/biomath/tutorials/Linear/LinearModels.html), accessed: 6/12/20

1.

t	2	4	6	8	10	12
$f(t)$	6.0	4.6	3.2	1.8	0.4	-1.0

2.

x	0	3	6	9	12	15
$g(x)$	6	7.5	9.25	11.5	15.5	20

3.

x	0	1	2	3	4	5
$h(x)$	3	4.5	6	7.5	9	10.5

Solution.

1. The change in t between any two consecutive inputs in the table is given by $\Delta t = 2$. The change in $f(t)$ as t increases by Δt (or, equivalently, the change in f between each pair of consecutive points in the table) can be calculated as follows:

$$\begin{aligned}\Delta f &= f(4) - f(2) &= 4.6 - 6.0 &= -1.4 \\ \Delta f &= f(6) - f(4) &= 3.2 - 4.6 &= -1.4 \\ \Delta f &= f(8) - f(6) &= 1.8 - 3.2 &= -1.4 \\ \Delta f &= f(10) - f(8) &= 0.4 - 1.8 &= -1.4 \\ \Delta f &= f(12) - f(10) &= -1.0 - 0.4 &= -1.4\end{aligned}$$

We can see that each time t increases by $\Delta t = 2$, $f(t)$ changes by $\Delta f = -1.4$; that is, $f(t)$ decreases by 1.4.

Since changes in f corresponding to equal changes in t are all equal, the function $f(t)$ is linear, so $f(t) = mt + b$ for some constants m and b . The slope m is given by

$$m = \frac{\Delta f}{\Delta t} = \frac{-1.4}{2} = -0.7.$$

Hence

$$f(t) = -0.7t + b.$$

In Example 2.3.1, we could easily find the vertical intercept b as we were given the value of the function at 0. The table for $f(t)$ does not directly give us the value $f(0)$, so we will use another point given in the table to set up an equation and solve for b . Taking the point $(2, 6)$ from the table, we have that $f(2) = 6$

$$\begin{aligned}
 -0.7 \cdot 2 + b &= 6 \\
 -1.4 + b &= 6 \\
 -1.4 + 1.4 + b &= 6 + 1.4 \\
 b &= 7.4
 \end{aligned}$$

The final formula for the linear function matching the table is

$$f(t) = -0.7t + 7.4.$$

2. Consecutive values of x in the table are equally spaced by $\Delta x = 3$. We must check if all the corresponding changes in $g(x)$ are equal.

By the way... Here, we can see that each time x increases by $\Delta x = 3$, the corresponding value of Δg becomes larger and larger. That means that the function $g(x)$ increases faster and faster—by more and more at each next step—so $g(x)$ increases at an increasing rate unlike a linear function that increases or decreases at a constant rate.

Because the change in the output, Δg , is not the same each time x increases by $\Delta x = 3$, the function is *not* linear.

3. All changes in x are equal between consecutive inputs in the table with $\Delta x = 1$. The corresponding changes in $h(x)$ are also equal: at every step $h(x)$ increases by 1.5, so $\Delta h = 1.5$. Hence, $h(x)$ is a linear function. We find the slope to be

$$m = \frac{\Delta h}{\Delta x} = \frac{1.5}{1} = 1.5.$$

We are given the value of h at 0: $h(0) = 3$. Hence, $b = 3$. The formula for $h(x)$ is:

$$h(x) = 1.5x + 3.$$

Example 2.3.3

A company purchased a computer system for \$20300. The company accountant decided to depreciate the item over 5 years of its “useful life” for tax purposes. The depreciated value of the system, $V(t)$, in dollars, reported to the IRS t years after the purchase (the so-called “carrying value”) is given by:

t (years after purchase)	0	1	2	3	4	5
$V(t)$ (dollars)	20300	17240	14180	11120	8060	5000

1. What is the amount of depreciation over each of the five years?
2. What is an appropriate formula for the carrying value $V(t)$?

Solution.

1. We begin by calculating the change in depreciated value each year:

$$\begin{aligned}
 \Delta V &= 17240 - 20300 &= -3060 \\
 \Delta V &= 14180 - 17240 &= -3060 \\
 \Delta V &= 11120 - 14180 &= -3060 \\
 \Delta V &= 8060 - 11120 &= -3060 \\
 \Delta V &= 5000 - 8060 &= -3060
 \end{aligned}$$

From the above, we conclude that the computer system depreciates by \$3060 each of the five years.

2. Since equal changes of $\Delta t = 1$ in t correspond to equal changes $\Delta V = -3060$ in the depreciated value of the computer system, $V(t)$ can be modeled by a linear function with formula:

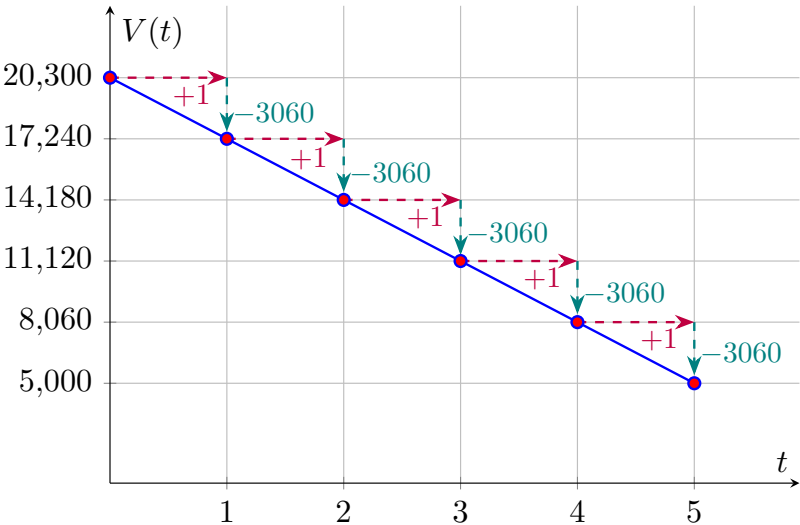
$$V(t) = -3060t + 20300.$$

The depreciation method used by the accountant in Example 2.3.3 is called “straight-line depreciation”. The 5 years of useful life is prescribed by the IRS for each type of asset. Information about how the value at the end of the asset’s useful life, called the “salvage value”, is prescribed can be found in accounting books.

In the straight-line depreciation method, the fixed annual depreciation amount is calculated by taking the difference between the purchase cost and the salvage value and dividing by the number of years of useful life of the asset. For the computer system from Example 2.3.3, this calculation results in

$$\frac{\text{purchase cost} - \text{salvage cost}}{\text{years of useful life}} = \frac{20300 - 5000}{5} = 3060.$$

The computer system is depreciating by \$3060 each year, so the slope of the associated line is -3060 . The graph of the depreciated value $V(t)$ for $0 \leq t \leq 5$ is the straight line shown below.



Exercises

1. Decide if each table could represent a linear function and explain your reasoning. For each table that could represent a linear function, find a formula that matches the data given in the table.

- (a)

x	0	1	2	3
y	12.6	9.3	6.0	2.7
- (b)

t	-1	1	3	5
$f(t)$	0.1	0.4	0.7	1.0
- (c)

n	0	1.5	3	4.5	6
$g(n)$	12	18	12	6	12
- (d)

t	0	2	4	6	8
$h(t)$	6.4	1.6	0.4	0.1	0.025
- (e)

z	-2	1	4	7
$w(z)$	6	4.8	3.6	2.4

2. The table below shows for each temperature F in degrees Fahrenheit the corresponding approximate temperature C in degrees Celsius. Is it appropriate to model C by a linear function of F ? If it is, find a formula for C as a function of F .

F (degrees Fahrenheit)	32	42	52	62
C (degrees Celsius)	0	5.55	11.12	16.67

3. The height of a human individual can be estimated by the length of the femur, as shown for males in the following table.

L (length of femur in cm)	42	45	48	51
H (height of human male in cm)	162.97	169.93	176.89	183.85

- (a) Use the information in this table to find a possible formula for the height of a human male H as a function of his femur length L .
- (b) What would the approximate height of a human male with femur length 46 cm be?
- (c) Fernando is 175.4 cm tall. What would you expect the approximate length of his femur to be?
4. The weight of water above a scuba diver as well as the air above the diver exerts pressure on their bodies. The pressure the diver experiences at sea level is 14.7 PSI (pounds per square inch), and this pressure increases by 0.4 PSI per each foot of depth.
- (a) Write a linear equation expressing the pressure P on a diver at a depth of d feet below sea level.
- (b) The deepest a recreational scuba diver typically dives is 130 feet. What is the pressure on a diver at this depth?
5. A hiker is at a trailhead about to climb a mountain. The temperature at the trailhead is 65°F . According to the standard atmosphere model ² the temperature drops by 0.00356°F per each 1 foot increase in altitude. Let $T = T(h)$ be the temperature, in $^{\circ}\text{F}$, h feet above the hiker.
- (a) Write a formula for the function $T(h)$.
- (b) Find the temperature at the mountaintop that is 6210 feet above the hiker.
- (c) Is the function $T(h)$ increasing or decreasing?
6. A man goes to a gym to exercise. After t minutes on a treadmill, his pulse (heart rate), H , in beats per minute, is:

t (minutes)	0	3	6	9	12	15
H (bpm)	85	87	89	91	93	95

- (a) Is it appropriate to model $H(t)$ by a linear function? If yes, find a formula for $H(t)$.
- (b) If the function $H(t)$ can be modeled by a linear function, give units for the slope and the vertical intercept of $H(t)$. Then explain their meaning in practical terms.

²<https://www.grc.nasa.gov/WWW/K-12/airplane/atmos.html> (www.grc.nasa.gov/WWW/K-12/airplane/atmos.html), accessed: 6/25/20

Chapter 3

Quadratic Functions

Describing the path of a projectile, designing lenses to bend and focus light, and maximizing revenue in certain business situations are just a few of the places that *quadratic functions* can be utilized. With “U”-shaped graphs known as parabolas, these functions allow us to model and understand a remarkable array of real-world phenomena.

3.1 Introduction to Quadratic Functions

Objectives

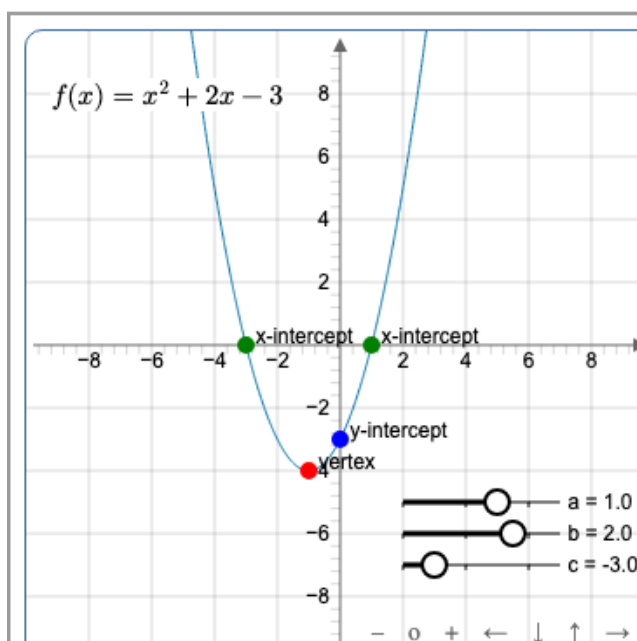
After completing this section, you should be able to do the following.

1. First objective.
 2. Second objective.
 3. Third objective.
-

Consider the function $f(x) = x^2 + 2x - 3$. A table of values for this function is provided below.

x	$y = f(x)$	point
-4	$(-4)^2 + 2(-4) - 3 = 5$	$(-4, 5)$
-3	$(-3)^2 + 2(-3) - 3 = 0$	$(-3, 0)$
-2	$(-2)^2 + 2(-2) - 3 = -3$	$(-2, -3)$
-1	$(-1)^2 + 2(-1) - 3 = -4$	$(-1, -4)$
0	$0^2 + 2(0) - 3 = -3$	$(0, -3)$
1	$1^2 + 2(1) - 3 = 0$	$(1, 0)$
2	$2^2 + 2(2) - 3 = 5$	$(2, 5)$

We can use this table of values to construct the plot of $f(x) = x^2 + 2x - 3$ or we can rely on a graphing utility. In either case, the result is the “U”-shaped graph in Figure 3.1.1. In the figure below, change the values of a, b, c to see how the function changes.



Standalone

Figure 3.1.1

From the graph, we can see that $f(x) = x^2 + 2x - 3$ has two x -intercepts, $x = -3$ and $x = 1$, located at the points $(-3, 0)$ and $(1, 0)$, respectively. The y -intercept $y = -3$ is located at $(0, -3)$. Furthermore, there is an additional special point on this graph at which it “turns around” that is referred to as its **vertex**. The vertex is located at $(-1, -4)$.

The function $f(x) = x^2 + 2x - 3$ is an illustration of a **quadratic function**.

Definition 3.1.2 Standard Form for a Quadratic Function.

A function that can be written in the form

$$f(x) = ax^2 + bx + c$$

where a , b , and c are real numbers with $a \neq 0$ is called a **quadratic function**.

The form $f(x) = ax^2 + bx + c$ is referred to as the **standard form** for a quadratic function.

Graphs of Quadratic Functions

The graph of a quadratic function is “U”-shaped and called a **parabola**. To be precise, we will often call it a **quadratic parabola** to distinguish it from parabola-like graphs of other functions.

By the way... You will encounter parabola-like graphs of other functions in Chapter 4.

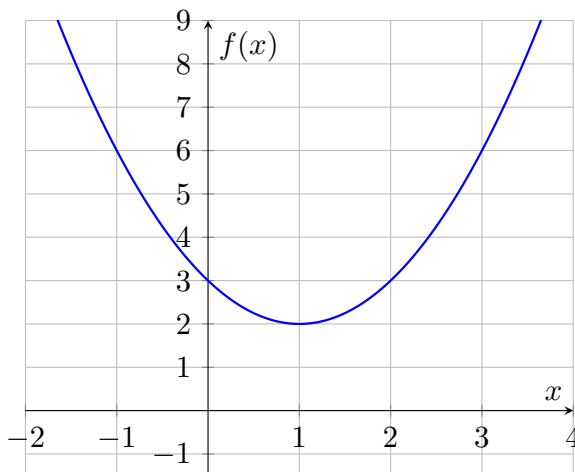
Example 3.1.3

Use the graph of each quadratic function to identify its intercept(s) and vertex.

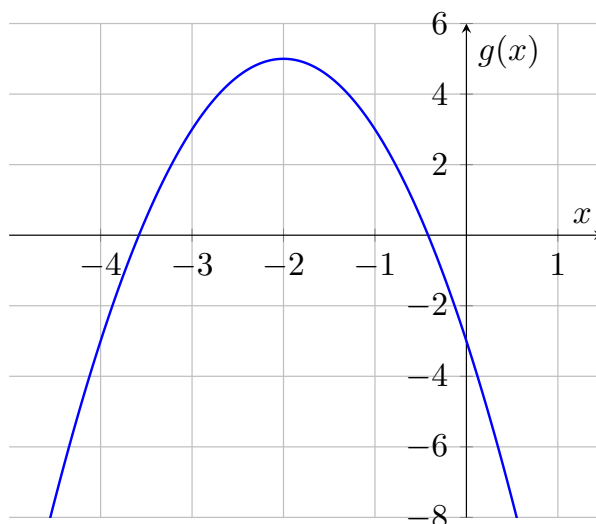
1. $f(x) = x^2 - 2x + 3$
2. $g(x) = -2x^2 - 8x - 3$
3. $h(x) = x^2 - 4x + 4$

Solution. The graph of each quadratic function can be obtained by making a table of values or by using a graphing calculator or utility.

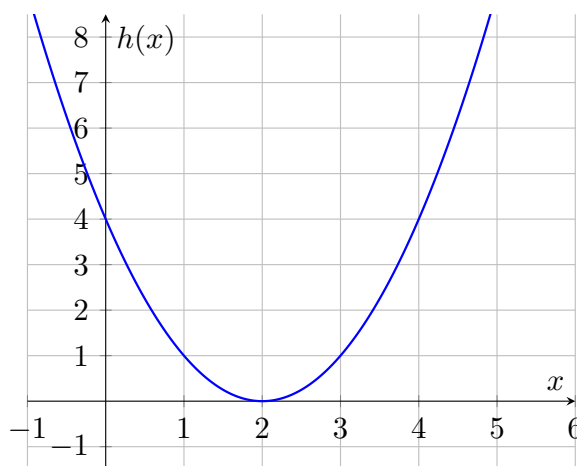
1. The y -intercept is where the graph crosses the y -axis; this occurs at $y = 3$. There are no x -intercepts, since the graph is located entirely above the x -axis. The “turning point” or vertex is the lowest point on the graph since this is an upward facing quadratic parabola; hence the vertex is $(1, 2)$.



2. The y -intercept of this quadratic function is $y = -3$. The graph crosses the x -axis at two points so there are two x -intercepts. Visual inspection reveals that the approximate values of x at which the x -intercepts are located are $x = -3.6$ and $x = -0.4$. The vertex is the highest point on the graph since this is a downward facing quadratic parabola; hence the vertex is $(-2, 5)$.



3. The y -intercept of this function is $y = 4$. The graph crosses the x -axis at one point only: $(2, 0)$. Hence, $h(x)$ has one horizontal intercept $x = 2$. The vertex is the lowest point on the graph since this is an upward facing quadratic parabola; hence the vertex is $(2, 0)$.



Example 3.1.3 indicates that a quadratic function $f(x) = ax^2 + bx + c$ may have two horizontal intercepts, one horizontal intercept, or no horizontal intercepts. Recall that horizontal intercepts of $f(x)$ are also called the real **zeros** of $f(x)$ since they are values of x where $f(x) = 0$. With this in mind, our observation about the number of horizontal intercepts a quadratic function may have can be used to conclude that the quadratic equation $ax^2 + bx + c = 0$ may have two real solutions, one real solution, or no real solution.

The three parts of Example 3.1.3 illustrate several other features of quadratic functions. Observe that the parabolas obtained by graphing $f(x) = x^2 - 2x + 3$ and $h(x) = x^2 - 4x + 4$ opened up while the

parabola obtained by graphing $g(x) = -2x^2 - 8x - 3$ opened down. Whether the parabola associated with a quadratic function $f(x) = ax^2 + bx + c$ will open up or open down is determined entirely by the sign of the leading coefficient a . Additionally, it can be seen that each of the quadratic parabolas obtained from graphing the functions listed in Example 3.1.3 is symmetric about the vertical line passing through its vertex.

Result 3.1.4 Special Features of the Graphs of Quadratic Functions.

The “U”-shaped graph of a quadratic function $f(x) = ax^2 + bx + c$ is called a **parabola**.

- If a is positive the “U” opens up and the vertex occurs at the lowest point on the parabola.
- If a is negative the “U” opens down and the vertex occurs at the highest point on the parabola.

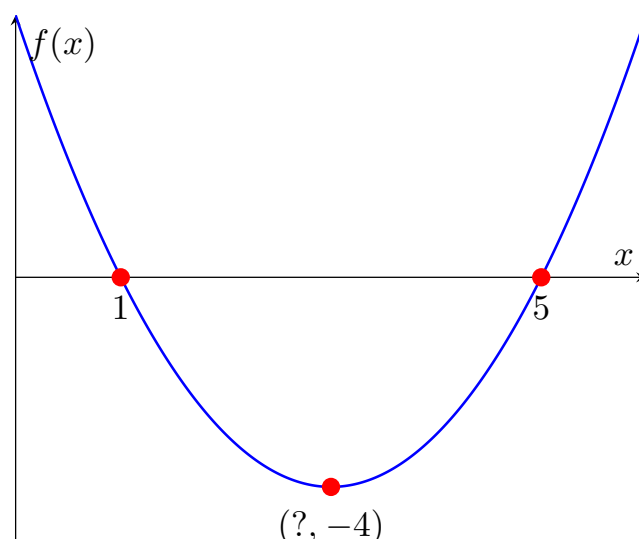
Furthermore, a quadratic parabola is symmetric about the vertical line passing through its vertex, which is referred to as the **axis of symmetry**. Because of this symmetry, the x -coordinate of the vertex of a quadratic function with two distinct horizontal intercepts occurs at the midpoint of those horizontal intercepts.

- The x -coordinate of the vertex of a quadratic function with distinct horizontal intercepts $x = x_1$ and $x = x_2$ has formula:

$$\text{the } x \text{ coordinate of the vertex} = \frac{x_1 + x_2}{2}.$$

Example 3.1.5

Below is the graph of a quadratic function $f(x) = ax^2 + bx + c$. The horizontal intercepts and the y -coordinate of the vertex are given. Find the x -coordinate of the vertex and the vertex itself.



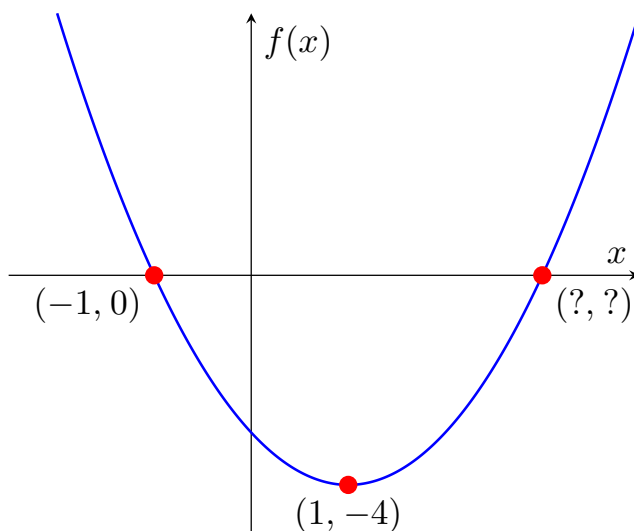
Solution. The x -coordinate of the vertex is the midpoint between the horizontal intercepts $x_1 = 1$ and $x_2 = 5$. Visually, we may be able to “guess” that the midpoint is $x = 3$. To ensure precision, we can use the formula for the midpoint of the x -intercepts:

$$\frac{x_1 + x_2}{2} = \frac{1 + 5}{2} = 3.$$

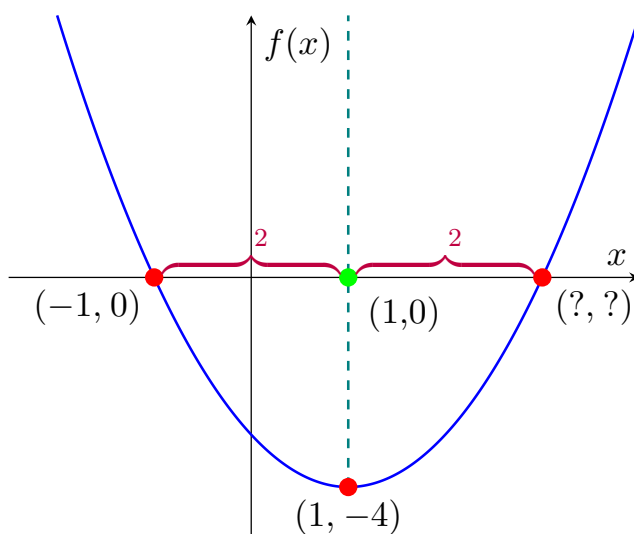
Hence the x -coordinate of the vertex is indeed 3. As the y -coordinate of the vertex is given, the vertex is the ordered pair $(3, -4)$.

Example 3.1.6

Below is the graph of a quadratic function $f(x) = ax^2 + bx + c$. One of the two horizontal intercepts and the vertex are given. Find the point on the x -axis that corresponds to the other horizontal intercept of $f(x)$.



Solution. The graph of every quadratic parabola is symmetric with respect to the vertical line passing through its vertex. In this case, we can see that the vertical line through the vertex $(1, -4)$ crosses the x -axis at the point $(1, 0)$.



By symmetry, each x -intercept must be the same horizontal distance from the point $(1, 0)$. As $(-1, 0)$ is a horizontal distance of 2 units from $(1, 0)$, the second horizontal intercept must be $(3, 0)$.

Alternatively, we could use the fact that the x -coordinate of the vertex, $x = 1$, is the midpoint between the given horizontal intercept $x_1 = -1$ and the second horizontal intercept x_2 that we are tasked with finding. Using the formula for the x -coordinate of the vertex:

$$\begin{aligned}
 x\text{-coordinate of the vertex} &= \frac{x_1 + x_2}{2} \\
 1 &= \frac{-1 + x_2}{2} \\
 2 \cdot 1 &= \frac{-1 + x_2}{2} \cdot 2 \\
 2 &= -1 + x_2 \\
 2+1 &= -1 + x_2+1 \\
 3 &= x_2
 \end{aligned}$$

Whether we use a visual approach or the formula, we arrive at the same conclusion: the second horizontal intercept is at the point $(3, 0)$.

Quadratic Functions and Their Graphs in Applications

Quadratic functions often model real-life scenarios for which the intercepts and vertex have important practical interpretations.

Example 3.1.7

A chair manufacturer finds that the number of chairs that it can sell depends on the price p (in dollars) that it charges per chair. Specifically, the number of chairs that will be sold if p dollars is charged per chair is given by the formula $1200 - 6p$.

1. Find the formula for the revenue function $R(p)$ and graph $R(p)$.
2. For what price(s) per chair is the manufacturer's revenue \$0?
3. What is the maximum revenue? What price should the chair manufacturer charge per chair in order to maximize revenue?

Solution.

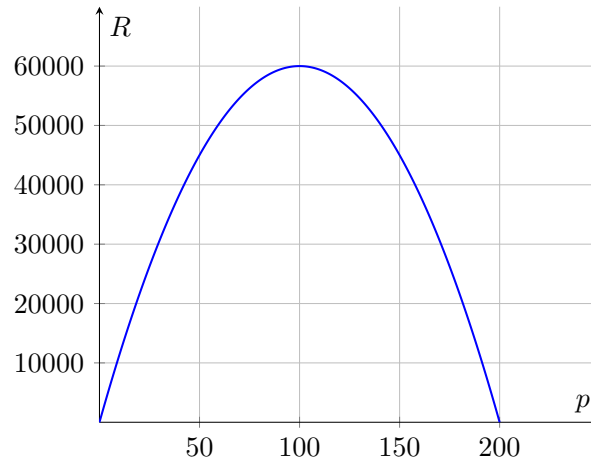
1. The revenue $R(p)$ is the income that the chair manufacturer makes from selling chairs. This means that

$$\begin{aligned}
 R(p) &= \underbrace{\left(\begin{array}{c} \text{price charged} \\ \text{per chair} \end{array} \right)}_p \times \underbrace{\left(\begin{array}{c} \text{number of chairs sold} \\ \text{at that price point} \end{array} \right)}_{1200-6p} \\
 &= p(1200 - 6p)
 \end{aligned}$$

Note that the revenue function is in fact a quadratic function, which can be seen more easily by using the distributive law to rewrite the formula we found above in the standard form $R(p) = ap^2 + bp + c$:

$$\begin{aligned}
 R(p) &= p(1200 - 6p) \\
 &= 1200p - 6p^2 \\
 &= -6p^2 + 1200p.
 \end{aligned}$$

Here, $a = -6$, $b = 1200$, and $c = 0$. The graph of the revenue function can be obtained using a table of values or a graphing utility and the result is the parabola below.



2. The price(s) for which the chair manufacturer's revenue will be \$0 can be found by setting the revenue function equal to 0 and solving for p :

$$p(1200 - 6p) = 0.$$

The only way that a product of two factors can be 0 is if one of the factors itself is 0, so the above breaks into the two equations:

$$p = 0 \quad \text{and} \quad 1200 - 6p = 0.$$

Solving the latter,

$$\begin{aligned}
 1200 - 6p &= 0 \\
 1200 - 1200 - 6p &= 0 - 1200 \\
 \frac{-6p}{-6} &= \frac{-1200}{-6} \\
 p &= 200.
 \end{aligned}$$

These two values can also be found by visually inspecting the graph of $R(p)$. The revenue is \$0 where the graph crosses the horizontal axis, which occurs at the values of $p = 0$ and $p = 200$. Hence if either \$0 or \$200 is charged per chair, the revenue will be \$0. In the first case, if the chair manufacturer charges nothing for a product, it will of course receive no income. In the second case, charging \$200 per chair results in the chair manufacturer pricing itself out of the market; they have set the price too high for customers to be willing to purchase the item.

3. Since the graph of $R(p)$ is a parabola that opens down, answering questions about the maximum revenue involves the vertex. We can find the vertex by visual inspection or by using the midpoint of the two horizontal intercepts $p_1 = 0$ and $p_2 = 200$. The formula

$$\frac{p_1 + p_2}{2} = \frac{0 + 200}{2} = 100$$

tells us that the p -value of the vertex is 100. The R -value of the vertex can be found by substituting $p = 100$ into the formula $R(p) = -6p^2 + 1200p$:

$$R(100) = -6(100)^2 + 1200(100) = 60000.$$

Since the vertex is $(100, 60000)$, the maximum revenue occurs when a price of \$100 is charged per chair. The maximum revenue is \$60,000.

Exercises

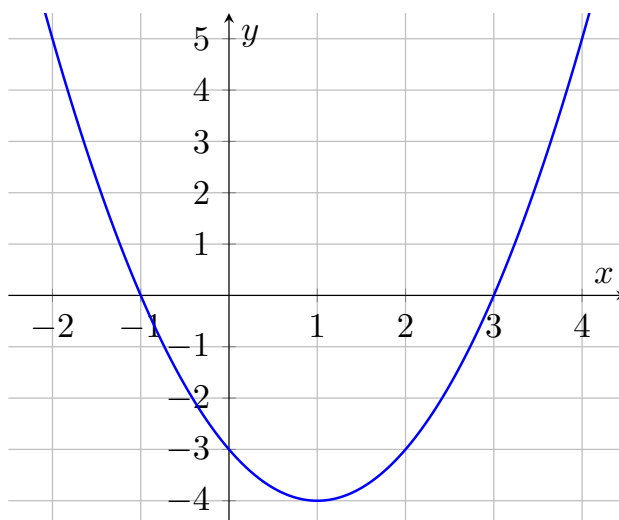
Standard Form. Write each quadratic function in standard form. Then determine whether its graph will open up or open down without graphing the function.

1. $f(x) = x(4x - 7)$
2. $h(t) = 17 + 3t - 5t^2 - 2(1.5t - 2)$
3. $g(x) = 4x(x - 4) - 2x(x - 1)$

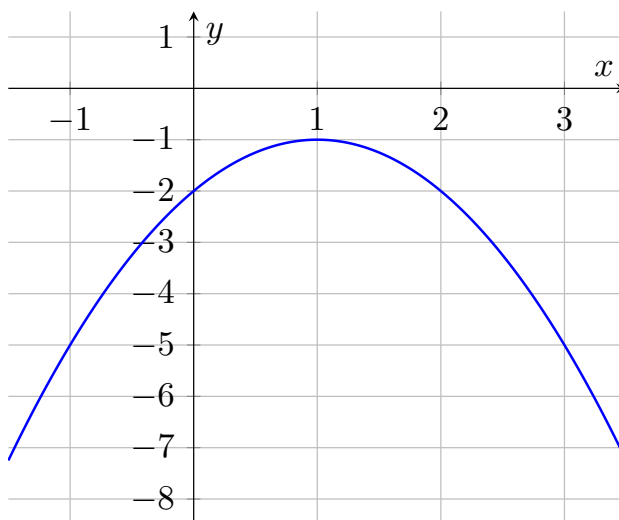
4. $y = -4(x + 5)^2 + 80$

Intercepts and Vertex. Use the graph of each quadratic function to identify its intercept(s) and vertex.

5.



6.



Horizontal Intercepts. The horizontal intercepts of a quadratic parabola are given. Find the x -coordinate of its vertex.

7. $x = -1.5$ and $x = 1.5$

8. $x = -2$ and $x = 5$

9. A quadratic parabola crosses the x -axis at the point $(2, 0)$ and has its vertex at the point $(4, 4)$. Find the second point on the x -axis where the parabola crosses the x -axis. Does the parabola open up or open down?

10. A lighting company can sell $1000 - 2p$ units of its specialty chandelier if it charges $\$p$ per specialty chandelier.
- (a) Find the formula for the revenue $R(p)$ generated from specialty chandelier sales and graph it.
 - (b) At what price will the lighting company sell zero of its specialty chandelier?
 - (c) What is the maximum revenue? What price should the lighting company charge per specialty chandelier in order to maximize revenue?

3.2 Factoring Quadratic Expressions

Objectives

After completing this section, you should be able to do the following.

1. First objective.
2. Second objective.
3. Third objective.

In order to algebraically solve for the horizontal intercepts of a quadratic function such as

$$f(x) = x^2 + 5x + 6$$

one must set $f(x)$ equal to 0 and then solve for x ; in other words, one must solve the quadratic equation

$$x^2 + 5x + 6 = 0.$$

This equation can be solved by **factoring**. This process will be explained below.

When we multiply together

$$(x + M)(x + N)$$

for some real numbers M and N , we obtain:

So if one begins with a quadratic expression of the form $x^2 + Bx + C$ and wants to find two numbers M and N such that

$$x^2 + Bx + C = (x + M)(x + N),$$

it must be the case that $M + N = B$ and $M \cdot N = C$.

Result 3.2.1 Factoring Quadratics of the Form $x^2 + Bx + C$.

In order to factor a quadratic expression of the form $x^2 + Bx + C$ (where B , and C are constants and x is a variable), find two numbers M and N whose product is C and whose sum is B . Then

$$x^2 + Bx + C = (x + M)(x + N).$$

Example 3.2.2

Factor each quadratic expression.

1. $x^2 + 5x + 6$

2. $t^2 - 16t + 64$

3. $w^2 + 8w - 20$

Solution.

1. We need to find two numbers whose product is $C = 6$ and whose sum is $B = 5$. Since $2 \cdot 3 = 6$ and $2 + 3 = 5$, we have

$$x^2 + 5x + 6 = (x + 2)(x + 3).$$

2. We need to find two numbers whose product is $C = 64$ and whose sum is $B = -16$. Since $-8 \cdot (-8) = 64$ and $-8 + (-8) = -16$, we have

$$t^2 - 16t + 64 = (t - 8)(t - 8) = (t - 8)^2.$$

3. We need to find two numbers whose product is $C = -20$ and whose sum is $B = 8$. Since $-2 \cdot 10 = -20$ and $-2 + 10 = 8$, we have

$$w^2 + 8w - 20 = (w - 2)(w + 10).$$

Example 3.2.3

Find the horizontal intercepts of each quadratic function.

1. $f(x) = x^2 + 5x + 6$

2. $y = z^2 - 49$

Solution.

1. To find the horizontal intercepts, we set the function equal to 0 and solve for x :

$$x^2 + 5x + 6 = 0.$$

We factored the left-hand side of this equation in Example 3.2.2 (a), so

$$(x + 2)(x + 3) = 0.$$

$$\begin{array}{ll} x + 2 = 0 & x + 3 = 0 \\ x + 2 - 2 = 0 - 2 & x + 3 - 3 = 0 - 3 \\ x = -2 & x = -3 \end{array}$$

Hence the horizontal intercepts of $f(x) = x^2 + 5x + 6$ are $x = -3$ and $x = -2$.

2. Again, we set the quadratic function equal to 0 and solve:

$$z^2 - 49 = 0.$$

In order to factor the left-hand side, we note that $z^2 - 49 = z^2 + 0z - 49$, so we need two numbers whose product is $C = -49$ and whose sum is $B = 0$. Since $-7 \cdot 7 = -49$ and $-7 + 7 = 0$, we obtain

$$(z - 7)(z + 7) = 0$$

$$\begin{array}{ll} z - 7 = 0 & z + 7 = 0 \\ z - 7 + 7 = 0 + 7 & z + 7 - 7 = 0 - 7 \\ z = 7 & z = -7 \end{array}$$

Hence the horizontal intercepts of $y = z^2 - 49$ are $z = -7$ and $z = 7$.

The quadratic function $z^2 - 49$ in the above example is an example of a ***difference of squares***.

Result 3.2.4 Difference of Squares.

The difference of squares $A^2 - B^2$ factors into $(A + B)(A - B)$:

$$A^2 - B^2 = (A + B)(A - B).$$

The above formula can be verified quickly through multiplication:

Example 3.2.5

Factor each difference of squares:

1. $16x^2 - 81$

2. $100y^2 - \frac{1}{4}$

Solution.

1. Here,

$$16x^2 - 81 = (4x)^2 - (9)^2$$

so

$$16x^2 - 81 = (4x + 9)(4x - 9).$$

2. Here,

$$100y^2 - \frac{1}{4} = (10y)^2 - \left(\frac{1}{2}\right)^2$$

so

$$100y^2 - \frac{1}{4} = \left(10y + \frac{1}{2}\right)\left(10y - \frac{1}{2}\right).$$

Each of the above factoring strategies are special cases of a more general approach to factoring quadratic functions.

Result 3.2.6 Factoring Quadratics of the Form $ax^2 + bx + c$.

*To factor a trinomial of the form $ax^2 + bx + c$ (where a , b , and c are constants with $a \neq 0$ and x is a variable), **rewrite the term** bx to factor by grouping. This is done by finding two numbers **whose product is $a \cdot c$ and whose sum is b** . These two numbers can be used to rewrite the term bx .*

Example 3.2.7

Factor each of the following quadratic expressions.

1. $x^2 + 5x - 36$

2. $6x^2 + 7x - 20$

Solution.

1. Here $a = 1$, $b = 5$, and $c = -36$. List all **factors of** $a \cdot c = -36$. Notice that the pair **9, -4 multiply to** -36 and **sum to** 5. Now

$$\begin{aligned} x^2 + 5x - 36 &= x^2 + 9x - 4x - 36 \\ &= x(x + 9) - 4(x + 9) \\ &= (x + 9)(x - 4). \end{aligned}$$

2. Here $a = 6$, $b = 7$, and $c = -20$. List all **factors of** $a \cdot c = -120$. Notice that the pair **15, -8 multiply to** -120 and **sum to** 7. Now

$$\begin{aligned} 6x^2 + 7x - 20 &= 6x^2 + 15x - 8x - 20 \\ &= 3x(2x + 5) - 4(2x + 5) \\ &= (2x + 5)(3x - 4). \end{aligned}$$

Example 3.2.8

Find the horizontal intercepts of the quadratic function $y = -3t^2 - t + 2$.

Solution. We set the quadratic function equal to 0 and solve:

$$-3t^2 - t + 2 = 0.$$

In order to factor the left-hand side, we observe that $a = -3$, $b = -1$, and $c = 2$. List all **factors of** $a \cdot c = -6$. Notice that the pair **-3, 2 multiply to** -6 and **sum to** -1 . Now

$$\begin{aligned} -3t^2 - t + 2 &= -3t^2 - 3t + 2t + 2 \\ &= -3t(t + 1) + 2(t + 1) \\ &= (t + 1)(-3t + 2). \end{aligned}$$

so we have that

$$(t + 1)(-3t + 2) = 0$$

$$t + 1 = 0$$

$$-3t + 2 = 0$$

$$t + 1 - 1 = 0 - 1$$

$$t = -1$$

$$-3t + 2 - 2 = 0 - 2$$

$$-3t = -2$$

$$\frac{-3t}{-3} = \frac{-2}{-3}$$

$$t = \frac{2}{3}$$

Hence, the horizontal intercepts of $y = -3t^2 - t + 2$ are $t = -1$ and $t = \frac{2}{3}$. The function $y = -3t^2 - t + 2$ can be factored in the form $y = (t + 1)(-3t + 2)$. By factoring out -3 from the second term, we can factor the function even further as

$$y = -3(t + 1) \left(t - \frac{2}{3} \right).$$

In this form, the real zeros (horizontal intercepts) of the function are readily apparent.

It is worth mentioning another special case of factoring a quadratic expression $ax^2 + bx + c$: the case when c is zero.

Result 3.2.9 Factoring Quadratics of the Form $ax^2 + bx$.

Consider an expression of the form $ax^2 + bx$ (where a and b are constants with $a \neq 0$ and x is a variable). To factor the expression, factor out x from both terms:

$$ax^2 + bx = x(ax + b)$$

or, alternatively, factor out ax from both terms:

$$ax^2 + bx = ax \left(x + \frac{b}{a} \right).$$

If we rewrite a given expression $ax^2 + bx$ as

$$ax \left(x + \frac{b}{a} \right),$$

the horizontal intercepts of the function $y = ax^2 + bx$ are straightforward to find. We want to solve the equation:

$$ax \left(x + \frac{b}{a} \right) = 0.$$

The product is equal to 0 if $ax = 0$ or $x + \frac{b}{a} = 0$. Since $a \neq 0$, $ax = 0$ when $x = 0$. The second term

$x + \frac{b}{a} = 0$ when $x = -\frac{b}{a}$. Hence, we have two x -intercepts for $ax^2 + bx$:

$$x = 0 \quad \text{and} \quad x = -\frac{b}{a}.$$

Example 3.2.10

Find horizontal intercepts of each of the quadratic functions given below.

1. $y = -2x^2 + 3x$

2. $y = x^2 - 5x$

Solution.

1. Here, $a = -2$, $b = 3$, and $c = 0$. We factor $-2x$ from both terms:

$$-2x^2 + 3x = -2x \left(x + \left(-\frac{3}{2} \right) \right) = -2x \left(x - \frac{3}{2} \right).$$

The solutions to the equation

$$-2x \left(x - \frac{3}{2} \right) = 0$$

are $x = 0$ and $x = \frac{3}{2}$, which are the two horizontal intercepts of the function.

2. Here, $a = 1$, $b = -5$, and $c = 0$. We factor out x :

$$x^2 - 5x = x(x - 5).$$

The horizontal intercepts are $x = 0$ and

$$x = -\frac{b}{a} = 5.$$

Quadratic Functions and Projectile Motion

Consider a projectile in a vertical motion straight up or straight down. The height of the projectile above the ground, $H(t)$, in feet, t seconds after it has been thrown can be modeled by the quadratic function

$$H(t) = -16t^2 + V_0t + H_0$$

where V_0 is the initial velocity of the object in feet per second and H_0 is the initial height of the object in feet.

Example 3.2.11

A model rocket is launched from the ground with an initial velocity of 180 feet per second. Let $H(t)$ be the height of the rocket above the ground t seconds after launch.

1. Write a formula for the function $H(t)$.
2. When will the model rocket hit the ground?
3. Find the maximum height that the model rocket reaches.

Solution.

1. The model rocket is launched from the ground. Hence, the initial height H_0 is 0 feet; that is, $H_0 = 0$. The initial velocity is 180 ft/sec. Hence, $V_0 = 180$. The function that gives the height of the rocket after t seconds is:

$$H(t) = -16t^2 + 180t.$$

2. The rocket travels vertically upwards, reaches its maximum height, and then falls back to the ground. It hits the ground at the time t when its height above the ground is zero; that is, when $H(t) = 0$. To find such t , we must solve the equation

$$H(t) = -16t^2 + 180t = 0.$$

Note that the solutions to this equation are the horizontal intercepts of the quadratic parabola associated with $H(t)$. Factoring the left-hand side of the equation results in $t(-16t + 180) = 0$, or, equivalently,

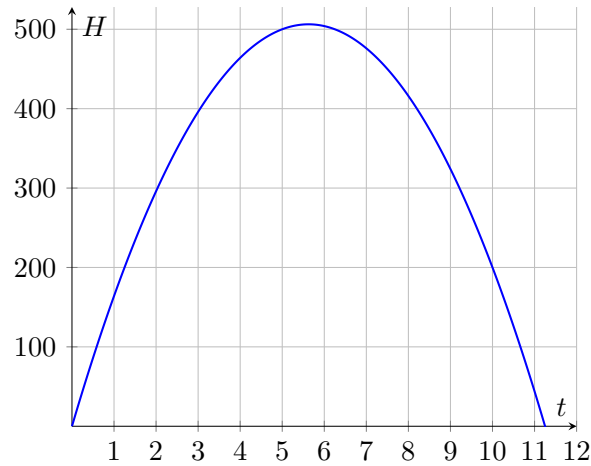
$$-16t(t - 11.25) = 0.$$

The horizontal intercepts are thus $t = 0$ and $t = 11.25$. So the height of the rocket is 0 feet above the ground at both time $t = 0$ — that is, at the moment of launch — and also at time $t = 11.25$, which is when the rocket comes back and hits the ground. Hence, the rocket hits the ground 11.25 seconds after launch.

3. The rocket reaches its maximum height when $H(t)$ is at its maximum. The graph of $H(t) = -16t^2 + 180t$ is a parabola opening down. Hence, $H(t)$ is at its maximum at the vertex of the parabola. Remember that the t -coordinate of the vertex is the midpoint of the horizontal intercepts, $t_1 = 0$ and $t_2 = 11.25$. Hence, the t -coordinate of the vertex is $\frac{t_1+t_2}{2} = \frac{0+11.25}{2} = 5.75$. Thus, the maximum height is reached at time $t = 5.75$. The height at $t = 5.75$ is

$$H(5.75) = -16(5.75)^2 + 180(5.75) = 506$$

feet. So at its highest, the rocket reaches 506 feet. The graph of the function $H(t)$ explains everything:



Note that the graph shows the height of the rocket $H(t)$ as a function of time t . It does *not* show the path of the rocket, which travels straight up and down along a vertical line.

Exercises

Finding Intercepts. Find the intercepts of each quadratic function.

1. $f(x) = x^2 + 3x + 2$
2. $y = x^2 - x - 6$
3. $g(t) = t^2 + t - 6$
4. $y = x^2 - 8x + 16$
5. $f(w) = w^2 + 14w + 49$
6. $f(t) = 64t^2 - 36$
7. $h(x) = 2x^2 + 10x + 8$
8. $y = 2w^2 - w - 3$
9. $f(x) = -x^2 - 6x + 16$
10. $g(x) = \frac{1}{49} - 4x^2$
11. $p(x) = 2x^2 - 6x$
12. $m(t) = -5t^2 + 2t$
13. A model rocket is launched vertically from the ground with an initial velocity of 168 feet per second. Let $H(t)$ be the height, in feet, of the rocket t seconds after launch.

- (a) Write a formula for the function $H(t)$.
- (b) Find t when the model rocket hits the ground.
- (c) Find the maximum height that the model rocket reaches.

3.3 Vertex Form and Completing the Square

Objectives

After completing this section, you should be able to do the following.

1. First objective.
 2. Second objective.
 3. Third objective.
-

Recall that the standard form of a quadratic function is

$$f(x) = ax^2 + bx + c,$$

where a , b , and c are real numbers with $a \neq 0$.

When one first looks upon the function

$$f(x) = (x - 1)^2 + 2$$

it may not immediately be apparent that this is in fact a quadratic function. This can be seen by expanding the right-hand side:

$$\begin{aligned} f(x) &= (x - 1)^2 + 2 \\ &= (x - 1)(x - 1) + 2 \\ &= x^2 - x - x + 1 + 2 \\ &= x^2 - 2x + 3 \end{aligned}$$

Hence,

$$f(x) = (x - 1)^2 + 2$$

can be written in the standard form of a quadratic function as

$$f(x) = x^2 - 2x + 3.$$

This is the same quadratic function that appeared in Example 3.1.3 where we saw that the vertex was $(1, 2)$. Notice that these numbers appear in the alternate formula for $f(x) = x^2 - 2x + 3$:

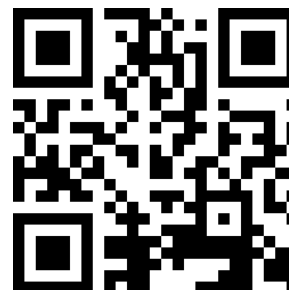
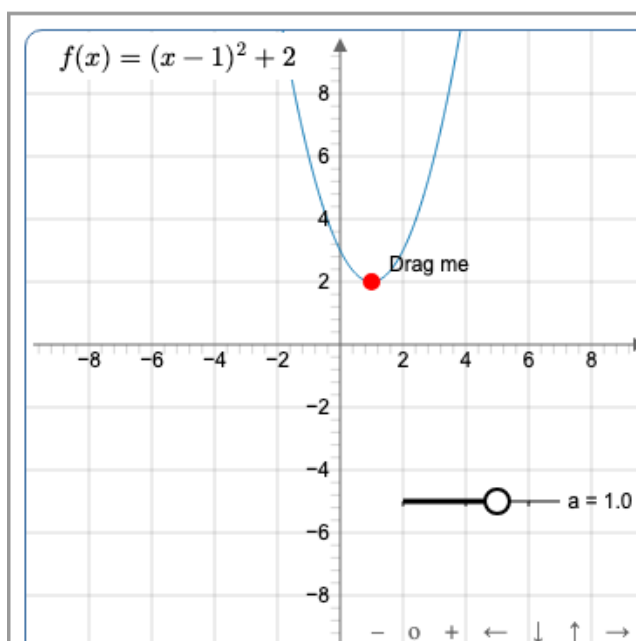
$$f(x) = (x - 1)^2 + 2.$$

This is not a coincidence and brings us to the **vertex form** of a quadratic function.

Definition 3.3.1 Vertex Form.

The *vertex form* of a quadratic function with vertex (h, k) is given by

$$f(x) = a(x - h)^2 + k$$



Standalone

Figure 3.3.2

Example 3.3.3

Identify the vertex of each of the following quadratic functions.

1. $f(t) = -3(t - 6)^2 - 4$
2. $y = (x + 7)^2 + 3$

Solution.

1. The function can be written as

$$f(t) = -3(t - 6)^2 + -4.$$

It can be seen that $h = 6$ and $k = -4$, so its vertex is $(6, -4)$.

2. Note that for the function

$$y = (x + 7)^2 + 3$$

addition is being performed inside the parentheses instead of subtraction, as is required to correctly identify h using the vertex form. Recalling that subtracting a negative number is the same as adding, we can rewrite the above as

$$y = (x - (-7))^2 + 3$$

which allows us to identify that $h = -7$ and $k = 3$. The vertex is $(-7, 3)$.

Converting from Vertex Form to Standard Form

In the beginning of this section, we converted $f(x) = (x - 1)^2 + 2$ from vertex form to its standard form $f(x) = x^2 - 2x + 3$ by expanding the right-hand side. This is what is done in general to convert a quadratic function from vertex form to standard form. To expedite this process, special product formulas may be used. Note that

$$\begin{aligned}(A + B)^2 &= (A + B)(A + B) \\ &= A^2 + AB + AB + B^2 \\ &= A^2 + 2AB + B^2\end{aligned}$$

$$\begin{aligned}(A - B)^2 &= (A - B)(A - B) \\ &= A^2 - AB - AB + B^2 \\ &= A^2 - 2AB + B^2\end{aligned}$$

Result 3.3.4 Special Product Formulas.

$$(A + B)^2 = A^2 + 2AB + B^2 \quad (A - B)^2 = A^2 - 2AB + B^2$$

Example 3.3.5

Convert each quadratic function from vertex form to standard form and identify a , b , and c .

$$1. y = \left(x - \frac{1}{2}\right)^2 + 5$$

$$2. g(t) = -3(t - 5)^2 - 10$$

Solution.

1. Note that

$$y = \left(x - \frac{1}{2}\right)^2 + 5$$

$$\begin{aligned}
&= \left((x)^2 - x + \left(\frac{1}{2}\right)^2 \right) + 5 \\
&= x^2 - x + \frac{1}{4} + 5 \\
&= x^2 - x + \frac{1}{4} + \frac{5}{1} \cdot \frac{4}{4} \\
&= x^2 - x + \frac{21}{4}
\end{aligned}$$

so the standard form is $y = x^2 - x + \frac{21}{4}$. Here $a = 1$, $b = -1$, and $c = \frac{21}{4}$.

2. Note that

$$\begin{aligned}
g(t) &= -3(t - 5)^2 - 10 \\
&= -3((t)^2 - 2(t)(5) + (5)^2) - 10 \\
&= -3(t^2 - 10t + 25) - 10 \\
&= -3(t^2b - 10ct + 25d) - 10 \\
&= -3t^2 + 30t - 75 - 10 \\
&= -3t^2 + 30t - 85
\end{aligned}$$

so the standard form is $g(t) = -3t^2 + 30t - 85$. Here $a = -3$, $b = 30$, and $c = -85$.

Converting from Standard Form to Vertex Form

Converting from standard form to vertex form requires ***completing the square***. This process will be explained below, but first an observation. Note that

$$\begin{aligned}
x^2 + Bx + \left(\frac{B}{2}\right)^2 &= x^2 + 2(x)\left(\frac{B}{2}\right) + \left(\frac{B}{2}\right)^2 \\
&= \left(x + \frac{B}{2}\right)^2
\end{aligned}$$

by the first special product formula (see Result 3.3.4).

Result 3.3.6 Completing the Square for $x^2 + Bx + C$.

To complete the square on a quadratic function of the form $x^2 + Bx + C$:

1. Identify B , the coefficient of the “ x ” term.
2. Compute $\left(\frac{B}{2}\right)^2$.
3. Add and subtract this quantity from the original quadratic function:

$$x^2 + Bx + \left(\frac{B}{2}\right)^2 + C - \left(\frac{B}{2}\right)^2$$

Note that by both adding and subtracting the same number from the original function, you have in fact added 0, which does not change the function.

4. Factor $x^2 + Bx + \left(\frac{B}{2}\right)^2$ into $\left(x + \frac{B}{2}\right)^2$ and combine constants to finish completing the square.

Example 3.3.7

Convert each of the following quadratic functions from standard form to vertex form. Then identify the vertex.

1. $f(x) = x^2 + 6x + 10$
2. $y = t^2 - 8t + 1$
3. $g(x) = x^2 + 5x - 3$

Solution.

1. The quadratic function $f(x) = x^2 + 6x + 10$ is of the form $x^2 + Bx + C$ with $B = 6$. Now

$$\left(\frac{B}{2}\right)^2 = \left(\frac{6}{2}\right)^2 = (3)^2 = 9.$$

Adding and subtracting this quantity from the quadratic yields:

$$\begin{aligned} x^2 + 6x + 10 &= \underbrace{x^2 + 6x + 9}_{\text{factors into } (x+3)^2} + 10 - 9 \\ &= (x + 3)^2 + 1. \end{aligned}$$

Hence, the vertex form of $f(x) = x^2 + 6x + 10$ is

$$f(x) = (x + 3)^2 + 1$$

and the vertex is $(-3, 1)$.

2. The quadratic function $y = t^2 - 8t + 1$ is of the form $t^2 + Bt + C$ with $B = -8$. Now

$$\left(\frac{B}{2}\right)^2 = \left(\frac{-8}{2}\right)^2 = (-4)^2 = 16.$$

Adding and subtracting this quantity from the quadratic yields:

$$\begin{aligned} t^2 - 8t + 1 &= \underbrace{t^2 - 8t + 16}_{\text{factors into } (t-4)^2} + 1 - 16 \\ &= (t - 4)^2 - 15. \end{aligned}$$

Hence, the vertex form of $y = t^2 - 8t + 1$ is

$$y = (t - 4)^2 - 15$$

and the vertex is $(4, -15)$.

3. The quadratic function $g(x) = x^2 + 5x - 3$ is of the form $x^2 + Bx + C$ with $B = 5$. Now

$$\left(\frac{B}{2}\right)^2 = \left(\frac{5}{2}\right)^2 = \frac{25}{4}$$

Adding and subtracting this quantity from the quadratic yields:

By the way... Don't forget that $x^2 + Bx + \left(\frac{B}{2}\right)^2$ always factors into $\left(x + \frac{B}{2}\right)^2$. This is particularly helpful to remember in more complicated exercises like this one.

$$\begin{aligned} x^2 + 5x - 3 &= \underbrace{x^2 + 5x + \frac{25}{4}}_{\text{factors into } \left(x + \frac{5}{2}\right)^2 = \left(x + \frac{5}{2}\right)^2} - 3 - \frac{25}{4} \\ &= \left(x + \frac{5}{2}\right)^2 - \frac{3}{1} \cdot \frac{4}{4} - \frac{25}{4} \\ &= \left(x + \frac{5}{2}\right)^2 - \frac{12}{4} - \frac{25}{4} \end{aligned}$$

$$= \left(x + \frac{5}{2}\right)^2 - \frac{37}{4}.$$

Hence, the vertex form of $g(x) = x^2 + 5x - 3$ is

$$g(x) = \left(x + \frac{5}{2}\right)^2 - \frac{37}{4}$$

and the vertex is $\left(-\frac{5}{2}, -\frac{37}{4}\right)$.

By the way... In short: when $a \neq 1$, create a “new” quadratic (the one in parentheses) that has its “ a ” coefficient equal to 1.

Result 3.3.8 Complete Square on $ax^2 + bx + c$.

To complete the square on a quadratic function of the form $ax^2 + bx + c$ where $a \neq 1$:

1. Factor a out of the first two terms to obtain an expression of the form

$$a \left(x^2 + \frac{b}{a}x\right) + c$$

2. Complete the square within the parentheses.
3. Distribute a and combine constants to finish completing the square on the original quadratic.

Example 3.3.9

Convert each of the following quadratic functions from standard form to vertex form. Then identify the vertex.

1. $y = 3x^2 - 6x + 1$
2. $f(t) = -2t^2 + t - 4$

Solution.

1. We begin by factoring the 3 out of the first two terms to obtain

$$3(x^2 - 2x) + 1.$$

We then complete the square inside the parentheses; in other words, we will complete the square on $x^2 - 2x$. Here $B = -2$ so

$$\left(\frac{B}{2}\right)^2 = \left(\frac{-2}{2}\right)^2 = (-1)^2 = 1$$

and

$$\begin{aligned} x^2 - 2x &= \underbrace{x^2 - 2x + 1}_{\text{factors into } (x-1)^2} - 1 \\ &= (x - 1)^2 - 1. \end{aligned}$$

Now

$$\begin{aligned} 3(x^2 - 2x) + 1 &= 3((x - 1)^2 - 1) + 1 \\ &= 3(x - 1)^2 - 3 + 1 \\ &= 3(x - 1)^2 - 2. \end{aligned}$$

Hence, the vertex form of $y = 3x^2 - 6x + 1$ is

$$y = 3(x - 1)^2 - 2$$

and the vertex is $(1, -2)$.

2. We begin by factoring the -2 out of the first two terms to obtain

$$-2\left(t^2 - \frac{1}{2}t\right) - 4.$$

We then complete the square inside the parentheses; in other words, we will complete the square on $t^2 - \frac{1}{2}t$. Here $B = -\frac{1}{2}$ so

$$\left(\frac{B}{2}\right)^2 = \left(-\frac{1/2}{2}\right)^2 = \left(-\frac{1}{4}\right)^2 = \frac{1}{16}$$

and

$$\begin{aligned} t^2 - \frac{1}{2}t &= \underbrace{t^2 - \frac{1}{2}t + \frac{1}{16}}_{\text{factors into } \left(t + \frac{B}{2}\right)^2 = \left(t - \frac{1}{4}\right)^2} - \frac{1}{16} \end{aligned}$$

$$= \left(t - \frac{1}{4}\right)^2 - \frac{1}{16}.$$

Now

$$\begin{aligned} -2\left(t^2 - \frac{1}{2}t\right) - 4 &= -2\left(\left(t - \frac{1}{4}\right)^2 - \frac{1}{16}\right) - 4 \\ &= -2\left(t - \frac{1}{4}\right)^2 + \frac{2}{16} - 4 \\ &= -2\left(t - \frac{1}{4}\right)^2 + \frac{1}{8} - \frac{4 \cdot 8}{1 \cdot 8} \\ &= -2\left(t - \frac{1}{4}\right)^2 + \frac{1}{8} - \frac{32}{8} \\ &= -2\left(t - \frac{1}{4}\right)^2 - \frac{31}{8}. \end{aligned}$$

Hence, the vertex form of $f(t) = -2t^2 + t - 4$ is

$$f(t) = -2\left(t - \frac{1}{4}\right)^2 - \frac{31}{8}$$

and the vertex is $\left(\frac{1}{4}, -\frac{31}{8}\right)$.

Using Vertex Form to Find Horizontal Intercepts

If a quadratic function is given in vertex form, then its horizontal intercepts can be found by following the process illustrated in the example below.

Example 3.3.10

Find the horizontal intercepts of each of the following functions.

1. $f(x) = x^2 - 9$
2. $g(x) = (x + 1)^2 - 10$
3. $y(t) = 4(t - 1)^2 - 64$
4. $h(x) = (x - 5)^2 + 8$

Solution.

1. To find the horizontal or x -intercept(s) of this quadratic function, we set $f(x)$ equal to 0 and solve for x . We could treat the left-hand side as a difference of squares and proceed by factoring, but we also could take an alternate approach:

$$\begin{aligned}x^2 - 9 &= 0 \\x^2 - 9 + 9 &= 0 + 9 \\x^2 &= 9.\end{aligned}$$

This equation can be solved by taking the square root of both sides, remembering that there will be both a positive solution and a negative solution since both $(\sqrt{9})^2 = 9$ and $(-\sqrt{9})^2 = 9$:

$$\begin{aligned}x^2 &= 9 \\x &= \pm\sqrt{9} \\x &= \pm 3.\end{aligned}$$

Hence, the horizontal intercepts of the function $f(x) = x^2 - 9$ and the solutions of the equation $x^2 - 9 = 0$ are $x = -3$ and $x = 3$.

2. To find the horizontal intercepts, we set $g(x)$ equal to 0 and solve for x :

$$\begin{aligned}(x + 1)^2 - 10 &= 0 \\(x + 1)^2 - 10 + 10 &= 0 + 10 \\(x + 1)^2 &= 10 \\x + 1 &= \pm\sqrt{10} \\x + 1 - 1 &= \pm\sqrt{10} - 1 \\x &= -1 \pm \sqrt{10}.\end{aligned}$$

We obtain two solutions to our equation and two horizontal intercepts of $g(x)$: $x = -1 - \sqrt{10}$ and $x = -1 + \sqrt{10}$.

3. To find the horizontal intercepts, we set $y(t)$ to 0 and solve for t :

$$\begin{aligned}
4(t-1)^2 - 64 &= 0 \\
4(t-1)^2 - 64 + 64 &= 0 + 64 \\
4(t-1)^2 &= 64 \\
\frac{4(t-1)^2}{4} &= \frac{64}{4} \\
(t-1)^2 &= 16 \\
t-1 &= \pm\sqrt{16} \\
t-1 &= \pm 4 \\
t-1+1 &= \pm 4+1 \\
t &= 1 \pm 4.
\end{aligned}$$

Hence, the two solutions and therefore the two horizontal intercepts are $t = 1 + 4 = 5$ and $t = 1 - 4 = -3$.

4. Once more, we set the function $h(x)$ equal to 0 and solve for x :

$$\begin{aligned}
(x-5)^2 + 8 &= 0 \\
(x-5)^2 + 8 - 8 &= 0 - 8 \\
(x-5)^2 &= -8.
\end{aligned}$$

In this case there is no real solution, since $\sqrt{-8}$ is not a real number. This means that the equation has no real solutions and the function $h(x) = (x-5)^2 + 8$ has no horizontal intercepts. If we take a moment to visualize the graph of the function, this makes sense. Its graph is an upward facing parabola with vertex $(5, 8)$. Since the lowest point (the vertex) of this parabola is above the x -axis, there cannot be any horizontal intercepts.

Exercises

Vertex. For each of the following, identify the vertex of the given quadratic function.

1. $f(x) = -2(x-3)^2 + 4$
2. $y = 5(x+2)^2 - 3$
3. $g(t) = 2(t-1)^2 - 7$

4. $h(x) = -(x + \frac{1}{2})^2 + \frac{5}{3}$

Vertex Form to Standard Form. Convert each quadratic function from vertex form to standard form and identify a , b , and c .

5. $f(x) = (x + 1)^2 + 2$

6. $g(x) = 5(x - 2)^2 + 6$

7. $y = -2(t - 4)^2 + 5$

8. $h(x) = -(x + \frac{1}{2})^2 + \frac{5}{3}$

Standard Form to Vertex Form. Convert each quadratic function from standard form to vertex form by completing the square. Then identify the vertex.

9. $f(x) = x^2 + 2x - 5$

10. $g(x) = x^2 - 6x + 10$

11. $h(x) = x^2 + 3x + \frac{5}{4}$

12. $F(x) = -x^2 + 10x + 15$

13. $G(x) = -3x^2 + 6x - 4$

14. $H(x) = -2x^2 - 16x - 17$

Horizontal Intercepts. Find the horizontal intercepts of each quadratic function. Give exact and approximate values rounded off to three decimal places.

15. $f(x) = (x + 3)^2 - 16$

16. $g(x) = (x - 1)^2 - 2$

17. $h(x) = (x - \frac{1}{2})^2 - 1$

18. $F(x) = x^2 + 4x - 6$

19. $G(x) = 2x^2 - 3x + 4$

20. $H(x) = -2x^2 - 16x - 27$

3.4 The Quadratic Formula

Objectives

After completing this section, you should be able to do the following.

1. First objective.
 2. Second objective.
 3. Third objective.
-

In the previous section, we saw that finding the horizontal intercepts of a quadratic function from the vertex form $f(x) = a(x - h)^2 + k$ requires setting $f(x) = 0$ and solving. If we wanted to find the horizontal intercepts of a quadratic function given in the standard form $f(x) = ax^2 + bx + c$, we could first complete the square to convert to vertex form, and then proceed as before. If this process is done in general; i.e., on $f(x) = ax^2 + bx + c$ (without replacing a , b , and c with numbers), we obtain the *quadratic formula*.

Result 3.4.1 Quadratic Formula.

Recall that a *quadratic equation in standard form* is written in the form

$$ax^2 + bx + c = 0.$$

The solution(s) to a quadratic equation in standard form are given by the *quadratic formula*:

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}.$$

By the way... Applying the quadratic formula when an equation is not in standard form is a common mistake!

Example 3.4.2

Use the quadratic formula to solve each quadratic equation.

1. $x^2 + x - 6 = 0$
2. $-2t^2 - 2t + 7 = 0$

3. $(x + 1)(x - 2) = 4$

Solution.

1. Since this quadratic equation is in standard form, $a = 1$, $b = 1$, and $c = -6$. Now by the quadratic formula

$$\begin{aligned} x &= \frac{-1 \pm \sqrt{1^2 - 4 \cdot 1 \cdot (-6)}}{2 \cdot 1} \\ &= \frac{-1 \pm \sqrt{25}}{2} \\ &= \frac{-1 \pm 5}{2}. \end{aligned}$$

So the two solutions are $x = \frac{-1-5}{2} = \frac{-6}{2} = -3$ and $x = \frac{-1+5}{2} = \frac{4}{2} = 2$.

2. Since this quadratic equation is in standard form, $a = -2$, $b = -2$, and $c = 7$. Now by the quadratic formula

$$\begin{aligned} t &= \frac{-(-2) \pm \sqrt{(-2)^2 - 4 \cdot (-2) \cdot 7}}{2 \cdot (-2)} \\ &= \frac{2 \pm \sqrt{60}}{-4} \\ &= \frac{2 \pm \sqrt{4 \cdot 15}}{-4} \\ &= \frac{2 \pm 2\sqrt{15}}{-4} \end{aligned}$$

So the two solutions are

$$t = \frac{2 - 2\sqrt{15}}{-4} = \frac{2}{-4} + \frac{2\sqrt{15}}{-4} = -\frac{1}{2} + \frac{\sqrt{15}}{2} \approx 1.4365$$

and

$$t = \frac{2 + 2\sqrt{15}}{-4} = \frac{2}{-4} - \frac{2\sqrt{15}}{4} = -\frac{1}{2} - \frac{\sqrt{15}}{2} \approx -2.4365.$$

3. Since this quadratic equation is **not** in standard form, we must first expand the left-hand side and then get all terms on one side before we are able to proceed:

$$(x + 1)(x - 2) = 4$$

$$\begin{aligned}
 x^2 - 2x + x - 2 &= 4 \\
 x^2 - x - 2 &= 4 \\
 x^2 - x - 2 - 4 &= 4 - 4 \\
 x^2 - x - 6 &= 0
 \end{aligned}$$

Now that the equation is in standard form, we can see that this is similar to part (a), so the solutions are $x = -3$ and $x = 2$.

Example 3.4.3

Find the horizontal intercepts of the quadratic function $g(x) = -2x^2 - 8x - 3$.

Solution. To find the horizontal intercepts, we set the quadratic function equal to 0 and solve for x :

$$-2x^2 - 8x - 3 = 0.$$

Here $a = -2$, $b = -8$, and $c = -3$, so by the quadratic formula,

$$\begin{aligned}
 x &= \frac{-(-8) \pm \sqrt{(-8)^2 - 4 \cdot (-2) \cdot (-3)}}{2 \cdot (-2)} \\
 &= \frac{8 \pm \sqrt{64 - 24}}{-4} \\
 &= \frac{8 \pm \sqrt{40}}{-4} \\
 &= \frac{8 \pm \sqrt{4 \cdot 10}}{-4} \\
 &= \frac{8 \pm 2\sqrt{10}}{-4}.
 \end{aligned}$$

Simplifying, the two solutions are

$$x = \frac{8 - 2\sqrt{10}}{-4} = \frac{8}{-4} + \frac{2\sqrt{10}}{-4} = -2 + \frac{\sqrt{10}}{2} \approx -0.4189$$

and

$$x = \frac{8 + 2\sqrt{10}}{-4} = \frac{8}{-4} - \frac{2\sqrt{10}}{4} = -2 - \frac{\sqrt{10}}{2} \approx -3.5811.$$

Hence, the horizontal intercepts are $-2 + \frac{\sqrt{10}}{2}$ and $-2 - \frac{\sqrt{10}}{2}$.

The Discriminant

How many solutions a quadratic equation in standard form will have is determined entirely by the quantity under the radical in the quadratic formula; we call this quantity the ***discriminant***.

Definition 3.4.4 The Discriminant.

For the quadratic equation in standard form $ax^2 + bx + c = 0$, the ***discriminant*** D is given by

$$D = b^2 - 4ac.$$

Result 3.4.5

Given the discriminant D of the quadratic equation $ax^2 + bx + c = 0$:

- If $D > 0$, then the quadratic equation has two distinct real solutions.
- If $D = 0$, then the quadratic equation has one real solution.
- If $D < 0$, then the quadratic equation has no real solutions.

By the way... Note: The discriminant allows you to state the number of solutions to the quadratic equation without actually *finding them*.

Example 3.4.6

Use the discriminant to determine how many solutions each quadratic equation has.

1. $x^2 - 4x + 4 = 0$
2. $-3t^2 + t - 10 = 0$
3. $-2t^2 - 2t + 7 = 0$

Solution.

1. Here, $a = 1$, $b = -4$, and $c = 4$, so $D = (-4)^2 - 4 \cdot 1 \cdot 4 = 0$. Hence this quadratic equation has ***one real solution***. By the quadratic formula, that solution is

$$x = \frac{-(-4) \pm \sqrt{0}}{2 \cdot 1} = \frac{4}{2} = 2.$$

2. Here, $a = -3$, $b = 1$, and $c = -10$, so $D = (1)^2 - 4 \cdot (-3) \cdot (-10) = -119$. Since $D < 0$, this quadratic equation has **no real solutions**. This can be seen by using the quadratic formula as well, yielding

$$x = \frac{-1 \pm \sqrt{-119}}{2 \cdot (-3)}$$

which is not a real quantity due to the negative number underneath the square root.

3. Here, $a = -2$, $b = -2$, and $c = 7$, so $D = (-2)^2 - 4 \cdot (-2) \cdot 7 = 60$. Since $D > 0$, this quadratic equation has **two distinct real solutions**. We found these solutions in part (b) of Example 3.4.2; they are $t = -\frac{1}{2} - \frac{\sqrt{15}}{2}$ and $t = -\frac{1}{2} + \frac{\sqrt{15}}{2}$.

Using Horizontal Intercepts to Find the Vertex

In Section 3.1, we noted that the graph of a quadratic function is symmetric about the vertical line passing through the vertex, so the horizontal position of the vertex must be exactly halfway between the horizontal intercepts, or real zeros, of the quadratic function. This was stated in Result 3.1.4.

The zeros of a quadratic function $f(x) = ax^2 + bx + c$ are found by solving the equation $f(x) = ax^2 + bx + c = 0$. By the quadratic formula, the solutions of this equation and therefore the zeros of the quadratic function are $x_1 = \frac{-b + \sqrt{b^2 - 4ac}}{2a}$ and $x_2 = \frac{-b - \sqrt{b^2 - 4ac}}{2a}$. As the x -coordinate of the vertex must be located at the average of x_1 and x_2 , denoting the x -coordinate of the vertex by h results in

$$h = \frac{x_1 + x_2}{2}.$$

Now

$$\begin{aligned} h &= \left(\frac{-b + \sqrt{b^2 - 4ac}}{2a} + \frac{-b - \sqrt{b^2 - 4ac}}{2a} \right) / 2 \\ &= \left(\frac{-b}{2a} + \frac{\cancel{\sqrt{b^2 - 4ac}}}{2a} + \frac{-b}{2a} - \frac{\cancel{\sqrt{b^2 - 4ac}}}{2a} \right) / 2 \\ &= \left(\frac{-2b}{2a} \right) / 2 \\ &= -\frac{b}{a} \cdot \frac{1}{2} \\ &= -\frac{b}{2a}. \end{aligned}$$

By the way... As is frequently the case in mathematics, many paths may lead to the same answer. There is another way to show $h = -b/2a$ that you may encounter if you go on to study critical points in calculus.

Note that the above formula for h is valid when $x_1 = x_2$, (i.e., if the quadratic function has a single horizontal intercept). In this case, the single horizontal intercept must also be the vertex. It is also valid when the quadratic function $f(x) = ax^2 + bx + c$ has no real horizontal intercepts, or, equivalently, no real zeros.

Result 3.4.7 Vertex Formula.

The vertex (h, k) of a quadratic function in the standard form

$$f(x) = ax^2 + bx + c$$

is given by

$$h = -\frac{b}{2a}, \quad k = f(h) = f\left(-\frac{b}{2a}\right).$$

Example 3.4.8

A baseball is thrown straight up with initial speed 40 ft/sec by a player who is 6 feet tall. Let $H(t)$ be the height of the ball above the ground t seconds later.

1. Find a formula for $H(t)$.
2. When does the ball reach its maximum height, and what is its maximum height?

Solution.

1. We use the formula for the vertical motion of a projectile with initial velocity V_0 and initial height H_0 that was originally presented in Section 3.3:

$$H(t) = -16t^2 + V_0t + H_0.$$

In our example, $H_0 = 6$ and $V_0 = 40$. Hence

$$H(t) = -16t^2 + 40t + 6.$$

This is a quadratic function in standard form with $a = -16$, $b = 40$, and $c = 6$.

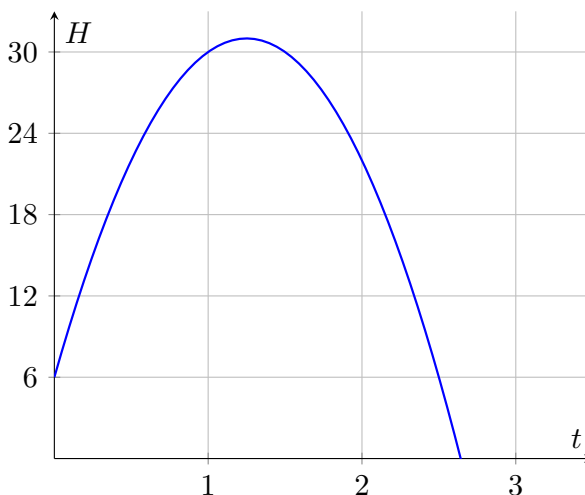
2. The parabola corresponding to the function $H(t)$ opens down so $H(t)$ is at its maximum at the vertex of the parabola. Given $a = -16$ and $b = 40$, we can calculate h , the t -coordinate of the vertex:

$$h = -\frac{b}{2a} = -\frac{40}{2 \cdot (-16)} = 1.25.$$

The ball will reach its maximum height at $t = 1.25$; that is, 1.25 seconds into the motion. The height of the ball at $t = 1.25$ is:

$$H(1.25) = -16 \cdot (1.25)^2 + 40 \cdot 1.25 + 6 = 31.$$

The maximum height of the ball is 31 feet. Here is the graph of $H(t)$.



The graph shows the part of the parabola that is relevant to the motion in our example.

Factored Form of a Quadratic Function

We have previously established that if we can factor a given quadratic function, we can quickly find its zeros (or, equivalently, horizontal intercepts). For example, the zeros of the function:

$$f(x) = 2(x - 1)(x + 1.5)$$

are $x = 1$ and $x = -1.5$.

The other way around is also true: if we have real zeros, x_1 and x_2 , of a given quadratic function $f(x) = ax^2 + bx + c$, we can write the function in factored form:

$$f(x) = a(x - x_1)(x - x_2).$$

Definition 3.4.9 Factored Form.

Let $f(x) = ax^2 + bx + c$ be a quadratic function with distinct real zeros x_1 and x_2 . The **factored form** of $f(x)$ is

$$f(x) = a(x - x_1)(x - x_2).$$

If $f(x)$ has a single real zero, then x_1 is called a *double root* of $f(x)$ and we use $x_1 = x_2$ in the formula for factored form defined above.

By the way... Remember, the real zeros of $f(x)$ are the real values of x for which $f(x) = 0$.

Note: Every quadratic function can be written in standard form and in vertex form. However, not every quadratic function can be written in factored form for real numbers x_1, x_2 . In particular, a quadratic function that has no real zeros, or equivalently no horizontal intercepts, cannot be factored over the real numbers and therefore cannot be written in the factored form we have defined.

Example 3.4.10

Find the zeros for a given quadratic function using any method you wish. Express the function in factored form.

1. $f(x) = 6x^2 - 27x + 12$

2. $g(x) = x^2 - 2x - 11$

3. $h(x) = -x^2 - 2$

Solution.

1. To find the zeros of $f(x)$, we have to solve the quadratic equation:

$$6x^2 - 27x + 12 = 0.$$

Notice that $a = 6$, $b = -27$, and $c = 12$. So by the quadratic formula:

$$\begin{aligned} x &= \frac{-(-27) \pm \sqrt{(-27)^2 - 4 \cdot 6 \cdot 12}}{2 \cdot 6} \\ &= \frac{27 \pm \sqrt{441}}{12} \\ &= \frac{27 \pm 21}{12} \end{aligned}$$

We simplify and obtain the two solutions:

$$x_1 = \frac{27 - 21}{12} = \frac{6}{12} = \frac{1}{2}$$

and

$$x_2 = \frac{27 + 21}{12} = \frac{48}{12} = 4.$$

The function $f(x)$ in factored form is

$$f(x) = 6 \left(x - \frac{1}{2} \right) (x - 4).$$

2. We set up the equation to find the zeros of $g(x)$:

$$x^2 - 2x - 11 = 0.$$

We have $a = 1$, $b = -2$, and $c = -11$. So by the quadratic formula:

$$\begin{aligned} x &= \frac{-(-2) \pm \sqrt{(-2)^2 - 4 \cdot 1 \cdot (-11)}}{2 \cdot 1} \\ &= \frac{2 \pm \sqrt{48}}{2} \\ &= \frac{2 \pm \sqrt{16 \cdot 3}}{2} \\ &= \frac{2 \pm 4\sqrt{3}}{2} \\ &= 1 \pm 2\sqrt{3} \end{aligned}$$

Hence, the two solutions are

$$x_1 = 1 + 2\sqrt{3}$$

and

$$x_2 = 1 - 2\sqrt{3}.$$

The function $g(x)$ in factored form is

$$g(x) = \left(x - (1 + 2\sqrt{3}) \right) \left(x - (1 - 2\sqrt{3}) \right).$$

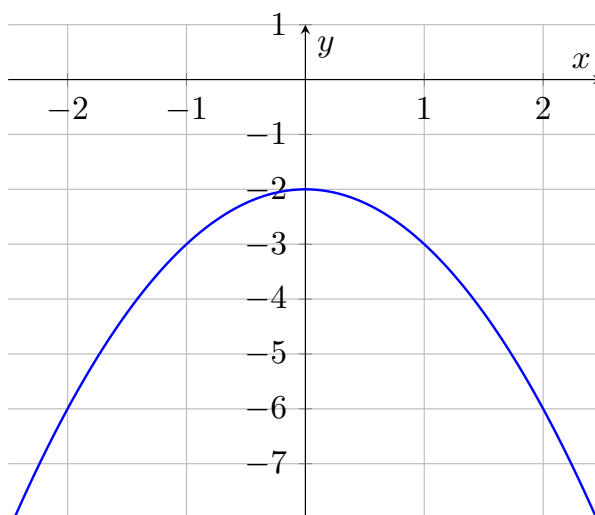
3. The equation we have to solve is:

$$-x^2 - 2 = 0.$$

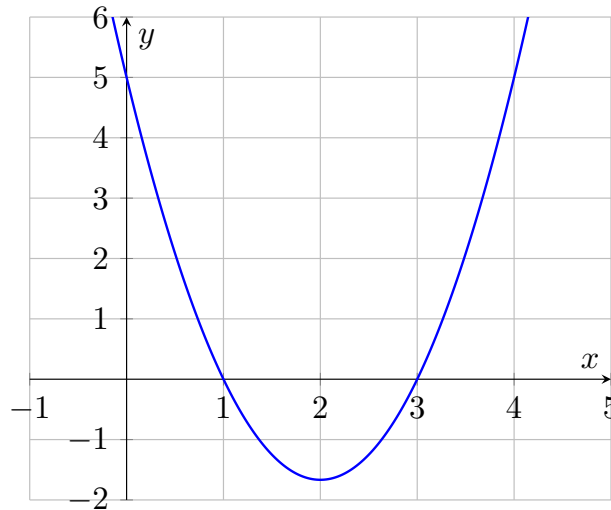
Since $a = -1$, $b = 0$, and $c = -2$, the discriminant is negative: $D = 0^2 - 4(-1)(-2) = -8 < 0$. Hence, the equation has no real solutions and $h(x)$ cannot be written in factored form. However, the vertex form of $h(x)$ is

$$h(x) = -1 \cdot (x - 0)^2 - 2.$$

The vertex is below the x -axis at $(0, -2)$ and the parabola opens down. So there cannot be any horizontal intercepts. Here is the graph of $h(x)$.

**Example 3.4.11**

Use the graph of a quadratic function $y = f(x)$ given below to find a formula for the function in factored form and in standard form.



Solution. The graph gives us the horizontal intercepts or equivalently the zeros of the function $f(x)$: $x = 1$ and $x = 3$. Hence, $f(x)$ in factored form is:

$$f(x) = a(x - 1)(x - 3)$$

We still don't have the value of the leading coefficient a . To find a , we will use the vertical intercept $y = 5$ of the function. The intercept is clearly seen on the graph. The vertical intercept is the value of the function at $x = 0$. Thus, $f(0) = 5$. We substitute $x = 0$ into factored form of $f(x)$ and obtain:

$$a(0 - 1)(0 - 3) = 5$$

We simplify the equation:

$$a \cdot (-1) \cdot (-3) = 5$$

which gives:

$$a \cdot 3 = 5$$

and finally:

$$a = \frac{5}{3}$$

The function $f(x)$ in factored form is:

$$f(x) = \frac{5}{3}(x - 1)(x - 3)$$

To obtain $f(x)$ in standard form, we multiply out all terms and simplify:

$$\begin{aligned}f(x) &= \frac{5}{3}(x-1)(x-3) \\&= \frac{5}{3}(x^2 - 3x - x + 3) \\&= \frac{5}{3}(x^2 - 4x + 3) \\&= \frac{5}{3}x^2 - \frac{20}{3}x + 5.\end{aligned}$$

Exercises

Quadratic Equations. Find all real solutions to each quadratic equation. Give exact and approximate values rounded to three decimal places.

1. $x^2 - 5x - 14 = 0$
2. $x(x + 3) = -2$
3. $6x^2 + x - 1 = 0$
4. $x^2 - 2x - 2 = 0$
5. $-4x^2 - 17x - 15 = 0$
6. $2x(3 - x) = 4(x - 5)$

Intercepts and Vertex. Find the intercepts and vertex of each quadratic function. Give exact and approximate values rounded off to three decimal places.

7. $y = x^2 + 2x - 3$
8. $f(x) = x^2 + 8x + 10$
9. $g(x) = 3x^2 - 6x - 1$
10. $y = t^2 - 4t + 5$

Discriminant. Calculate the discriminant of each quadratic equation and use it to determine whether the equation will have one real solution, two distinct real solutions, or no real solutions.

11. $3x^2 - 12x + 12 = 0$
12. $8x^2 + 14x + 2 = 0$
13. $3x^2 - 8x + 15 = 0$
14. $x(x - 4) = 10$
15. A baseball is thrown straight up with the initial speed of 50 ft/sec by a player who is 6 feet tall. Let $H(t)$ be the height of the ball above the ground t seconds later.

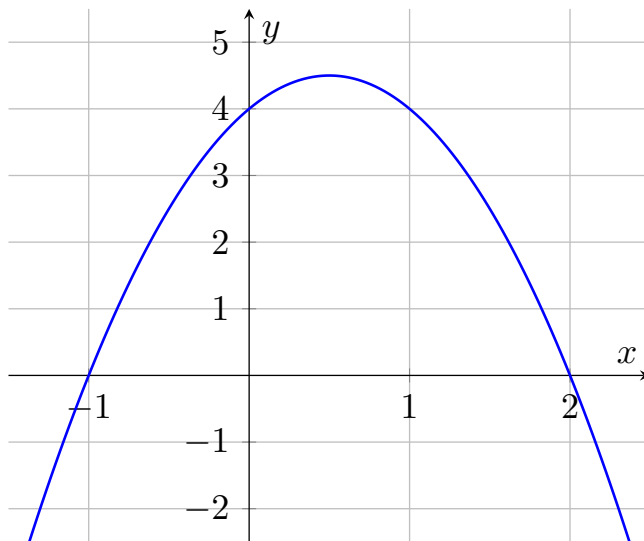
- (a) Find a formula for $H(t)$.
- (b) When does the ball reach its maximum height and what is its maximum height? Round your answers to two decimal places.

Factored Form. Rewrite the given quadratic function in factored form. Be sure to use the exact values of the zeros of each function.

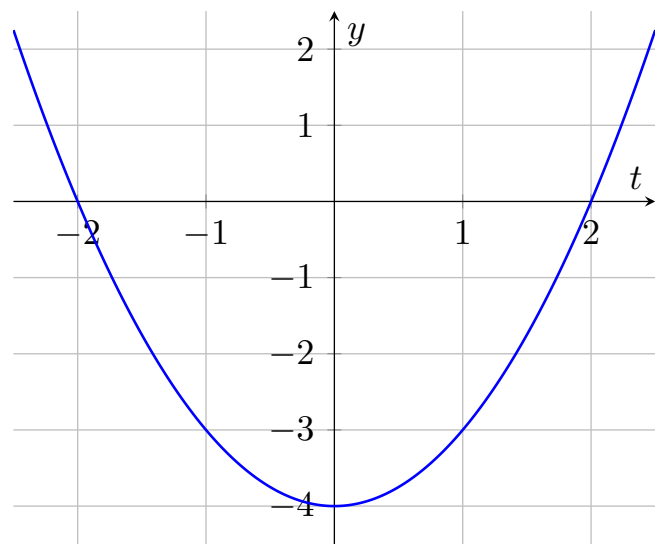
16. $f(x) = 3x^2 - 3x - 6$

17. $g(x) = x^2 + 6x + 7$

18. The graph of a quadratic function $f(x)$ is given below. Write the function in factored form. Then rewrite the function in standard form.



19. The graph of a quadratic function $g(t)$ is given below. Write the function in factored form. Then rewrite the function in standard form.



Chapter 4

Power Functions

Intro about Chapter 4.

4.1 Algebra of Powers: Integral Exponents

Objectives

After completing this section, you should be able to do the following.

1. First objective.
 2. Second objective.
 3. Third objective.
-

In the next two sections we review algebra of power expressions of the form:

$$a^p$$

where a and p are given numbers. The number a is called the *base* in the expression and the power p that the base is being raised to is called the *exponent*.

We begin with integral exponents; that is, exponents that are integers. If p is a positive integer $p = 1, 2, 3, \dots$, then a^p is simply a short way of writing repeated multiplication:

$$a^p = \underbrace{a \cdot a \cdot \dots \cdot a}_{p \text{ times}}$$

In particular, for $p = 1, 2, 3, \dots$,

$$0^p = 0, \quad 1^p = 1.$$

We expand this simple definition to the exponent $p = 0$ by defining for any number a :

$$a^0 = 1.$$

In particular, by definition:

$$0^0 = 1.$$

Next, we extend the definition to negative integer exponents by defining for every $a \neq 0$ and every $p = 0, 1, 2, 3, \dots$,

$$a^{-p} = \frac{1}{a^p}$$

In particular:

$$a^{-1} = \frac{1}{a}$$

These simple definitions easily imply the basic properties of power expressions.

Result 4.1.1 Rules of Exponents — Integral Exponents.

Let a, b be given numbers, p and r be integers. Then the following equalities hold provided both sides are defined:

$$1. a^0 = 1$$

$$2. a^{-p} = \frac{1}{a^p}$$

$$3. a^p \cdot a^r = a^{p+r}$$

$$4. \frac{a^p}{a^r} = a^{p-r}$$

$$5. (a^p)^r = a^{p \cdot r}$$

$$6. (a \cdot b)^p = a^p \cdot b^p$$

$$7. \left(\frac{a}{b}\right)^p = \frac{a^p}{b^p}$$

$$8. \left(\frac{a}{b}\right)^{-p} = \left(\frac{b}{a}\right)^p$$

An expression in the formulas above may not be defined if there is a zero in the denominator. For example, in Property 2, $\frac{1}{a^p}$ is not defined if $a = 0$ and $p \neq 0$. So Property 2 holds for $a \neq 0$.

The properties of exponents listed above follow very easily from our definitions. For example, to illustrate Property 3 observe:

$$a^p \cdot a^r = \underbrace{a \cdot a \cdot \dots \cdot a}_{p \text{ times}} \cdot \underbrace{a \cdot a \cdot \dots \cdot a}_{r \text{ times}} = \underbrace{a \cdot a \cdot \dots \cdot a}_{p+r \text{ times}} = a^{p+r}.$$

Example 4.1.2

Rewrite each expression given below as a power of 3; that is, in the form 3^p for some constant p .

$$1. \frac{1}{3^2}$$

$$2. 3^5 \cdot 3^{-2}$$

$$3. \frac{(3^2)^3}{3^8}$$

$$4. \left(\frac{3^{-3}}{3^4}\right)^2$$

5. $\left(\frac{3^5}{27}\right)^2$

Solution.

1. All we need is Property 2:

$$\frac{1}{3^2} = 3^{-2}.$$

We rewrote the expression as 3^p for $p = -2$.

2. By Property 3:

$$3^5 \cdot 3^{-2} = 3^{(5+(-2))} = 3^3.$$

3. By Property 5 and Property 4:

$$\frac{(3^2)^3}{3^8} = \frac{3^6}{3^8} = 3^{(6-8)} = 3^{-2}.$$

4. As $3^{-3} = \frac{1}{3^3}$ we have:

$$\left(\frac{3^{-3}}{3^4}\right)^2 = \left(\frac{1}{3^3 \cdot 3^4}\right)^2 = \left(\frac{1}{3^7}\right)^2 = (3^{-7})^2 = 3^{-14}.$$

5. Note that $27 = 3^3$. Using Property 7:

$$\left(\frac{3^5}{27}\right)^2 = \left(\frac{3^5}{3^3}\right)^2 = \frac{3^{10}}{3^6} = 3^4.$$

Example 4.1.3

Use Rules of Exponents to simplify the following expressions if possible:

1. $(x^2 \cdot x^3)^2$

2. $\left(\frac{x^2 y^5}{x^4}\right)^{-3}$

3. $\left(\frac{x^{-2} y^5}{x^{-3} y^2}\right)^{-1}$

4. $(a + b)^7$

Solution.

1. We use Property 3 to simplify the expression under the outside power 2. Then we use Property 5:

$$(x^2 \cdot x^3)^2 = (x^5)^2 = x^{10}.$$

2. We begin by simplifying the expression under the power -3 using Property 4:

$$\left(\frac{x^2 y^5}{x^4}\right)^{-3} = \left(\frac{y^5}{x^2}\right)^{-3}.$$

Now we use Property 8 and then Property 7:

$$\left(\frac{y^5}{x^2}\right)^{-3} = \left(\frac{x^2}{y^5}\right)^3 = \frac{(x^2)^3}{(y^5)^3} = \frac{x^6}{y^{15}}.$$

3. We simplify the expression under the power -1 using Property 2:

$$\left(\frac{x^{-2} y^5}{x^{-3} y^2}\right)^{-1} = \left(\frac{x^3 y^5}{x^2 y^2}\right)^{-1}.$$

By Property 4 and then Property 2:

$$\left(\frac{x^3 y^5}{x^2 y^2}\right)^{-1} = (xy^3)^{-1} = \frac{1}{xy^3}.$$

4. There is no rule for the power of a sum or a difference! We cannot simplify $(a + b)^7$ using Rules of Exponents. You certainly **cannot** distribute the power 7 and write the expression as $a^7 + b^7$.

Exercises

Rewriting Expressions. Rewrite the given expression as a power of 2; that is, express it in the form 2^m for some m .

1. $\frac{4}{2^4}$
2. $(2^3)^{-2}$
3. $\frac{8^2}{2}$
4. $\left(\frac{2^3}{2^4 \cdot 2^5}\right)^2$

Rewriting Expressions. Rewrite the given expression as a power of 5; that is, express it in the form 5^m for some m .

5. $5^2 \cdot 5^{-3}$

6. $\left(\frac{25^2}{5}\right)^{-1}$

7. $\left(\frac{5^{-1}}{5^{-2}}\right)^2$

8. $\left(\frac{5^3}{25^2}\right)^{-2}$

Simplifying Expressions. Simplify the given expression if possible. If not possible, state so.

9. $(x^2y^5)^0$

10. $\left(\frac{x^{-3}}{xy^2}\right)^{-1}$

11. $\frac{(x-y)^3}{x}$

12. $(2x^2y^4)^3$

13. $\frac{x+y}{y}$

14. $\left(\frac{\pi(x^2)^6}{x^2y^{-4}}\right)^2$

15. $\frac{x^2+y^3}{xy}$

16. Let p and r be positive integers. Use the definition of exponentiation

$$a^m = \underbrace{a \cdot a \cdot \dots \cdot a}_{m \text{ times}}$$

to explain why the following formula is valid:

$$(a^p)^r = a^{p \cdot r}.$$

4.2 Algebra of Powers: Fractional Exponents

Objectives

After completing this section, you should be able to do the following.

1. First objective.
2. Second objective.
3. Third objective.

In this section, we will review the concepts of roots and radicals. Please note that in this book we stay within real numbers, and we do not consider complex numbers. In particular, when we talk about roots, we mean real roots.

Roots and Radicals

We say that a number y is a root of order 2 (a square root) of a number a if $y^2 = a$. We say that a number y is a root of order 3 (a cube root) of a if $y^3 = a$ and so on.

Definition 4.2.1

Let a be a given number and let n be a positive integer. We say that y is a root of order n of a , or an n th root of a , if $y^n = a$.

For example, let $a = 4$, $n = 2$. The number 4 has two roots of order 2: $y = 2$ and $y = -2$, since $2^2 = (-2)^2 = 4$. On the other hand, let $a = -4$ and $n = 2$. The number -4 has no roots of order 2, since there are no numbers whose square is -4 . We say that the square root of -4 *does not exist*.

Take $a = -64$ and $n = 3$. The number -64 has *exactly one* root of order 3 which is $y = -4$. Indeed, $(-4)^3 = -64$. Observe that $4^3 = 64$ so 4 is *not* a cube root of -64 .

Result 4.2.2 Roots and Radicals.

Let a be a given number and n be a positive integer.

- Suppose n is even.
 - If $a < 0$, then a has no n th order roots.
 - If $a > 0$, then a has two n th roots, one positive and one of the same magnitude but

negative. We denote the **positive** n th root as:

$$\sqrt[n]{a}$$

(We use the “radical” symbol $\sqrt{}$.) The two n th roots can then be written as

$$\sqrt[n]{a} \quad \text{and} \quad -\sqrt[n]{a}.$$

By the way... Typically, we might use the notation $\pm\sqrt[n]{a}$ to denote the positive and negative roots.

- Suppose n is odd. Then a has exactly one n th root which we denote as

$$\sqrt[n]{a}$$

If a is negative, the root is negative; if a is positive the root is positive.

- If $a = 0$, then a has one root of any order: $\sqrt[n]{0} = 0$.

By the way... When $n = 2$, instead of writing $\sqrt[2]{a}$ we simply drop the 2 and write $\sqrt{a}$. This is the only exception to this notation.

Example 4.2.3

Find all (real) roots specified below and write them in terms of radicals.

1. All roots of order 4 of 81
2. All roots of order 2 of 7
3. All roots of order 3 of -27
4. All roots of order 4 of -81

Solution.

1. We are looking for all numbers y such that $y^4 = 81$. The order, 4, is even. Hence, we have two roots, one positive and one negative. The positive root is denoted by $\sqrt[4]{81}$ and the two roots are:

$$\sqrt[4]{81} \quad \text{and} \quad -\sqrt[4]{81}.$$

The positive number whose 4th power is 81 is 3. That is:

$$\sqrt[4]{81} = 3.$$

Hence, the two roots of order 4 of 81 are 3 and -3 . Indeed, $3^4 = (-3)^4 = 81$.

2. 7 has two roots of order 2: $\sqrt{7}$ and $-\sqrt{7}$. We cannot easily guess them as they are not integers. We can use our calculator, though, and calculate $\sqrt{7} \approx 2.65$. The two roots are then approximately 2.65 and -2.65 .

3. The order, 3, is odd. Hence, there is only one root of order 3 of -27 denoted as $\sqrt[3]{-27}$. As $(-3)^3 = -27$, we have

$$\sqrt[3]{-27} = -3.$$

4. A negative number $a = -81$ has no roots of even order, and 4 is even. Hence, -81 has no roots of order 4.

By definition, x is a root of order n of a if x is a solution to the equation:

$$x^n = a$$

Hence, roots and radicals appear naturally when solving equations containing powers of the unknown. We saw plenty of radicals in Chapter 3 in the context of quadratic equations.

Example 4.2.4

Solve for x . Find all (real) solutions and give their exact as well as approximate values.

1. $2x^2 - 5 = 7$

2. $-3x^3 = 27$

3. $2x^4 + 3 = 4$

4. $2(x - 1)^2 + 3 = 11$

Solution.

1. We add 5 to both sides of the equation and then divide both sides by 2 to obtain

$$x^2 = 6.$$

Solutions to the equation are roots of order 2 of 6. There are two such roots:

$$x = \sqrt{6} \approx 2.449 \quad \text{and} \quad x = -\sqrt{6} \approx -2.449.$$

2. We divide both sides of the equation by -3 and obtain:

$$x^3 = -9.$$

There is one cube root of -9 . Hence, our equation has one solution:

$$x = \sqrt[3]{-9} \approx -2.08.$$

3. We subtract 3 from both sides of the equation, then divide by 2. The equation becomes:

$$x^4 = 0.5.$$

We are looking for all roots of order 4 of 0.5. As 4 is even, we have two such roots:

$$x = \sqrt[4]{0.5} \approx 0.841 \quad \text{and} \quad x = -\sqrt[4]{0.5} \approx -0.841.$$

4. We begin by “solving” for $(x - 1)$. Subtract 3 from both sides and divide both sides by 2:

$$(x - 1)^2 = 4.$$

Hence, $(x - 1)$ is a root of order 2 of 4. There are two such roots:

$$(x - 1) = \sqrt{4} \quad \text{and} \quad (x - 1) = -\sqrt{4}.$$

This gives $x - 1 = 2$ and $x - 1 = -2$. We solve each of the two equations for x and obtain two solutions:

$$x = 3 \quad \text{and} \quad x = -1.$$

Here are a few important properties of roots.

Result 4.2.5 Properties of Radicals.

Let a, b be given numbers. Let n, m be positive integers. Then the following equalities hold provided that the roots involved exist, and both sides are defined:

- $\sqrt[n]{a \cdot b} = \sqrt[n]{a} \cdot \sqrt[n]{b}$
- $\sqrt[n]{\frac{a}{b}} = \frac{\sqrt[n]{a}}{\sqrt[n]{b}}$
- $(\sqrt[n]{a})^m = \sqrt[n]{a^m}$
- $\sqrt[n]{a^n} = |a|$ if n is even

- $\sqrt[n]{a^n} = a$ if n is odd

Powers with Fractional Exponents and Arbitrary Exponents

So far, we have defined powers a^p for all **integer** exponents p .

To extend the definition to fractional exponents $a^{\frac{m}{n}}$, we will use roots. In the first step, for every positive integer n , we define $a^{\frac{1}{n}}$ as:

$$a^{\frac{1}{n}} = \sqrt[n]{a}$$

Does this make sense? Recall that $(\sqrt[n]{a})^n = a$. Therefore,

$$(a^{\frac{1}{n}})^n = a$$

which is what Rules of Exponents would dictate. In the next step, we define $a^{\frac{m}{n}} = (a^{\frac{1}{n}})^m = (\sqrt[n]{a})^m$ which seems to make sense. Here is a precise definition of a power with a fractional exponent.

Definition 4.2.6

Let a be a given number and let m and n be positive integers. Assume that $\sqrt[n]{a}$ exists. We define:

$$\begin{aligned} a^{\frac{1}{n}} &= \sqrt[n]{a} \\ a^{\frac{m}{n}} &= (\sqrt[n]{a})^m = \sqrt[n]{a^m} \\ a^{-\frac{m}{n}} &= \frac{1}{a^{m/n}} \text{ provided } a \neq 0. \end{aligned}$$

Note that $\sqrt[n]{a}$ exists unless n is even and a is negative. The combination of negative radicands (numbers under radicals) with even roots causes a number of problems for the behavior of fractional exponents, and the Rules of Exponents do not always hold.

For example, is the following equality *always* true?

$$\left(a^{\frac{1}{n}}\right)^m = (a^m)^{\frac{1}{n}}$$

It seems reasonable at first, but consider the following:

$$\left((-4)^{\frac{1}{4}}\right)^2 \quad \text{and} \quad ((-4)^2)^{\frac{1}{4}}.$$

The first number is $\left((-4)^{\frac{1}{4}}\right)^2 = (\sqrt[4]{-4})^2$ which is undefined. However,

$$((-4)^2)^{\frac{1}{4}} = 16^{\frac{1}{4}} = \sqrt[4]{16} = 2.$$

Therefore, when we talk about fractional powers, we will most often assume that bases are positive except for some very simple cases like:

$$a^{\frac{1}{3}} = \sqrt[3]{a}.$$

Example 4.2.7

Find:

1. $4^{\frac{1}{2}}$
2. $(-4)^{\frac{1}{2}}$
3. $27^{\frac{1}{3}}$
4. $(-27)^{\frac{1}{3}}$
5. $(1700)^{\frac{1}{20}}$

Solution.

1. By definition $4^{\frac{1}{2}} = \sqrt[2]{4} = \sqrt{4} = 2$. Note that $4^{\frac{1}{2}}$ is the **positive** of the two square roots of 4 as is $\sqrt{4}$.
2. $(-4)^{\frac{1}{2}} = \sqrt{-4}$ is undefined. An even-order root of a negative number is undefined. Hence, $(-4)^{\frac{1}{2}}$ is undefined.
3. $27^{\frac{1}{3}} = \sqrt[3]{27} = 3$.
4. $(-27)^{\frac{1}{3}} = \sqrt[3]{-27} = -3$.
5. $(1700)^{\frac{1}{20}} = \sqrt[20]{1700} \approx 1.451$.

Depending on your calculator, it may be easier to calculate a fractional power than to enter the corresponding radical. You can simply enter $1700^{(1/20)}$. Be sure to use parentheses!

We have defined powers a^p for integer and fractional exponents p . As you may know, not all real numbers can be expressed as fractions (irrational numbers, for example). Can we define powers a^p for all real numbers p ? The answer is affirmative provided the base a is positive. The construction falls outside the scope of this course. It suffices to know that a^p can be defined for all exponents and Rules of Exponents are preserved. Therefore, we have the following result.

Result 4.2.8 Rules of Exponents — Arbitrary Exponents.

Let a, b be given **positive** numbers, p and r be real numbers. Then the following equalities hold:

$$1. a^{-p} = \frac{1}{a^p}$$

$$2. a^p \cdot a^r = a^{p+r}$$

$$3. \frac{a^p}{a^r} = a^{p-r}$$

$$4. (a^p)^r = a^{p \cdot r}$$

$$5. (a \cdot b)^p = a^p \cdot b^p$$

$$6. \left(\frac{a}{b}\right)^p = \frac{a^p}{b^p}$$

$$7. \left(\frac{a}{b}\right)^{-p} = \left(\frac{b}{a}\right)^p$$

Example 4.2.9

Rewrite each of the following expressions as a power of 5; that is, in the form 5^p for some p .

$$1. \sqrt[3]{5}$$

$$2. 5\sqrt{5}$$

$$3. \frac{5\sqrt{5}}{25}$$

$$4. \left(\frac{5}{\sqrt[3]{5}}\right)^2$$

Solution.

1. By definition of fractional powers:

$$\sqrt[3]{5} = 5^{\frac{1}{3}}.$$

2. By definition of fractional powers and Property 2:

$$5\sqrt{5} = 5 \cdot 5^{\frac{1}{2}} = 5^{1+\frac{1}{2}} = 5^{\frac{3}{2}}.$$

3. Note that $25 = 5^2$. Using Property 2 and Property 3:

$$\frac{5\sqrt{5}}{25} = \frac{5 \cdot 5^{\frac{1}{2}}}{5^2} = \frac{5^{\frac{3}{2}}}{5^2} = 5^{\frac{3}{2}-2} = 5^{-\frac{1}{2}}.$$

4. Note that $\sqrt[3]{5} = 5^{\frac{1}{3}}$. Using Rules of Exponents we get:

$$\left(\frac{5}{\sqrt[3]{5}}\right)^2 = \left(\frac{5}{5^{\frac{1}{3}}}\right)^2 = \left(5^{\frac{2}{3}}\right)^2 = 5^{\frac{2}{3} \cdot 2} = 5^{\frac{4}{3}}.$$

Example 4.2.10

Use Rules of Exponents to simplify the following. Write your answers in terms of powers and not radicals. Assume a , b , x , and y are all positive.

1. $\sqrt{\frac{36}{a^5}}$
2. $\sqrt[3]{8x^2y^6}$
3. $\frac{b^{\frac{3}{2}}\sqrt{a^3}}{ba^2}$
4. $\frac{14x^{-2}}{21\sqrt{x^7}}$

Solution.

1. $\sqrt{\frac{36}{a^5}} = \frac{\sqrt{36}}{\sqrt{a^5}} = \frac{6}{a^{5/2}} = 6a^{-(5/2)}.$
2. By computing $\sqrt[3]{8x^2y^6} = \sqrt[3]{8} \cdot \sqrt[3]{x^2} \cdot \sqrt[3]{y^6} = 2 \cdot (x^2)^{1/3} \cdot (y^6)^{1/3}$ first, then we have $2 \cdot x^{2/3} \cdot y^{6/3} = 2x^{2/3}y^2.$
3. $\frac{b^{\frac{3}{2}}\sqrt{a^3}}{ba^2} = \frac{b^{\frac{3}{2}}a^{\frac{3}{2}}}{ba^2} = b^{\frac{3}{2}-1}a^{\frac{3}{2}-2} = b^{1/2}a^{-(1/2)}.$
4. We have $14 = 2 \cdot 7$, $21 = 3 \cdot 7$, $x^{-2} = \frac{1}{x^2}$, $\sqrt{x^7} = x^{7/2}$. Therefore, $\frac{14x^{-2}}{21\sqrt{x^7}} = \frac{2}{3x^2x^{7/2}} = \frac{2}{3}x^{-(11/2)}.$

Exercises

Solving Equations. Solve the equation for x or for b . Be sure to list all solutions. Give exact and approximate values rounded off to three decimal places. If there are no solutions say so.

1. $\frac{2}{5}x^4 + 3 = 11$

2. $1500b^{30} = 17000$
3. $5x^3 - 6 = -8$
4. $\frac{5}{3}x^6 + 9 = 4$
5. $3(x - 1)^2 + 1 = 10$
6. $(x + 2)^3 = -8$

Rewriting Expressions. Rewrite the expression as a power of 3; that is, in the form 3^r for some r . If it is not possible, say so.

7. $\frac{9^2}{3\sqrt{3}}$
8. $\frac{27}{\sqrt{3}}$
9. $\frac{3^2 \sqrt[3]{9}}{\sqrt{3}}$
10. $\frac{3^{1/5} \sqrt[4]{3}}{\sqrt{3}}$

Simplifying. Simplify the given expression. If it is not possible, say so. x and y are assumed to be positive.

11. $\frac{\sqrt{x} + \sqrt{y}}{xy}$
12. $(\sqrt[4]{y} \sqrt[3]{x})^4$
13. $\frac{x^{\frac{2}{3}} y^3}{xy}$
14. $\frac{x^2 y^3}{\sqrt{x} + \sqrt[3]{y}}$

4.3 Power Functions: Positive Integral Exponents

Objectives

After completing this section, you should be able to do the following.

1. First objective.
2. Second objective.
3. Third objective.

In the next two sections we look at the properties and the graphs of the so-called power functions.

Definition 4.3.1 Power Function.

A function $y = f(x)$ is called a **power function** if $f(x)$ can be expressed in the form:

$$f(x) = kx^p$$

where k and p are constants and $k \neq 0$. The constant k is called the *coefficient* of the power function $f(x)$ and p is called the *exponent*.

Note: When y depends on x according to the formula $y = kx^p$, we say that y is **directly proportional** (or proportional) to x^p with the coefficient of proportionality k . If $y = \frac{k}{x^p}$, we say that y is **inversely proportional** to x^p with the coefficient of proportionality k . So power functions express proportionality of the dependent variable to powers of the independent variable.

As the title suggests, in this section we look at the case when the exponent p is a positive integer.

Example 4.3.2

Which of the functions below are power functions? For those which are, rewrite in the standard form $y = kx^p$. Identify the coefficient and the exponent.

1. $f(x) = 2x^2 \cdot 4x^3$

2. $g(x) = -(x^3)^5$

3. $h(x) = \frac{(x^2)^4}{7x^3}$

4. $r(x) = 2x^3 + 7x^2$

$$5. m(x) = \frac{8x}{\pi^2}$$

$$6. l(x) = 3 \cdot 2^x$$

Solution.

1. The function $f(x)$ is a power function. Using Rules of Exponents from Section 4.1, we can rewrite:

$$f(x) = 2x^2 \cdot 4x^3 = (2 \cdot 4)(x^2x^3) = 8x^5.$$

The coefficient is $k = 8$ and the exponent is $p = 5$.

2. $g(x)$ is a power function:

$$g(x) = (-(x^3))^5 = (-1 \cdot (x^3))^5 = (-1)^5(x^3)^5 = -1 \cdot x^{15} = -x^{15}.$$

The coefficient is $k = -1$ and the exponent is $p = 15$.

3. $h(x)$ is a power function as well:

$$h(x) = \frac{(x^2)^4}{7x^3} = \frac{x^8}{7x^3} = \frac{1}{7}x^{8-3} = \frac{1}{7}x^5.$$

The coefficient is $k = \frac{1}{7}$ and the exponent is $p = 5$.

4. The function $r(x)$ is not a power function. It cannot be rewritten as $r(x) = kx^p$. The function $r(x)$ is a sum of two power functions, $2x^3$ and $7x^2$. A function that is a sum of power functions is called a polynomial function.

5. The function $m(x)$ is a power function. Remember that π is just a constant.

$$m(x) = \frac{8x}{\pi^2} = \frac{8}{\pi^2}x.$$

The coefficient is $k = \frac{8}{\pi^2}$ and the exponent is $p = 1$.

6. $l(x)$ is not a power function. Observe that in $l(x)$ the base of the power expression 2^x is constant and equal to 2. The exponent x is a variable. In a power function, it is the other way around: the base is a variable and the exponent is constant.

By the way... Functions like $l(x)$ where the variable x is in the *exponent* is the topic of discussion in Chapter 5.

Example 4.3.3

A ball dropped from the Empire State Building has traveled down the distance of $d(t)$ feet after t seconds where:

$$d(t) = 16t^2.$$

1. Is the function $d(t)$ a power function? If yes, identify the exponent and the coefficient.
2. The Empire State Building is 1250 ft tall. How long will it take for the ball to hit the ground?

Solution.

1. $d(t)$ is a power function with $k = 16$ and $p = 2$.
2. The ball will hit the ground when it has traveled 1250 feet. That is, for a positive t such that:

$$d(t) = 16t^2 = 1250.$$

To solve the equation for t we divide both sides by 16:

$$t^2 = \frac{1250}{16}.$$

The two solutions to this quadratic equation are:

$$t = \pm \sqrt{\frac{1250}{16}}.$$

Since t has to be positive, then the solution is

$$t = \sqrt{\frac{1250}{16}}.$$

Hence, the ball will hit the ground after:

$$t = \sqrt{\frac{1250}{16}} \approx 8.8 \text{ seconds.}$$

Note that in terms of proportionality, we can say that the distance d is directly proportional to t^2 with the coefficient of proportionality 16.

Example 4.3.4

Let $V(r)$ be the volume of a sphere of radius r . We know from elementary geometry that:

$$V(r) = \frac{4}{3}\pi r^3.$$

1. Is $V(r)$ a power function? If yes, find the coefficient and the exponent.
2. What radius is required for the volume to be 25 cm^3 ?

Solution.

1. Yes, $V(r)$ is a power function. The coefficient k is $\frac{4\pi}{3}$ and the exponent p is 3.
2. We are looking for the radius r , in centimeters, such that:

$$\frac{4}{3}\pi r^3 = 25$$

Dividing both sides by 4π and multiplying by 3, then

$$r^3 = \frac{3}{4\pi}25$$

Now we take the power $\frac{1}{3}$ of both sides (in other words, the cube root) and obtain:

$$(r^3)^{\frac{1}{3}} = \left(\frac{3}{4\pi}25\right)^{\frac{1}{3}}$$

Since $(r^3)^{\frac{1}{3}} = r$, then

$$r = \left(\frac{3}{4\pi}25\right)^{\frac{1}{3}} \approx 1.814 \text{ cm}$$

The volume is 25 cm^3 when the radius is 1.814 cm.

Graphs of Power Functions: Positive Integral Exponents

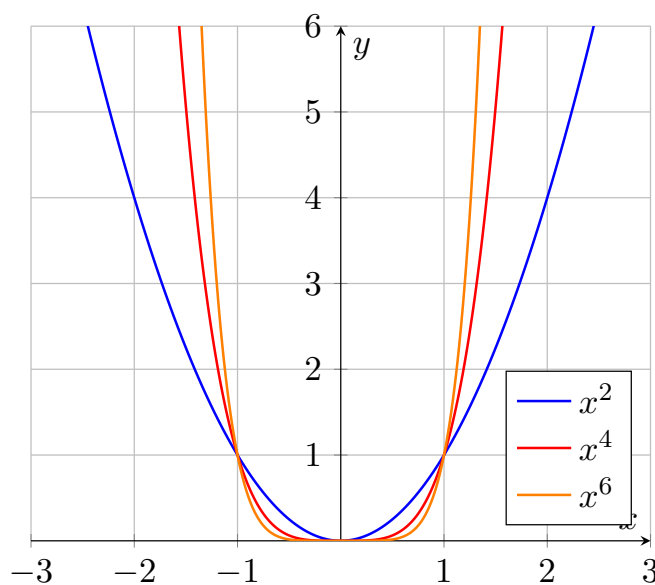
Graphs of power functions $y = kx^p$ with exponents p that are positive integers are different for p even and for p odd.

Even Positive Exponents

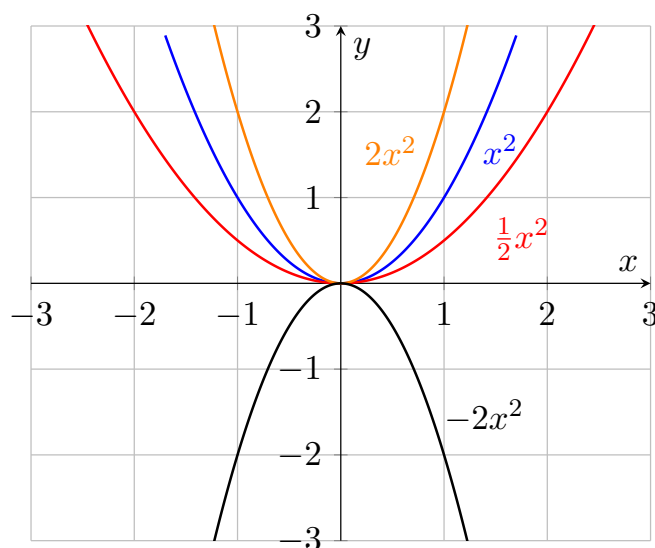
If p is even, then x^p is positive for all x except for $x = 0$ where $x^p = 0$. For example, if $p = 6$ then $1^6 = 1$, $(-1)^6 = 1$, $2^6 = 64$, $(-2)^6 = 64$, and so on. Hence, the graph of the power function $f(x) = kx^p$

for any even p is symmetric about the y -axis as $f(-x) = f(x)$.

Here are the graphs of the functions $y = kx^p$ for $p = 2, 4, 6$ and $k = 1$. All graphs are U-shaped and reminiscent of the quadratic parabola $y = x^2$.



The coefficient k in $y = kx^p$, stretches or shrinks the graph of $y = x^p$ vertically. Additionally, if $k < 0$, the graph is reflected over the x -axis.

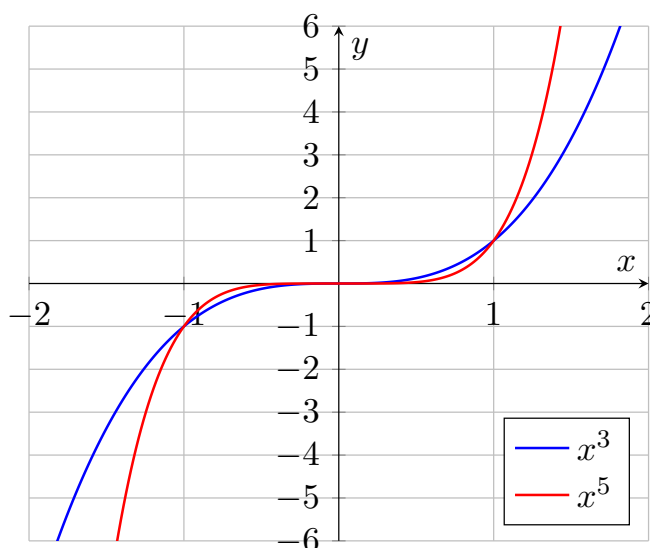


Odd Positive Exponents

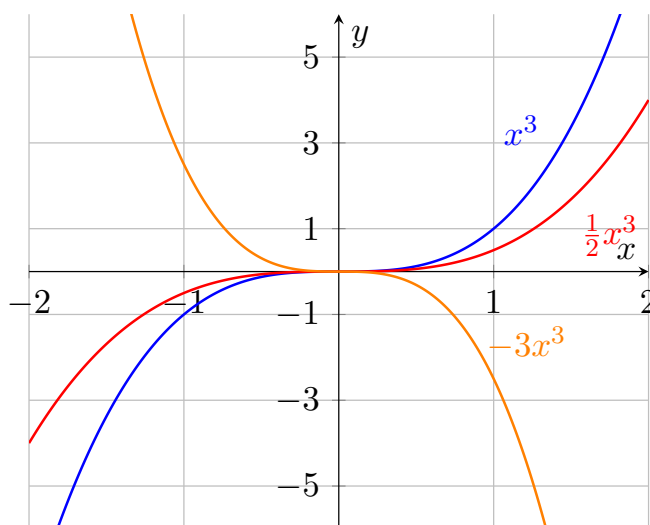
If p is odd, then x^p is positive for $x > 0$, it is 0 for $x = 0$, and negative for $x < 0$. For example, let $p = 3$. Then $1^3 = 1$, $(-1)^3 = -1$, $2^3 = 8$, $(-2)^3 = -8$, and so on. Hence, the graph of the power

function $f(x) = kx^p$ for any odd p is symmetric about the origin $(0, 0)$ as $f(-x) = -f(x)$.

Here are the graphs of the functions $y = kx^p$ for $p = 3, 5$ and $k = 1$. This time, the graphs are S-shaped.



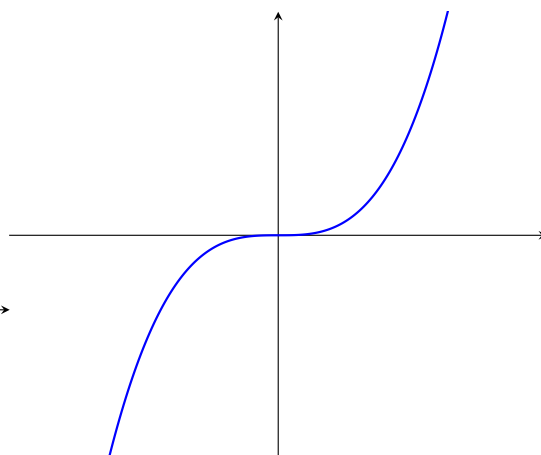
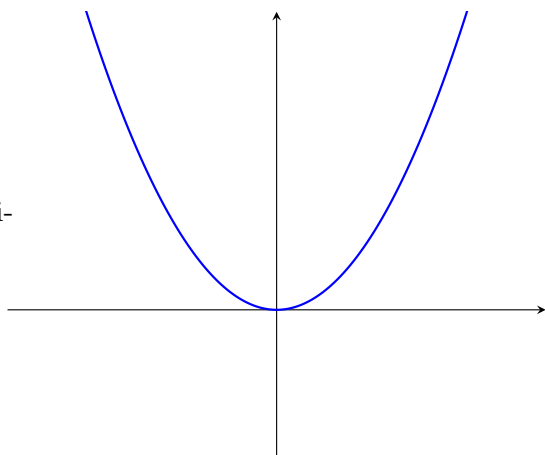
The coefficient k in $y = kx^p$, stretches or shrinks the graph of $y = x^p$ vertically. Additionally, if $k < 0$, the graph is reflected over the x -axis.



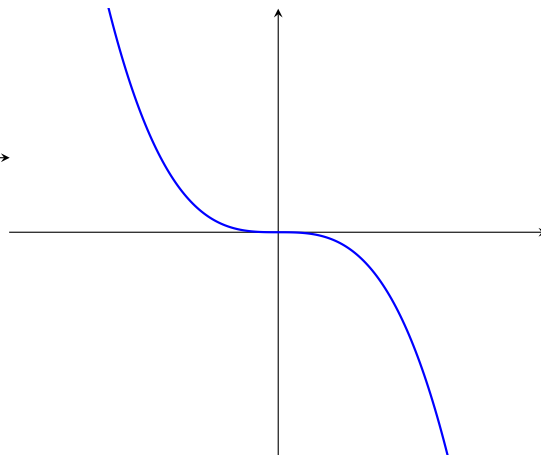
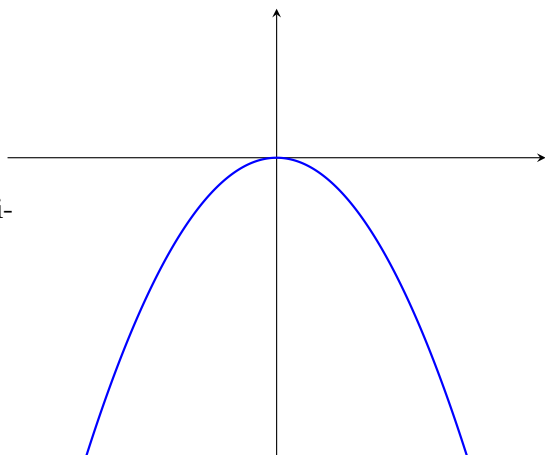
Here is a summary of how graphs of power functions look for positive integer exponents p , even and odd, and for coefficients k positive and negative.

$$f(x) = kx^p \quad \begin{array}{cc} p \text{ even} & p \text{ odd} \end{array}$$

Positive
coefficient, $k > 0$



Negative
coefficient, $k < 0$



Exercises

Power Functions. Determine whether a given function is a power function. If yes, identify the coefficient k and the exponent p . If not, say so.

1. $y = \frac{0.8x^4}{0.2x}$

2. $y = \frac{\sqrt{9x^8}}{x}$

3. $f(x) = \frac{1}{3} \cdot 3^{x+1}$

4. $g(x) = \sqrt{0.01x^6}$

5. $h(x) = 0.01x^3 - 2x^2$

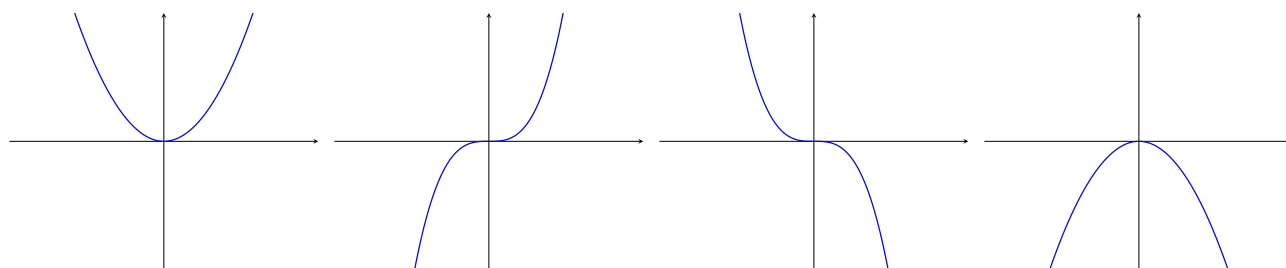
6. $y = 9\sqrt{\frac{x^5}{x}}$

Braking Distance. “Braking distance” or “stopping distance” refers to the distance a car will travel from the point when its brakes are fully applied to when it comes to a complete stop ¹. The braking distance is proportional to the square of the car’s speed and it depends on the coefficient of friction, μ , between the tires and the road surface. Let D denote distance, in feet, and S speed in mph. The formula for the braking distance is:

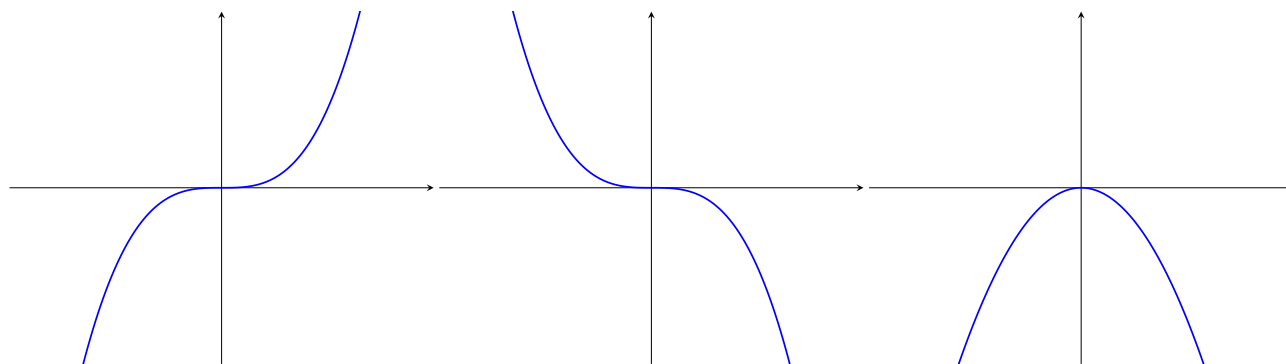
$$D = \frac{0.034}{\mu} S^2$$

Note that the braking distance does not include a driver’s reaction time ².

7. Is the braking distance a power function of speed? If yes, give the coefficient k and the exponent p . Assume that μ is a given constant.
8. The coefficient of friction under normal conditions when the road is dry is $\mu = 0.7$. What is the braking distance of a car that travels on a dry road at 35 mph? What is the braking distance at 70 mph?
9. By what factor does the braking distance increase when speed S doubles?
10. The coefficient of friction on a wet road is $\mu = 0.4$. Calculate the braking distance of a car traveling at 70 mph on a wet road.
11. Below are the graphs of four power functions $y = kx^p$ where p is a positive integer. In each of the graphs, is the exponent p even or odd? Is the coefficient k positive or negative?



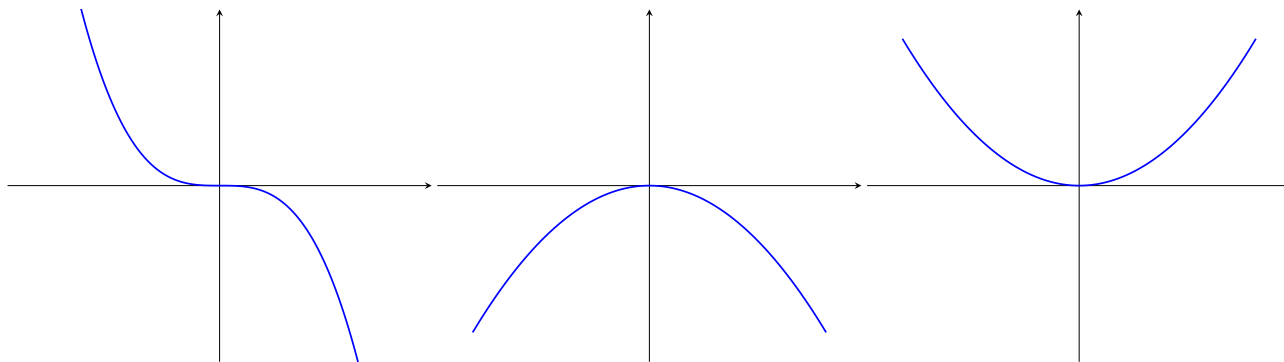
12. Which of the following is a graph of $f(x) = -4x^5$?



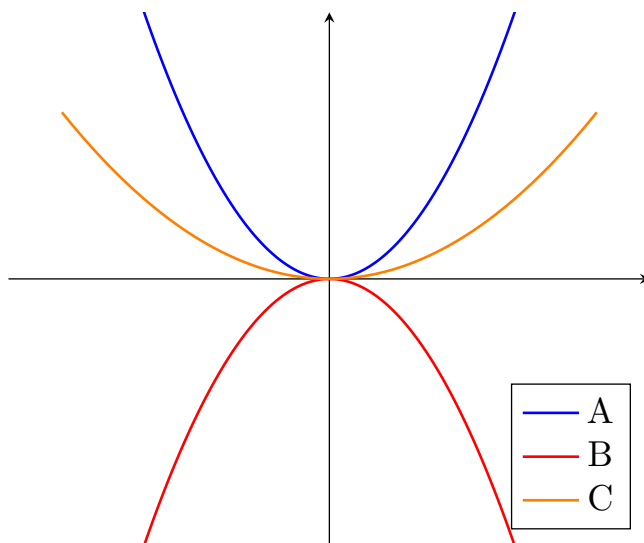
¹<http://hyperphysics.phy-astr.gsu.edu/hbase/crstp.html> (hyperphysics.phy-astr.gsu.edu/hbase/crstp.html), accessed: 7/5/20

²https://en.wikipedia.org/wiki/Braking_distance (en.wikipedia.org/wiki/Braking_distance), accessed: 7/5/20

13. Which of the following is a graph of $f(x) = -3x^4$?



14. Below you see graphs of the functions $y = -x^4$, $y = \frac{1}{4}x^4$, and $y = x^4$. Decide which is which.



15. The area, A , of a unilateral triangle whose sides have length a is given by:

$$A = \frac{\sqrt{3}}{4}a^2.$$

Find the side length a , in cm, which gives the area A equal to 8 cm^2 . Round off your answer to three decimal places.

4.4 Power Functions: Fractional and Negative Integral Exponents

Objectives

After completing this section, you should be able to do the following.

1. First objective.
 2. Second objective.
 3. Third objective.
-

In the previous section, we considered power functions $f(x) = kx^p$ in the case when p is a positive integer. When p is a negative integer, power functions behave quite differently.

Negative Integral Exponents

Consider a power function

$$f(x) = kx^p$$

in the case when the exponent p is a negative integer. The behavior and graphs of such functions are very different from those of power functions with positive integral exponents.

A negative exponent x^{-m} means the reciprocal $\frac{1}{x^m}$. So now the independent variable is in the denominator. The domain of a power function in this case is no longer the set of all real numbers as x cannot be 0.

We now look more closely at these graphs.

Example 4.4.1

Which of the functions below are power functions? Rewrite those which are in standard form $y = kx^p$. Identify the coefficient and the exponent.

1. $f(x) = \frac{3x^2}{2x^3}$
2. $g(x) = \frac{-\pi}{x^4}$
3. $h(x) = \frac{70}{3x}$

Solution.

1. The function $f(x)$ is a power function. Using Rules of Exponents from Section 4.1, we can rewrite:

$$f(x) = \frac{3x^2}{2x^3} = \frac{3}{2x} = \frac{3}{2} \cdot \frac{1}{x} = \frac{3}{2} \cdot x^{-1}.$$

The coefficient is $k = \frac{3}{2}$ and the exponent is $p = -1$.

2. $g(x)$ is a power function:

$$g(x) = \frac{-\pi}{x^4} = -\pi x^{-4}.$$

The coefficient is $k = -\pi$ and the exponent is $p = -4$.

3. $h(x)$ is not a power function. Using rules of exponents for arbitrary exponents, we can rewrite:

$$h(x) = \frac{70}{3^x} = 70 \cdot 3^{-x}.$$

The function $h(x) = 70 \cdot 3^{-x}$ is not a power function. Indeed, the base of the power expression 3^{-x} is constant at 3, meanwhile the exponent $-x$ varies.

Example 4.4.2

Body mass index (BMI) is an easy screening method for weight category — underweight, healthy weight, overweight, and obesity. BMI is calculated as follows:

$$\text{BMI} = \frac{(\text{weight in lb}) \cdot 703}{(\text{height in inches})^2}$$

A person who weighs 170 lb may be underweight or obese—it depends on the person's height. Let $I(h)$ denote body mass index of a person who weighs 170 lb and whose height is h . Then, according to the formula,

$$I(h) = \frac{170 \cdot 703}{h^2}$$

Note that with the weight fixed at 170 lb, body mass index is a power function of height: $I = 119510 \cdot h^{-2}$. (In terms of proportionality, BMI is inversely proportional to the square of height.) The obese category is defined as BMI of 30.0 or above. The normal weight category corresponds to BMI between 18.5 and 24.9.

Find the height h at and below which a person weighing 170 lb is obese.

Solution. We want to find h such that $I(h) = 30$:

$$\frac{170 \cdot 703}{h^2} = 30$$

After multiplying both sides by h^2 and dividing by 30, then

$$h^2 = \frac{170 \cdot 703}{30}$$

which gives:

$$h = \pm \sqrt{\frac{170 \cdot 703}{30}}$$

Since h in our problem has to be positive, the solution is:

$$h = \sqrt{\frac{170 \cdot 703}{30}} \approx 63$$

At the height $h = 63$ inches and below a person weighing 170 lb is obese.

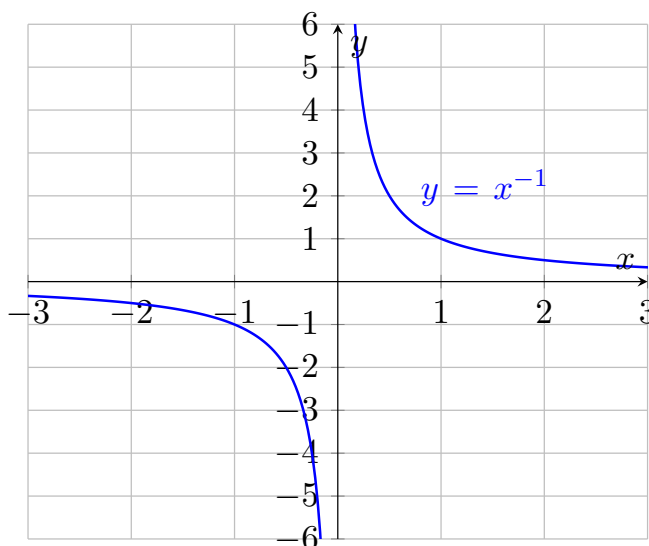
Note that when h increases $I(h)$ decreases as the denominator of the expression $\frac{170 \cdot 703}{h^2}$ becomes larger. On the other hand, when h gets smaller, $I(h)$ gets larger. Hence, every person shorter than 63 inches who weighs 170 lb is also obese.

Graphs of Power Functions: Negative Integral Exponents

Graphs of power functions $y = kx^p$ with exponents p that are negative integers have a different shape depending on whether p is even or p is odd.

Odd Negative Exponents

We begin with the case $k = 1$ and $p = -1$. The graph of the function $y = x^{-1} = \frac{1}{x}$ looks as follows:



Observe that when x gets larger and larger, say:

$$x = 1, 2, 3, 4, 5, \dots 1000 \dots,$$

the values $y = \frac{1}{x}$ become very close to 0. Indeed, they are:

$$y = 1, \frac{1}{2}, \frac{1}{3}, \frac{1}{4}, \frac{1}{5}, \dots \frac{1}{1000} \dots$$

In terms of the graph, this behavior translates to the graph getting very close, arbitrarily close, to the x -axis. We say that the x -axis, or equivalently the horizontal line $y = 0$, is a **horizontal asymptote** of the function $y = \frac{1}{x}$.

Observe that as x is getting close to 0 from the right, the values are becoming very large. We test a few positive inputs close to 0:

$$x = \frac{1}{2}, \frac{1}{3}, \frac{1}{4}, \frac{1}{5}, \dots \frac{1}{1000} \dots$$

The corresponding values $y = \frac{1}{x}$ are:

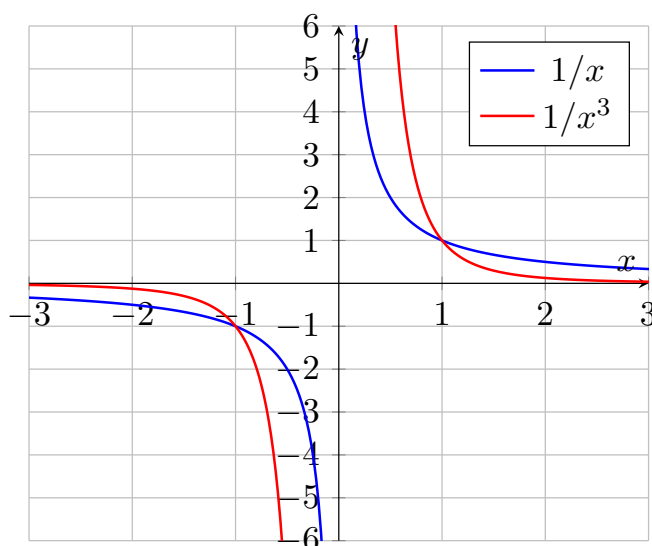
$$y = 2, 3, 4, 5, \dots 1000 \dots$$

In terms of the graph, this behavior translates to the graph getting very close to the y -axis with the values of the function becoming arbitrarily large. We say that the y -axis, or equivalently the vertical line $x = 0$, is a **vertical asymptote** of the function $y = \frac{1}{x}$.

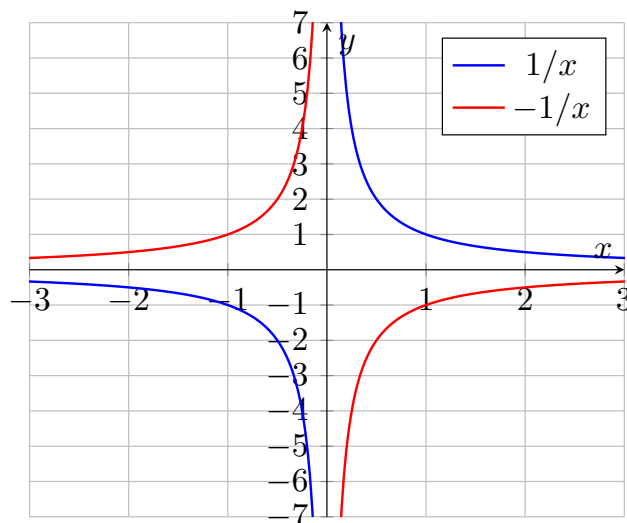
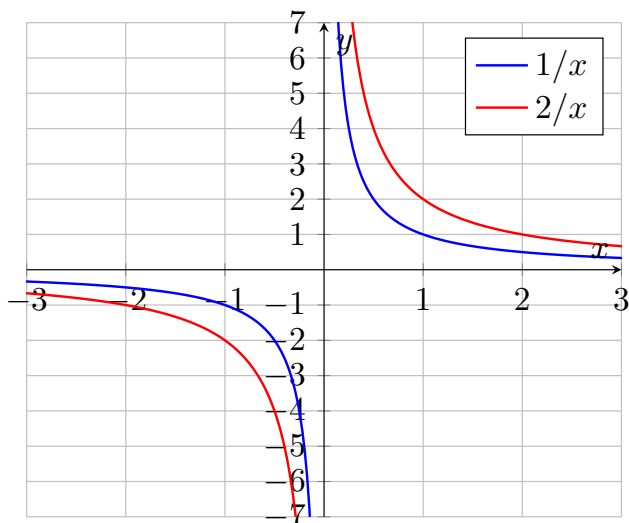
The portion of the graph corresponding to negative inputs x , is symmetric about the origin to the portion of the graph corresponding to positive inputs. Indeed, when x changes sign $y = \frac{1}{x}$ changes sign:

$$\frac{1}{-x} = -\frac{1}{x}.$$

Graphs of other power functions $y = kx^p$ in which the exponent p is an odd negative integer have a similar shape to the graph of $y = \frac{1}{x}$. For example:

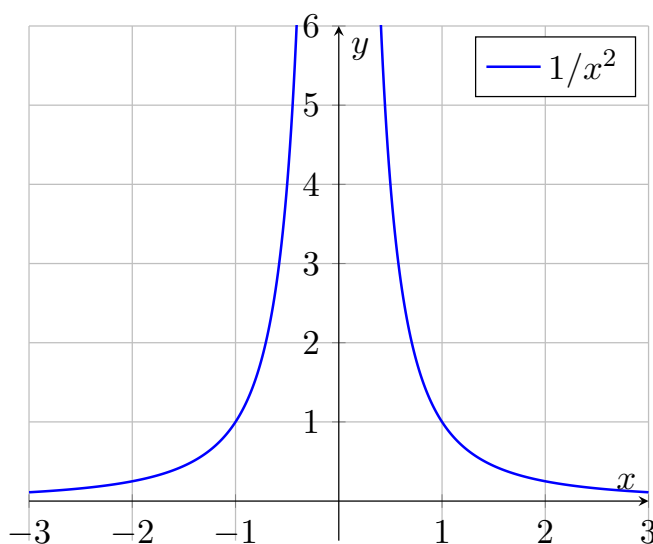


As always, the coefficient k in $y = kx^p$ stretches or shrinks the graph of $y = x^p$ vertically. Additionally, if k is negative, the graph is reflected about the x -axis.

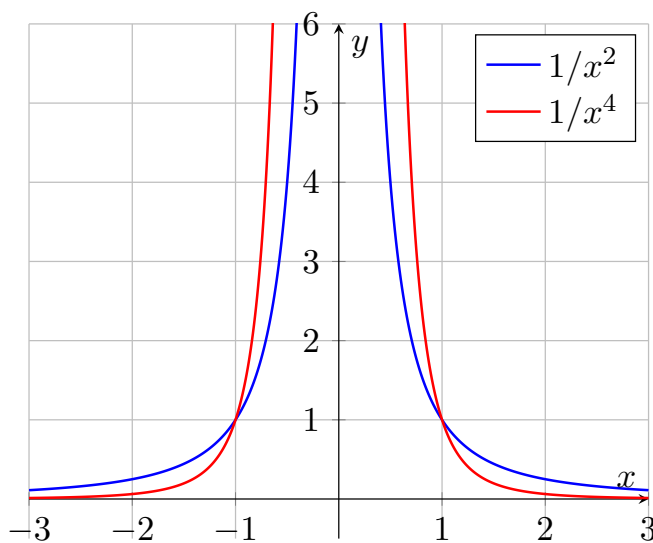


Even Negative Exponents

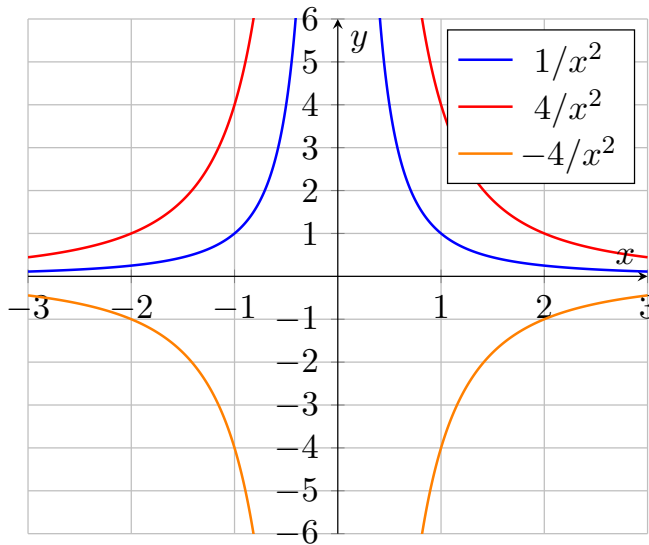
When p is a negative even integer, the graph of $y = x^p$ is entirely above the x -axis and it is symmetric about the y -axis, since $(-x)^p = x^p$ when p is even.



The graphs of the functions $y = x^p$ for other even negative integers p have a similar shape:



The x -axis is still a horizontal asymptote and the y -axis is a vertical asymptote. The coefficient k is responsible for vertical scaling and a reflection about the x -axis if k is negative. For example:



Summary

Here is a summary of how graphs of power functions look for negative integer exponents p , even and odd, and for coefficients k positive and negative.

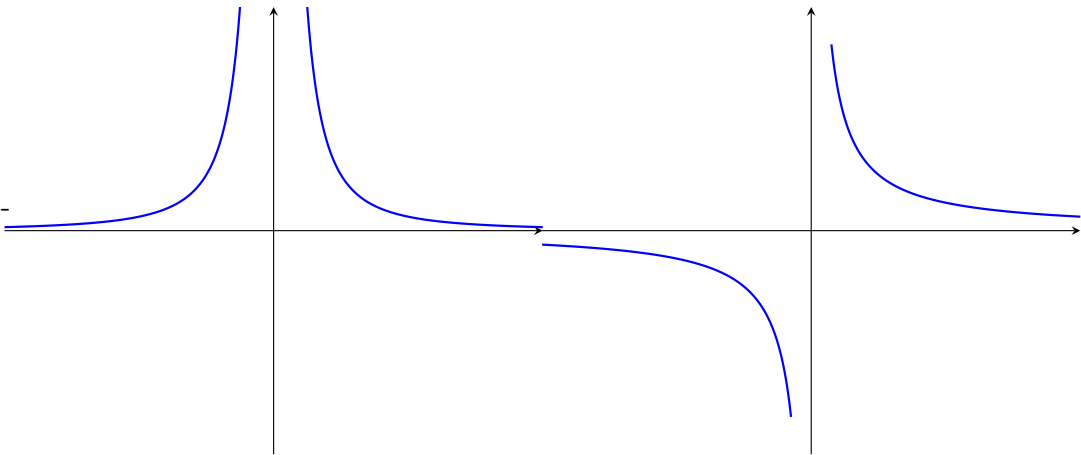
$f(x) = kx^p$

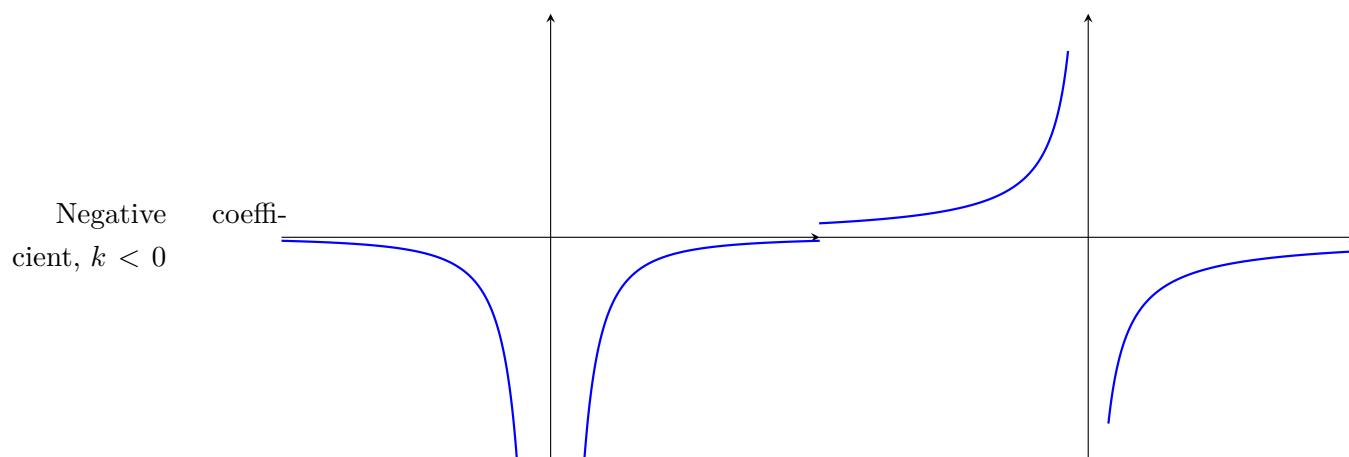
p even and negative

p odd and negative

Positive
coefficient, $k > 0$

coefficient





Fractional Exponents

Example 4.4.3

Which of the functions below are power functions? Those which are, rewrite in standard form $y = kx^p$. Identify the coefficient and the exponent.

1. $f(x) = \frac{3x^3}{7\sqrt{x}}$

2. $g(x) = \frac{2x^{1/3}}{x^4}$

3. $h(x) = \sqrt{\frac{2x}{x^{-2}}}$

Solution.

1. The function $f(x)$ is a power function. Using rules of exponents for arbitrary exponents, we can rewrite:

$$f(x) = \frac{3x^3}{7\sqrt{x}} = \frac{3x^3}{7x^{1/2}} = \frac{3x^{5/2}}{7} = \frac{3}{7}x^{5/2}$$

The coefficient is $k = \frac{3}{7}$ and the exponent is $p = \frac{5}{2}$.

2. $g(x)$ is a power function:

$$g(x) = \frac{2x^{1/3}}{x^4} = 2x^{\frac{1}{3}-4} = 2x^{-\frac{11}{3}}$$

The coefficient is $k = 2$ and the exponent is $p = -\frac{11}{3}$.

3. $h(x)$ is a power function as well. Using rules of exponents for arbitrary exponents and properties of the square root, we can rewrite:

$$\begin{aligned} h(x) &= \sqrt{\frac{2x}{x^{-2}}} = \frac{\sqrt{2x}}{\sqrt{x^{-2}}} = \frac{\sqrt{2} \cdot x^{1/2}}{(x^{-2})^{1/2}} \\ &= \frac{\sqrt{2} \cdot x^{1/2}}{x^{-1}} = \sqrt{2} \cdot x^{(1/2)-(-1)} \\ &= \sqrt{2} \cdot x^{\frac{3}{2}} \end{aligned}$$

The coefficient is $k = \sqrt{2}$ and the exponent is $p = \frac{3}{2}$.

Example 4.4.4

Body surface area (BSA) is the total surface area of the human body. The body surface area is used in many measurements in medicine, including the calculation of drug dosages and the amount of fluids to be administered intravenously. There are several accepted formulas to calculate BSA. One of the most commonly used is the Du Bois formula:

$$\text{BSA} = 0.007184 \cdot w^{0.425} \cdot h^{0.725}$$

where w is weight in kilograms (kg) and h is height in centimeters (cm). The formula gives BSA in square meters, m^2 .

1. Calculate BSA of a female who stands 158 cm tall and weighs 60 kg. Give units with your answer.
2. Let weight be fixed at 70 kg. Then BSA depends on height h only— with weight fixed at 70, BSA is a function of h only. Denote this function by $S(h)$. Find a formula for $S(h)$. Is it a power function? If yes, identify the coefficient and the exponent.
3. Use your formula for $S(h)$ to calculate BSA for a female that weighs 70 kg and is 158 cm tall. Give units with your answer.

Solution.

1. We want to calculate BSA for $w = 60$ and $h = 158$. We substitute the values into the Du Bois formula:

$$\text{BSA} = 0.007184 \cdot 60^{0.425} \cdot 158^{0.725} \approx 1.61.$$

The total body surface area of a person who weighs 60 kg and is 158 cm tall is 1.61 m^2 .

2. Fix $w = 70$. Then BSA as a function of h is:

$$S(h) = 0.007184 \cdot 70^{0.425} \cdot h^{0.725} = 0.043705 \cdot h^{0.725}$$

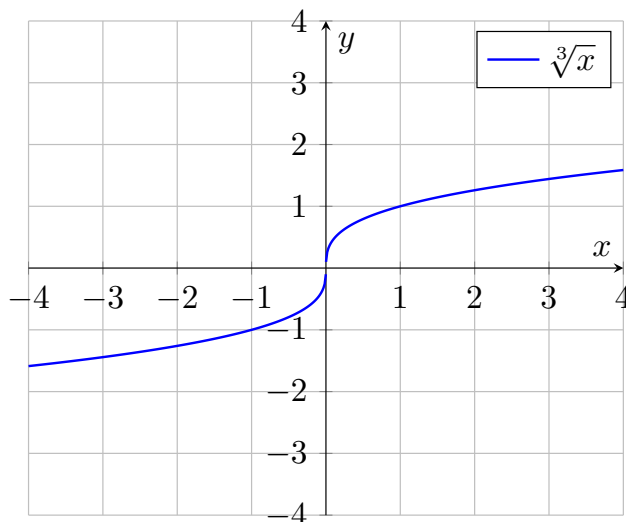
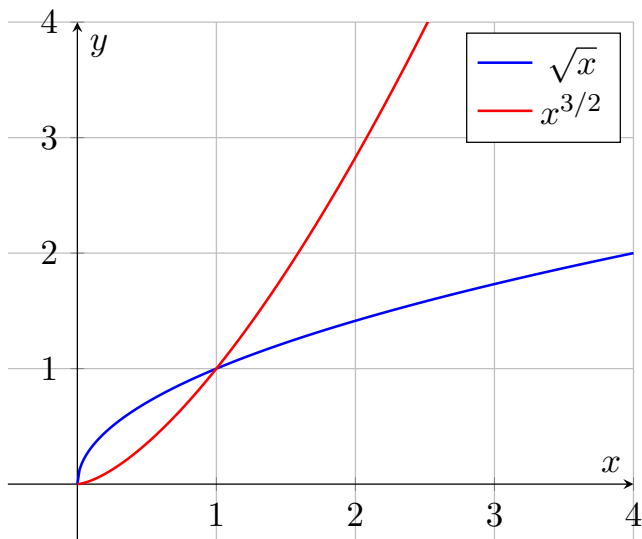
$S(h)$ is a power function of h with the coefficient 0.043705 and exponent 0.725.

3. BSA of a female who weighs 70 kg and is 158 cm tall is $S(158) = 0.043705 \cdot 158^{0.725} \approx 1.72$.

The total body surface area of a person who weighs 70 kg and is 158 cm tall is 1.72 m².

Graphs of Power Functions: Fractional Exponents

When considering power functions with fractional exponents or, more general, with exponents which are not integers, we restrict the domain to $x \geq 0$ (or to $x > 0$ if a non-integral exponent is negative). The graphs of $x^{1/2}$ and $x^{3/2}$ are given below as an example. Traditionally, roots of odd orders like $\sqrt[3]{x}$ or $\sqrt[5]{x}$ are considered and graphed for all x .



Exercises

Power Functions. Determine whether the given function is a power function. If yes, identify the coefficient k and the exponent p . If not, say so.

1. $y = \frac{\sqrt{x^3}}{0.1x}$
2. $y = \frac{\sqrt{0.01x^3}}{\sqrt{x}}$

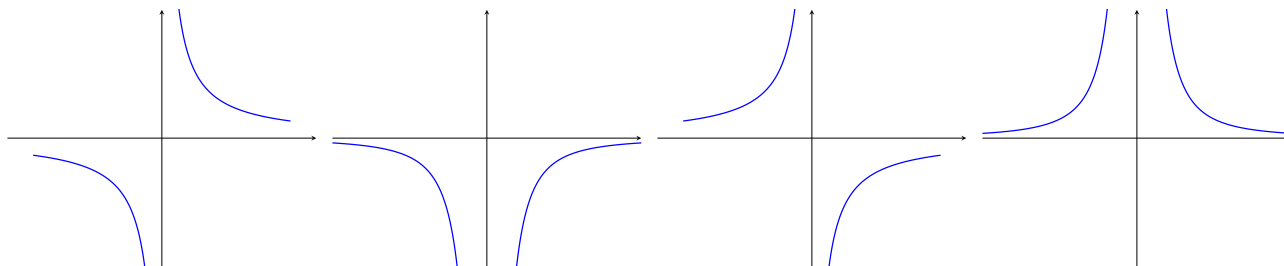
3. $g(x) = \frac{x^{\frac{5}{3}}}{4x^{\frac{1}{4}}}$

4. $h(x) = 2x^{\frac{3}{2}} - x^{-2}$

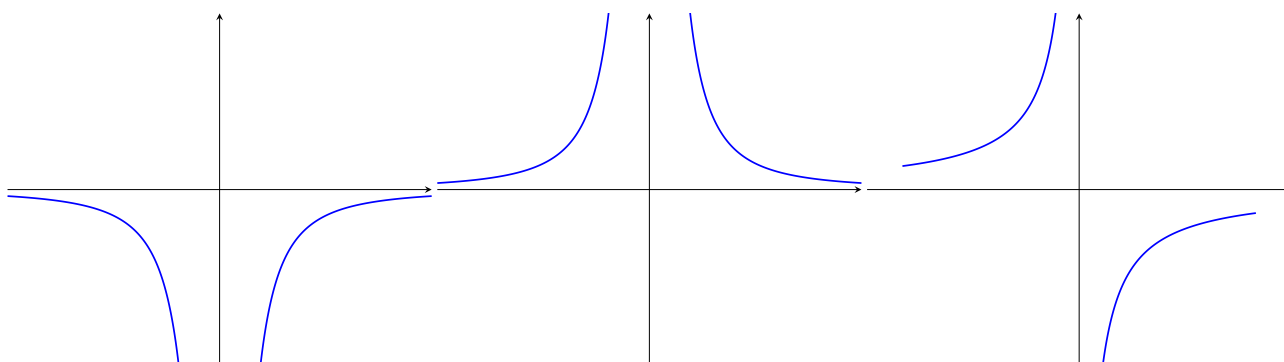
5. $y = \sqrt{\frac{x^5}{9x}}$

6. $f(x) = (\sqrt{2x})^2$

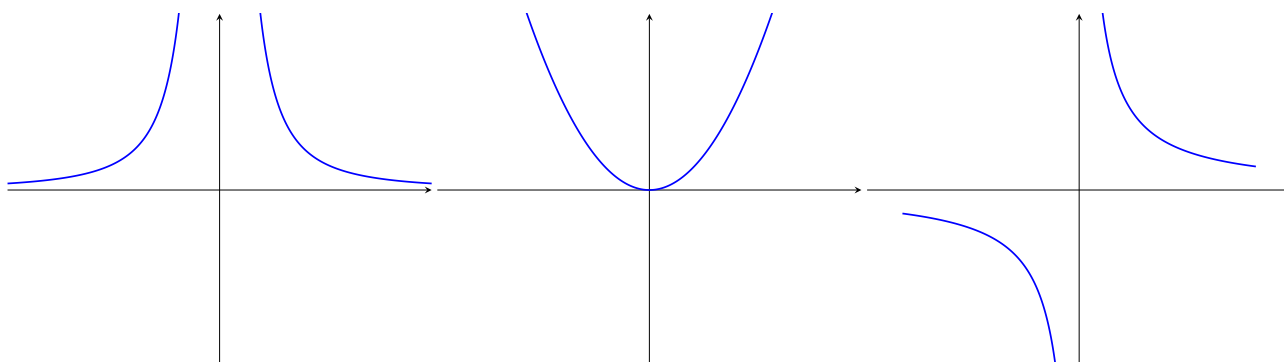
7. Four power functions $y = kx^p$ are graphed below, where p is a negative integer. In each of the graphs, is the exponent p even or odd? Is the coefficient k positive or negative?



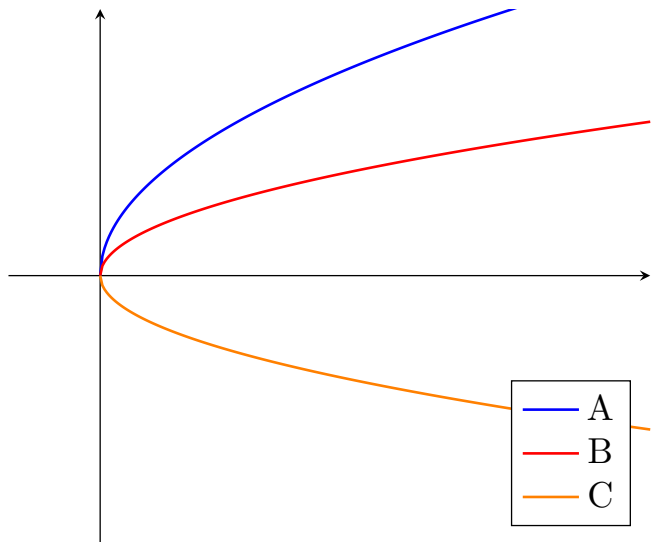
8. Which of the following is a graph of $f(x) = -2x^{-5}$?



9. Which of the following is a graph of $f(x) = 2x^{-4}$?



10. Below you see graphs of the functions $y = x^{\frac{1}{2}}$, $y = 2x^{\frac{1}{2}}$, and $y = -x^{\frac{1}{2}}$. Decide which is which.



11. Find a formula for a power function $f(x) = kx^p$ given numerically by:

Table 4.4.5

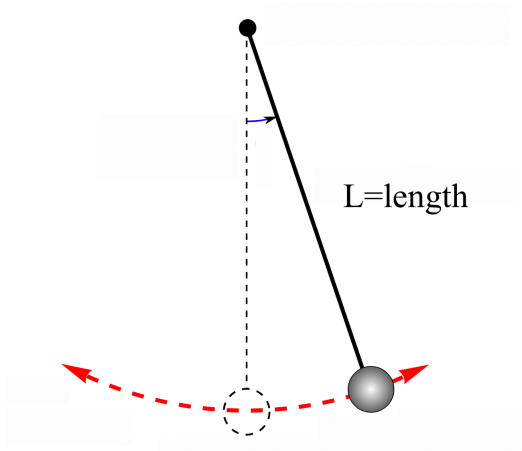
x	0	1	2	3	5
$f(x)$	0	3	$3\sqrt{2}$	$3\sqrt{3}$	$3\sqrt{5}$

12. Find a formula for a power function $g(x) = kx^p$ given numerically by:

Table 4.4.6

x	0	1	2	3	5
$g(x)$	undefined	2	1	$\frac{2}{3}$	$\frac{2}{5}$

13. Consider a pendulum depicted below ¹.



When the pendulum is displaced sideways from its resting position—called the equilibrium position—the force due to gravity will cause the pendulum to oscillate back and forth about the equilibrium position. The time, T , needed to execute one full cycle—a left swing and the right

swing—is called the pendulum’s period. The period T depends on the length L of the pendulum ² and the local acceleration due to gravity g :

$$T = 2\pi\sqrt{\frac{L}{g}}.$$

- (a) With g fixed, is T a power function of L ? In other words, is $T = kL^p$? If yes, find the coefficient and the exponent.
- (b) On the surface of the Earth, the acceleration due to gravity, g , is equal to 9.81 m/sec^2 where m stands for meters. Calculate the period of a pendulum of length 0.5 m that happily oscillates in Kingston, RI.
- (c) An astronaut is standing on the surface of a faraway asteroid wondering about the acceleration due to gravity on the asteroid. The astronaut has a watch and a pendulum whose length is 0.3 m. The astronaut measures the period of the pendulum which turns out to be 5 seconds. What is the acceleration due to gravity on the asteroid?

¹Modified from a public domain image at <https://en.wikipedia.org/wiki/Pendulum> (en.wikipedia.org/wiki/Pendulum), accessed: 7/8/20

²<https://en.wikipedia.org/wiki/Pendulum> (en.wikipedia.org/wiki/Pendulum), accessed: 7/8/20

Appendix A

Selected Solutions

1 • Introduction to Functions

1.1 • The Concept of a Function

• Practice Problems

1.

a. $f(1) = 5$

b. $f(-1.5) = 0$

c. $f(0) = 3$

2. The independent variable is t and has units of years. The dependent variable is V and has units of dollars.

• Exercises

1.

(a) The amount of caffeine in a person's body 0 hours after drinking a cup of coffee is 96 mg.

(b) The amount of caffeine in a person's body 5 hours after drinking a cup of coffee is 48 mg.

(c) The amount of caffeine in a person's body 24 hours after drinking a cup of coffee is approximately 0 mg.

2.

(a) A total of 10 units of the product are produced in 1 hour.

(b) A total of 0 units of the product are produced in 0 hours.

- (c) A total of approximately 29 units of the product are produced in 3 hours.

3.

(a) p ; dollars

(b) C ; dollars

(c) \$30

4.

(a) The independent variable is t , which represents the time spent baking in hours.

(b) The dependent variable is B , which represents the number of batches of cookies made by the bakery after t hours have passed.

(c) $f(3) = 18$; The bakery makes 18 batches of cookies (or approximately 216 cookies) in 3 hours.

5.

(a) The amount of fuel left in the fuel tank 70 miles into the drive is 6 gallons.

(b) The amount of fuel left in the fuel tank 200 miles into the drive is 1 gallon.

6.

(a) $-9b + 4$

(b) $-3b + 1$

(c) $-3b$

7.

(a) 2

(b) -10

(c) $3\sqrt{2} - 2$

8.

(a) $h^2 - 2h + 4$

(b) $h^2 + 2h + 4$

(c) $b^2 + 2b$

9.

(a) 1

(b) undefined

(c) -1

10.

(a) $\frac{3}{h+1}$

(b) $\frac{3}{h+3}$

(c) $\frac{3}{b+2} - 4$

11.

(a) $\frac{9}{2}$

(b) $h^2 + 2h + 1 + \frac{1}{h+1}$

(c) $h^2 + 2h + \frac{1}{h+1} - \frac{7}{2}$

12.

(a) $\frac{17}{2}$

(b) $2x + \frac{23}{2}$

(c) $-2x^2 - \frac{31}{2}x - 35$

13.

(a) \$45

(b) 4.5 hours

14.

(a) \$16500

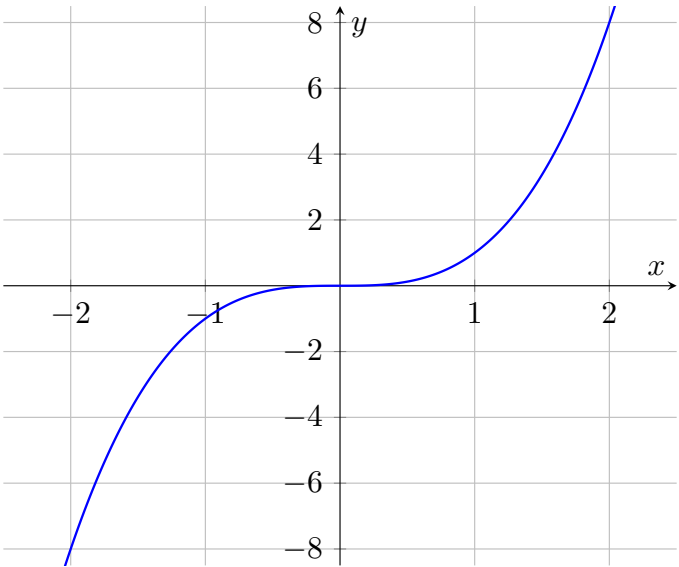
(b) \$9000

(c) 11 years

15.(a) S is the independent variable; D is the dependent variable(b) $\frac{4}{5}$ of a mile**16.** $f(x) = 1.75x + 10$ **17.**(a) $7x + 5$ (b) $7(x + 5)$ **18.**(a) $5x^2 - 8$ (b) $(5x)^2 - 8 = 25x^2 - 8$ (c) $5(x - 8)^2$ **19.**(a) $\frac{x}{3} + \sqrt{x} \cdot 4$ (b) $\frac{1}{\sqrt{x}} + 1 + x^2$ **20.**(a) all real numbers or $(-\infty, \infty)$ in interval notation(b) all real numbers $x \neq 1$ or $(-\infty, 1) \cup (1, \infty)$ in interval notation(c) all real numbers $x \neq -2, x \neq 2$ or $(-\infty, -2) \cup (-2, 2) \cup (2, \infty)$ in interval notation(d) all real numbers $x \geq 0$ or $[0, \infty)$ in interval notation(e) all real numbers or $(-\infty, \infty)$ in interval notation**21.**(a) all real numbers or $(-\infty, \infty)$ in interval notation(b) all real numbers $x \neq 0$ or $(-\infty, 0) \cup (0, \infty)$ in interval notation(c) all real numbers $x > 0$ or $(0, \infty)$ in interval notation

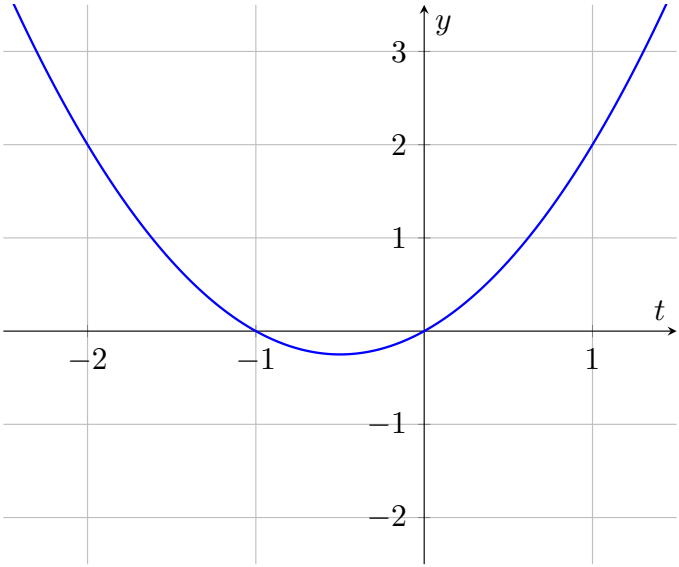
1.2 · The Graph of a Function
· Exercises
1.

x	-2	-1	0	1	2
y	-8	-1	0	1	8



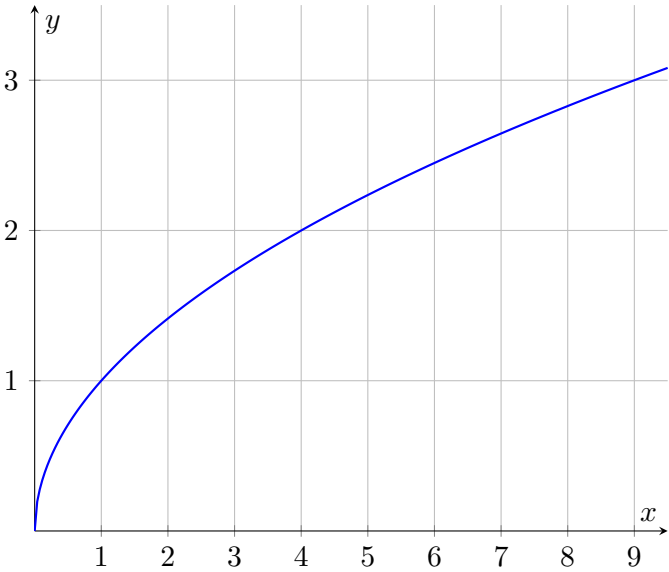
2.

t	-2	-1	0	1
$f(t)$	2	0	0	2



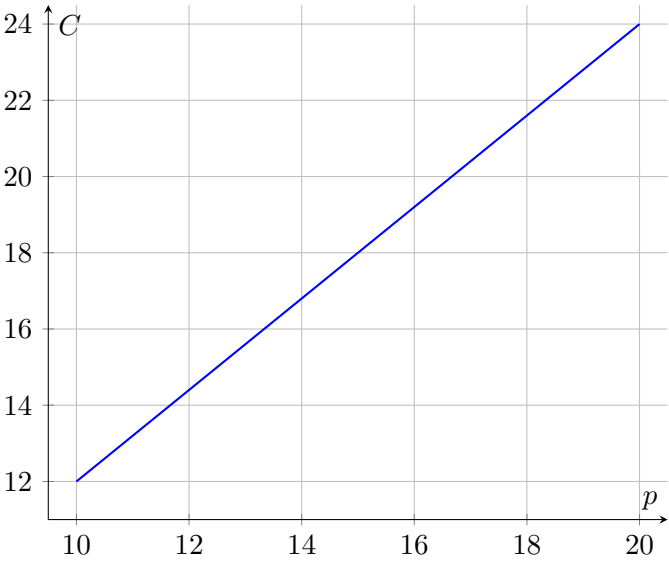
3.

x	0	1	4	9
$g(x)$	0	1	2	3



4.

p	10	12	14	16	18	20
C	12	14.4	16.8	19.2	21.6	24



5.

(a) $f(2) = -8$

(b) $f(3) = -10$

(c) $f(-1) = 10$

6. $x = -2$, $x = 1$, and $x = 4$.

7. $x \approx -1.5$, $x = 0$, and $x \approx 4.5$.

8.

(a) $f(0) = 9$

(b) $f(12) = 9$

(c) $f(10) \approx 10.5$

9. $t = 6$ and $t = 18$

10. $t = 3$, $t = 9$, $t = 15$, and $t = 21$

11.

(a) 13.2 gallons.

(b) 8 gallons.

(c) The gas tank is empty.

(d) Approximately 40 miles per gallon.

12.

(a) 96 mg.

(b) After 5 hours, there are 48 mg left. After 10 hours, 24 mg.

(c) Decreasing.

13.

(a) \$1000.

(b) After 10 years, \$2000. After 20 years, \$4000.

(c) Increasing.

14. No, because it fails the vertical line test (in many places!).

15.

(a) $f(0) = -4$.

(b) $x = -2$ and $x = 2$.

(c) $x = -3$ and $x = 3$.

16. Increasing on the intervals $(0, 3) \cup (9, 12)$. Or, write as $0 < t < 3$ and $9 < t < 12$. Decreasing on the interval $(3, 9)$. Or, write as $3 < t < 9$.

1.3 · Functions Given Numerically

· Exercises

1.3.1.

(a) 2.23

(b) 3.87

(c) 4.47

1.3.2.

(a) $x = 0$

(b) $x = 5$

1.3.3.

(a) $g(4) = 1.5$

(b) $g(2) = 3.7$

(c) $g(8) = 1.5$

1.3.4.

(a) $t = 0$, $t = 4$, and $t = 8$

(b) $t = 2$

1.3.5. $h(x) = x^3$ **1.3.6.** $F(t) = -2t$

t	5	6
$F(t)$	-10	-12

1.3.7.

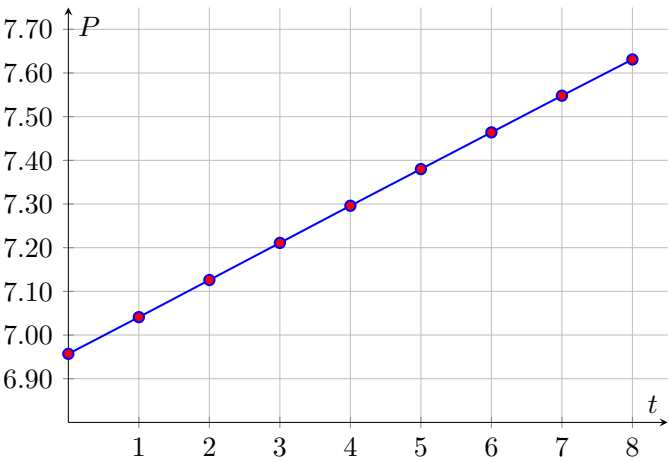
d	180	216	252
$G(d)$	11	9.5	8

The gas mileage of the driver’s Volkswagen Passat is 24 miles per gallon.

1.3.8.

t	0	1	2	3	4	5	6	7	8
$P(t)$	6.957	7.041	7.126	7.211	7.296	7.380	7.464	7.548	7.631

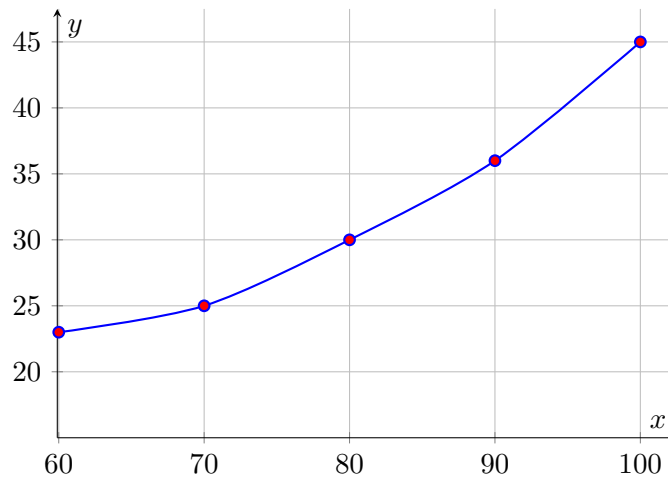
1.3.9.



1.3.10.

- (a) The amount of power required for the dragon to carry the knight is increasing as the dragon’s speed increases. This is evident from the table since the values associated with the power required increase as the speed increases.
- (b) The amount of power required for the dragon to carry the knight is increasing faster and faster as the dragon’s speed increases. This is evident from the table since each time the speed increases by 10 miles per hour starting at 60 miles per hour, the power required to carry the knight increases by more and more.

1.3.11.



- (a) The amount of power required for the dragon to carry the knight is increasing as the dragon's speed increases. This is evident from the graph since it climbs over the interval $60 \leq x \leq 100$.
- (b) The amount of power required for the dragon to carry the knight is increasing faster and faster as the dragon's speed increases. This is evident from the graph since each time the speed increases by 10 miles per hour starting at 60 miles per hour, the graph climbs more and more quickly.

1.4 · Vertical and Horizontal Intercepts

· Exercises

1.4.1.

- (a) *horizontal intercept:* 0.5; *vertical intercept:* -1.5
- (b) *horizontal intercepts:* $-1, 1$; *vertical intercept:* 1
- (c) *horizontal intercept:* 27; *vertical intercept:* 5.4
- (d) *horizontal intercept:* -2 ; *vertical intercept:* 16
- (e) *horizontal intercept:* $9/4$; *vertical intercept:* -3
- (f) *horizontal intercepts:* none; *vertical intercept:* none
- (g) *horizontal intercept:* none; *vertical intercept:* -1

1.4.2.

- (a) *horizontal intercepts:* $-1, 2$; *vertical intercept:* 2
- (b) *horizontal intercepts:* none; *vertical intercept:* 5

(c) *horizontal intercept*: 8; *vertical intercept*: none

1.4.3.

(a) *horizontal intercepts*: 0, 4, 8; *vertical intercept*: 0

(b) *horizontal intercepts*: none; *vertical intercept*: 3

1.4.4. *vertical intercept*: 300—this is the number of feet above the lake that the ball is dropped from; ***horizontal intercept***: $t \approx 4.33$ —this is the number of seconds that it takes after being dropped for the ball to hit the surface of the lake

1.4.5. *vertical intercept*: 21500—this is the price in dollars for which the car was purchased; ***horizontal intercept***: $t \approx 14.33$ —this is the number of years after purchase that it takes for the car's value to depreciate to \$0

1.5 · The Average Rate of Change

· Exercises

1.5.1.

(a) 6

(b) -3

(c) $-1/3$

(d) -3

(e) -3

1.5.2.

(a) 0.0845 billion people per year; 0.0835 billion people per year

(b) No.

1.5.3.

(a) -48 feet per second; -80 feet per second

(b) negative; the height of the ball is decreasing as time progresses (as it was dropped)

1.5.4.

(a) -9.6 mg/hr

(b) -2.4 mg/hr

- (c) The average rate of change from $t = 0$ to $t = 5$ hours is larger than the average rate of change from $t = 10$ to $t = 15$ hours. It can be seen that the rate at which the amount of caffeine in the bloodstream is changing gets smaller and smaller as time progresses.

1.5.5.

- (a) positive since the graph is increasing on this interval
 (b) negative since the graph is decreasing on this interval

1.5.6.

- (a) -0.325 pounds/minute
 (b) -0.035 pounds/minute
 (c) The rate of weight loss slows as the daily amount of exercise increases.

1.5.7.

- (a) approximately 2.124 million cars sold per year
 (b) -3.175 million cars sold per year
 (c) from 2017 to 2018 and from 2018 to 2019

2 · Linear Functions**2.1 · Properties of Linear Functions****· Exercises****Slope-Intercept Form.**

1. $f(x) = 3x - 6$; slope: 3; vertical intercept: -6
2. $g(x) = 8x - 2$; slope: 8; vertical intercept: -2
3. $h(t) = 9t - 2.5$; slope: 9; vertical intercept: -2.5
4. $m(x) = 2x$; slope: 2; vertical intercept: 0

Computing the Slope.

5. 2
6. $-\frac{1}{7}$
7. 26.5
8. $\frac{3}{8}$
9. 0
10. 1

Finding the Slope.

11. 1.5

12. $-\frac{4}{3} \approx -1.33$

13. 2

14. 0

15. Figure 2.1.17: slope 0.5; vertical intercept 1; equation $y = 0.5x + 1$ Figure 2.1.18: slope $-\frac{3}{2}$; vertical intercept -1 ; equation $y = -\frac{3}{2}x - 1$ Figure 2.1.19: slope $\frac{2}{3}$; vertical intercept 0; equation $y = \frac{2}{3}x$ Figure 2.1.20: slope $-\frac{2}{3}$; vertical intercept -1 ; equation $y = -\frac{2}{3}x - 1$ 16. 1. Line 1 is $y = 3x + 2$; Line 2 is $y = 3x - 1$; Line 3 is $y = 2$; Line 4 is $y = x - 1$; Line 5 is $y = -2x - 3$.

2.

- Line 1

- $y = 3x + 2$

- Line 2

- $y = 3x - 1$

- Line 3

- $y = 2$

- Line 4

- $y = x - 1$

- Line 5

- $y = -2x - 3$

17. $A = 1567n + 55683$

18.

- (a) The initial elevation of the hiker is 1350 feet.
- (b) The hiker's elevation is increasing at a rate of 35 feet per minute.

19.

- (a) The bicyclist's initial distance from the finish line is 50 miles.

- (b) The bicyclist's distance from the finish line is decreasing at a rate of 25 miles per hour.

2.2 • Working with Linear Functions and Linear Equations

• Exercises

Slope-Intercept Form.

1. $y = -2x + 10$

2. $y = \frac{5}{2}x + 17$

3. $y = 2.1x - 0.21$

4. $y = 6x + 32$

5. $y = -\frac{1}{2}x - \frac{4}{5}$

6. $y = 3$

7. $x = 5$

8. $y = -1$

Slope-Intercept Form.

9. $f(x) = \frac{5}{2}x - 10$

10. $f(x) = -\frac{3}{2}x - \frac{1}{2}$

11. $f(x) = 2.1x - 0.21$

12. $f(x) = -x + 1$

13. $f(x) = \frac{5}{6}x + \frac{23}{6}$

14. $f(x) = 3x + 6$

15. $f(x) = x + 2$

16. $f(x) = -\frac{3}{7}x + 3$

17. (c) is not the graph of a linear function.

18.

- (a) The vertical intercept 60 is the distance in miles that the biker is from the finish line at the start of the race.
- (b) The horizontal intercept 3 is the number of hours that it takes the biker to complete the race.
- (c) The slope -20 indicates that the distance between the biker and the finish line is decreasing at a rate of 20 miles per hour.

19.

- (a) \$1900

- (b) The lamp will have a value of 4000 dollars 8.4 years after its purchase.

20.

(a) $A(m) = 0.20m + 75$; $B(m) = 0.10m + 90$

(b) $m = 150$

(c) $B(m)$

Solving Linear Equations.

21. $x \approx 1.3158$

22. $x = \frac{1}{7}$

23. no solution

24. infinite number of solutions

2.3 · Modeling and Numerically Given Functions

· **Exercises**

2.3.1.

(a) linear with equation $y = -3.3x + 12.6$

(b) linear with equation $f(t) = 0.15t + 0.25$

(c) not linear

(d) not linear

(e) linear with equation $w(z) = -0.4z + 5.2$

2.3.2. approximately linear with approximate equation $C = 0.555F - 17.76$

2.3.3.

(a) $H = 2.32L + 65.53$

(b) 172.25 cm

(c) approximately 47.36 cm

2.3.4.

(a) $P = 0.4d + 14.7$

(b) 66.7 PSI

2.3.5.

- (a) $T(h) = -0.00356h + 65$
- (b) 42.89° F
- (c) The function $T(h)$ is decreasing.

2.3.6.

- (a) Linear; formula $H(t) = \frac{2}{3}t + 85$
- (b) The slope is $\frac{2}{3}$ bpm per minute; the practical meaning of the slope is that the man's bpm is increasing at a rate of $\frac{2}{3}$ bpm per minute. The vertical intercept is 85 bpm; the practical meaning of the vertical intercept is that the man's resting heart rate (heart rate when performing no exercise on the treadmill) is 85 bpm.

3 · Quadratic Functions**3.1 · Introduction to Quadratic Functions****· Exercises****Standard Form,**

- 1. $f(x) = 4x^2 - 7x$; opens up
- 2. $h(t) = -5t^2 + 21$; opens down
- 3. $g(x) = 2x^2 - 14x$; opens up
- 4. $y = -4x^2 - 40x - 20$; opens down

Intercepts and Vertex.

- 5. horizontal intercepts: $-1, 3$; vertical intercept: -3 ; vertex: $(1, -4)$
- 6. no horizontal intercepts; vertical intercept: -2 ; vertex: $(1, -1)$

Horizontal Intercepts.

- 7. 0
- 8. $\frac{3}{2}$
- 9. $(6, 0)$; opens down
- 10.

- (a) $R(p) = p(1000 - 2p)$ or $R(p) = -2p^2 + 1000p$
- (b) \$500

- (c) The maximum revenue is \$125,000. The lighting company should charge \$250 per specialty chandelier to maximize revenue.

3.2 • Factoring Quadratic Expressions

• Exercises

Finding Intercepts.

1. horizontal intercepts: $-1, -2$; vertical intercept: 2
2. horizontal intercepts: $-2, 3$; vertical intercept: -6
3. horizontal intercepts: $-3, 2$; vertical intercept: -6
4. horizontal intercepts: 4 ; vertical intercept: 16
5. horizontal intercept: -7 ; vertical intercept: 49
6. horizontal intercepts: $-\frac{3}{4}, \frac{3}{4}$; vertical intercept: -36
7. horizontal intercepts: $-4, -1$; vertical intercept: 8
8. horizontal intercepts: $-1, \frac{3}{2}$; vertical intercept: -3
9. horizontal intercepts: $-8, 2$; vertical intercept: 16
10. horizontal intercepts: $-\frac{1}{14}, \frac{1}{14}$; vertical intercept: $\frac{1}{49}$
11. horizontal intercepts: $0, 3$; vertical intercept: 0
12. horizontal intercepts: $0, \frac{2}{5}$; vertical intercept: 0

13.

- (a) $H(t) = -16t^2 + 168t$
- (b) 10.5 seconds
- (c) The maximum height that the model rocket reaches is 441 feet.

3.3 • Vertex Form and Completing the Square

• Exercises

Vertex.

1. $(3, 4)$
2. $(-2, -3)$
3. $(1, -7)$
4. $(-\frac{1}{2}, \frac{5}{3})$

Vertex Form to Standard Form.

5. $f(x) = x^2 + 2x + 3$; $a = 1, b = 2, c = 3$

6. $g(x) = 5x^2 - 20x + 26$; $a = 5$, $b = -20$, $c = 26$
7. $y = -2t^2 + 16t - 27$; $a = -2$, $b = 16$, $c = -27$
8. $h(x) = -x^2 - x + \frac{17}{12}$; $a = -1$, $b = -1$, $c = \frac{17}{12}$

Standard Form to Vertex Form.

9. $f(x) = (x + 1)^2 - 6$; vertex: $(-1, -6)$
10. $g(x) = (x - 3)^2 + 1$; vertex $(3, 1)$
11. $h(x) = (x + \frac{3}{2})^2 - 1$; vertex $(-\frac{3}{2}, -1)$
12. $F(x) = -(x - 5)^2 + 40$; vertex $(5, 40)$
13. $G(x) = -3(x - 1)^2 - 1$; vertex $(1, -1)$
14. $H(x) = -2(x + 4)^2 + 15$; vertex $(-4, 15)$

Horizontal Intercepts.

15. $-7, 1$
16. $1 - \sqrt{2} \approx -0.414$, $1 + \sqrt{2} \approx 2.414$
17. $-0.5, 1.5$
18. $-2 - \sqrt{10} \approx -5.162$, $-2 + \sqrt{10} \approx 1.162$
19. No real solutions.
20. $-4 - \sqrt{\frac{5}{2}} \approx -5.581$, $-4 + \sqrt{\frac{5}{2}} \approx -2.419$

3.4 • The Quadratic Formula**• Exercises****Quadratic Equations.**

1. $x = -2$ and $x = 7$
2. $x = -2$ and $x = -1$
3. $x = -\frac{1}{2} = -0.5$ and $x = \frac{1}{3} \approx 0.333$
4. $x = 1 - \sqrt{3} \approx -0.732$ and $x = 1 + \sqrt{3} \approx 2.732$
5. $x = -3$ and $x = -\frac{5}{4} = -1.25$
6. $x = \frac{1 - \sqrt{41}}{2} \approx -2.701$ and $x = \frac{1 + \sqrt{41}}{2} \approx 3.702$

Intercepts and Vertex.

7. horizontal intercepts: $-3, 1$; vertical intercept: -3 ; vertex: $(-1, -4)$
8. horizontal intercepts: $-4 - \sqrt{6} \approx -6.450, -4 + \sqrt{6} \approx -1.551$; vertical intercept: 10 ; vertex: $(-4, -6)$
9. horizontal intercepts: $\frac{3-2\sqrt{3}}{3} \approx -0.155, \frac{3+2\sqrt{3}}{3} \approx 2.155$; vertical intercept: -1 ; vertex: $(1, -4)$
10. no horizontal intercepts; vertical intercept: 5 ; vertex $(2, 1)$

Discriminant.

11. 0; one real solution
12. 132; two distinct real solutions
13. -116 ; no real solutions
14. 56; two distinct real solutions

15.

- (a) $H(t) = -16t^2 + 50t + 6$
- (b) The ball reaches its maximum height approximately 1.56 seconds after it is thrown straight up and that maximum height is approximately 45.06 feet.

Factored Form.

16. $f(x) = 3(x + 1)(x - 2)$
17. $g(x) = (x - (-3 - \sqrt{2}))(x - (-3 + \sqrt{2}))$
18. $f(x) = -2(x + 1)(x - 2)$; $f(x) = -2x^2 + 2x + 4$
19. $g(t) = (t - 2)(t + 2)$; $g(t) = t^2 - 4$

4 · Power Functions**4.1 · Algebra of Powers: Integral Exponents****· Exercises****Rewriting Expressions.**

- 4.1.1. 2^{-2}
- 4.1.2. 2^{-6}
- 4.1.3. 2^5
- 4.1.4. 2^{-12}

Rewriting Expressions.

4.1.5. 5^{-1}

4.1.6. 5^{-3}

4.1.7. 5^2

4.1.8. 5^2

Simplifying Expressions.

4.1.9. 1

4.1.10. $x^4 y^2$

4.1.11. Cannot be simplified.

4.1.12. $8x^6 y^{12}$

4.1.13. Cannot be simplified.

4.1.14. $\pi^2 x^{20} y^8$

4.1.15. Cannot be simplified.

4.1.16.

$$\begin{aligned}
 (a^p)^r &= \left(\underbrace{a \cdot a \cdot \dots \cdot a}_{p \text{ times}} \right)^r \\
 &= \underbrace{\left(\underbrace{a \cdot a \cdot \dots \cdot a}_{p \text{ times}} \right) \cdot \dots \cdot \left(\underbrace{a \cdot a \cdot \dots \cdot a}_{p \text{ times}} \right)}_{r \text{ times}} \\
 &= \underbrace{a \cdot a \cdot \dots \cdot a}_{p \cdot r \text{ times}} \\
 &= a^{p \cdot r}
 \end{aligned}$$

4.2 • Algebra of Powers: Fractional Exponents

• Exercises

Solving Equations.

1. $x = -\sqrt[4]{20} \approx -2.115$ and $x = \sqrt[4]{20} \approx 2.115$

2. $b = -\sqrt[30]{\frac{34}{3}} \approx -1.084$ and $b = \sqrt[30]{\frac{34}{3}} \approx 1.084$

3. $x = \sqrt[3]{-\frac{2}{5}} \approx -0.737$

4. No real solutions.

5. $x = 1 - \sqrt{3} \approx -0.732$ and $x = 1 + \sqrt{3} \approx 2.732$

6. $x = -4$

Rewriting Expressions.

7. $3^{5/2}$
8. $3^{5/2}$
9. $3^{13/6}$
10. $3^{-1/20}$

Simplifying.

11. Cannot be simplified.
12. $xy\sqrt[3]{x}$
13. $\frac{y^2}{\sqrt[3]{x}}$
14. Cannot be simplified.

4.3 · Power Functions: Positive Integral Exponents

· Exercises

Power Functions.

1. $y = 4x^3$; $k = 4$; $p = 3$
2. $y = 3x^3$; $k = 3$; $p = 3$
3. Not a power function.
4. $y = 0.1x^3$; $k = 0.1$; $p = 3$
5. Not a power function.
6. $y = 9x^2$; $k = 9$; $p = 2$

Braking Distance.

7. Yes; $k = \frac{0.034}{\mu}$; $p = 2$
8. 59.5 feet; 238 feet
9. Factor of 4.
10. 416.5 feet
11. From left to right:
 - p is even and k is positive.
 - p is odd and k is positive.
 - p is odd and k is negative.
 - p is even and k is negative.

12. The graph in the middle.

13. The second graph.

14. A is $y = x^4$; B is $y = -x^4$; C is $y = \frac{1}{4}x^4$.

15. 4.298 cm

4.4 • Power Functions: Fractional and Negative Integral Exponents • Exercises

Power Functions.

1. $y = 10x^{1/2}$; $k = 10$; $p = \frac{1}{2}$

2. $y = 0.1x$; $k = 0.1$; $p = 1$

3. $y = \frac{1}{4}x^{17/12}$; $k = \frac{1}{4}$; $p = \frac{17}{12}$

4. Not a power function.

5. $y = \frac{1}{3}x^2$; $k = \frac{1}{3}$; $p = 2$

6. Not a power function.

7. From left to right:

- p is odd; k is positive.
- p is even; k is negative.
- p is odd; k is negative.
- p is even; k is positive.

8. The graph on the right.

9. The graph on the left.

10. Graph A is $y = 2x^{1/2}$, graph B is $y = x^{1/2}$, and graph C is $y = -x^{1/2}$.

11. $f(x) = 3x^{\frac{1}{2}}$

12. $g(x) = 2x^{-1}$

13.

(a) yes; $T = \frac{2\pi}{\sqrt{g}}L^{\frac{1}{2}}$ where the coefficient is $\frac{2\pi}{\sqrt{g}}$ and the exponent is $\frac{1}{2}$.

(b) approximately 1.42 seconds.

(c) approximately 0.47 m/sec².

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Colophon

This book was authored in PreTeXt.